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Design and simulation of wide-band low-loss and fabrication tolerant silicon nitride interleavers

Mémoire

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Résumé

L'objectif de ce travail est d'augmenter la performance et la tolérance de fabrication des interleavers en nitrure de silicium en utilisant la conception inverse de sous-composants et l'utilisation de guides d'ondes multimodes pour réduire les erreurs de phase introduites par les variations de fabrication.

Les interleavers sont des filtres passifs largement utilisés dans les liaisons de transceivers photoniques. Un interleaver augmente l'espacement de fréquence de la série de lasers en entrée, permettant ainsi une capacité de liaison évolutive au-delà du schéma conventionnel à une seule gamme spectrale libre. Les interleavers contribuent également à réduire la perte d'insertion du modulateur. Nous avons présenté deux conceptions d'interleavers. Le premier présente une large bande passante et une empreinte compacte, tandis que le second est tolérant aux variations de fabrication.

Étant donné que les seuls éléments dépendant de la fréquence dans le circuit de l'interleaver sont les coupleurs, nous avons utilisé un algorithme d'optimisation (l'optimisation par essaims particulaires) pour optimiser les coupleurs de manière à améliorer leur bande passante et leur perte d'insertion. En utilisant des dispositifs d'interférence multimodes conçus de manière inverse, il a été démontré que l'interleaver présente un rapport d'extinction de 16 dB, une perte d'insertion de 1.27 dB, une bande passante de -3 dB de 70 GHz, et un facteur d'atténuation relativement marqué de 0.5 dB/GHz.

Dans la deuxième conception, de larges largeurs de guide d'onde ont été utilisées, réduisant les erreurs de fabrication puisque le mode optique présente moins de recouvrement avec les parois latérales. Les guides d'onde dans cette conception pouvant supporter des modes d'ordre supérieur, nous avons utilisé des courbes d'Euler pour éviter l'excitation de ces modes. La courbure des courbes d'Euler augmente de manière adiabatique, contrairement aux courbes à courbure constante, permettant des courbes serrées même avec des largeurs de guide d'onde plus importantes. Le rapport d'extinction et la perte d'insertion pour cette conception sont respectivement de 13 dB et 2.5 dB. Une réponse à sommet plat a été obtenue avec une amplitude de fluctuation de la bande passante de 0.012 dB. La bande passante de -3 dB pour cette conception est de 50 GHz.

Abstract

The goal of this work is to increase the performance and fabrication tolerance of silicon nitride interleavers using inverse design of sub-components and utilization of multimode waveguides to reduce the phase errors introduced by manufacturing variations.

Interleavers are widely used passive filters in photonic transceiver links. An interleaver increases the frequency spacing of the input laser comb, allowing for scalable link capacity beyond the conventional single-free-spectral-range scheme. Interleavers also help reducing the modulator insertion loss. We have presented two interleaver designs. The first one features a wide bandwidth and a compact footprint, while the second is fabrication-tolerant.

Since the only frequency dependent components in the interleaver circuit are the couplers, we employed an optimization algorithm (particle swarm optimization) to optimize couplers in the interleaver in such way to enhance their bandwidth as well as their insertion loss. By using inversely designed multimode interference devices, it was demonstrated that the interleaver has an extinction ratio of 16 dB, an insertion loss of 1.27 dB, a -3 dB bandwidth of 70 GHz, and a relatively sharp attenuation factor of 0.5 dB/GHz.

In the second design, wide waveguide widths were used, reducing fabrication error as the optical mode has less overlap with the sidewalls. Since the waveguides in this design can support higher-order modes, we utilized Euler bends to prevent the excitation of these modes. The curvature of Euler bends increases adiabatically, unlike radially curved bends with constant curvature, allowing for sharp bends even with wider waveguide widths. The extinction ratio and insertion loss for this design are reported as 13 dB and 2.5 dB, respectively. A flat-top response was achieved with a passband ripple amplitude of 0.012 dB. The -3 dB bandwidth for this design is 50 GHz.

Table des matières

Résumé ii

Abstract iii

Table des matières iv

Liste des tableaux vi

Liste des figures vii

Liste des abréviations x

Liste des symboles xii

Acknowledgments xiv

Introduction 1

1 Fundamentals of Silicon Nitride devices and principles 11
- 1.1 Silicon nitride material platform . . . 11
- 1.2 Silicon nitride passive devices . . . 13

2 Principles of interleaving using ring-assisted Mach-Zehnder interferometer 23
- 2.1 MZI interleavers . . . 23
- 2.2 Ring-assisted MZI interleavers . . . 27
- 2.3 Anatomy of optical interconnects . . . 31

3 Designing a wideband SiN interleaver 38
- 3.1 Requirements . . . 38
- 3.2 Inverse designed MMI . . . 40
- 3.3 Designing a compact Y-splitter . . . 47
- 3.4 Calculating the OPL of the tunable coupler . . . 48
- 3.5 Simulation results . . . 52
- 3.6 Layout for fabrication . . . 52

4 Designing a Fabrication Tolerant Interleaver 55
- 4.1 Wide SiN waveguides . . . 56
- 4.2 Group index and length of the ring . . . 58

4.3 Euler bend 59
4.4 MMI design 60
4.5 Simulation results 62
4.6 Sensitivity analysis 62

Conclusion **67**

A Code listing **70**
A.1 Listing 1 70
A.2 Listing 2 73
A.3 Listing 3 78
A.4 Listing 4 80
A.5 Listing 5 82

Bibliographie **84**

Liste des tableaux

1.1 Comparing different material properties of Si and SiN in telecommunication O-band. 12

3.1 Geometrical parameter range for the optimization of the 50/50 MMI. 43
3.2 Final values for the DOFs to achieve minimum FOM. 44
3.3 Optimized values for different widths of the Y-splitter. Values are in μm units. 48

4.1 Corner analysis for Design I 66
4.2 Corner analysis for Design II 66

Liste des figures

1.1 Cross section of different layers in AMF fabrication process (adopted from [56]). 12
1.2 Waveguide dimensions of a standard SiN waveguide in O-band (right) and mode profile of the first TE mode (left). 13
1.3 Variation of the effective index versus waveguide width (left) and versus waveguide thickness (right). 14
1.4 Wavelength dependence of (a) effective index, (b) propagation constant, (c) group index, and (d) dispersion coefficient in the O-band corresponding to the waveguide dimensions indicated in Fig. 1.2. 15
1.5 Schematics of a directional coupler with field coupling coefficient κ 15
1.6 Supermodes in a coupled two-state system such as two coupled waveguides or two connected pendulums by a spring. 16
1.7 Schematic of a generic N-by-M MMI coupler. 17
1.8 Index profile and mode distribution in the core of MMI. 17
1.9 Schematics of field propagation inside MMI core. At distances equal to odd multiples of $3L_\pi/2$ two-fold or duplicate image of the input is formed. At distances equal to even multiples of $3L_\pi/2$, we hve single images. 19
1.10 Comparative analysis of loss due to the transmission to higher order modes in an Euler bend versus a radial bend. (a) Transmission of the fundamental mode at the input to the first four modes at the output of a 90° radial bend. (b) Transmission of the fundamental mode at the input to the first four modes at the output of a 90° Euler bend. As shown, the transmission to TE_0 mode in an Euler bend is much larger compared to radial bend. (c) 2-D profile of a radial and en Euler bend. As could be observed, the curvature of the radial bend is constant while the curvature of the Euler bend increases linearly along the bend. (d) Electric field profile of the output fields at the output of the radial and the Euler bends and the electric field profile of the fundamental mode. As shown, the field profile at the output port of the Euler bend has more overlap with that of the fundamental mode. (e) Field profile of the first 4 modes inside a waveguide. 21

2.1 A simple MZI structure consisted of two directional couplers and one imbalanced section ΔL 25
2.2 Amplitude response of an imbalanced MZI at the cross and bar ports 26
2.3 Comparison between the response of single stage and two-stage MZI interleaver. Cascading more interleavers leads to creation of more zeros, which could potentially flatten the response if the coefficients are chosen properly. 27
2.4 Schematics of a RAMZI interleaver. 28

2.5 An all-pass system. The phase response in this figure indicates the phase difference at the output relative to the input. . . . 28
2.6 RAMZI interleaver response for a) $|\kappa_r|^2 = 0.29$ and b) $|\kappa_r|^2 = 0.89$. In (b), dashed lines specify the region where the phase difference between the upper and lower arm of the MZI have π phase difference, leading to maximum transmission in one port and null transmission in the other port. In (a), the ϕ phase difference is not satisfied, hence leading to a non flat-top amplitude response. Note how the value of κ determines the region where π phase difference holds in the phase response. A MATLAB script has been provided in Appendix A code listing 4 that generates these plots. . . . 30
2.7 Schematics of different optical link architectures. (a) a simple point-to-point link using a single frequency CW laser input source. (b) A WDM link using array of single frequency lasers as input. (c) A WDM link using a comb source. (d) A WDM link where MRMs perform the modulation and mux/demux funtionality at the same time. MOD : modulator. PD : photodetector. . . . 33
2.8 Three architectures of interconnect transmitters using MRMs. a) single bus transmitter. b) band interleaving configuration where input frequency combs are segmented into separate groups. In this configuration, frequency spacing between adjacent channels remains the same as the frequency spacing of the input comb lines. c) even-odd interleaving where the frequency spacing is increases and segmented as well. . . . 35
2.9 Channel arrangement of a single-FSR link architecture. . . . 36
2.10 An example of a multi-FSR link with two stages of interleavers, enabling a four-fold increase in comb line spacing and channel capacity. . . . 37

3.1 A summary of the design flow. The design is performed in two steps, namely, device simulation section and circuit simulations. The results of the device simulation (indicated by red arrow) are fed into the circuit simulation. In the device simulation, we will design the components used in the circuit simulation and obtain compact models that could be used in the circuit simulations. In this chapter, we will discuss each section of this flow. As shown in the subset of the boxes in this figure, the simulations are mainly performed in commercially available Lumerical MODE, FDTD, and INTERCONNECT packages. . . . 39
3.2 Calculated mode profile of the waveguide used in the interleaver design in linear (left) and logarithmic (right) scales. . . . 40
3.3 Mode convergence versus number of mesh cells in each transverse coordinates. Dimensions of the FDE solver is set to $8 \times 5\mu m^2$ . . . 41
3.4 A schematics of the interleaver under design. Δd is the extra path length compared with the top arm. L_{ring} is the length of the ring's waveguide. The shaded yellow region indicates the tunable coupler. . . . 42
3.5 Illustration of defined DOF for MMI PSO optimization. . . . 44
3.6 Convergence of FOM after 20 generations. . . . 44
3.7 Power transmission of the PSO-optimized MMI. . . . 45
3.8 Schematics of 89/11 splitting ratio tunable coupler. . . . 45
3.9 Coupling ratio of the tunable coupler versus phase shift on the top arm of the MZI. . . . 46

3.10 Transmission of the tunable coupler with $\phi_t = 0.6484$ rad at the two output ports (left axis) and the power ratio (right axis). As shown, the ratio is close to theoretical 89/11 splitting. 46
3.11 Power transmission of the PSO-optimized Y-Splitter at the two output ports (right). Phase difference in radians at the two output ports(left). 47
3.12 Magnitude of the electric field of the (a) MMI and (b) Y-splitter. 47
3.13 Illustration of DOFs defined for PSO optimization of the Y-splitter. 48
3.14 (a) Coupling into the ring using a tunable coupler. (b) the equivalent circuit of (a) using an ideal point coupler instead of a tunable coupler. 48
3.15 Transmission of the interleaver at one of the output ports when (a) OPL of the tunable coupler is not considered, (b) OPL considered assuming a rectangular core MMI, and (c) a final length sweep has been performed in INTERCONNECT to correct for the rectangular assumption of the MMI. 51
3.16 Transmission of the interleaver at the two output ports. 52
3.17 Different performance metrics of an interleaver 53
3.18 Fabrication layout of the designed interleaver. 53
3.19 Fabricated O-band transmitter with one stage of interleaver and two modulation buses each containing 6 MRMs. 54

4.1 Comparison of magnitude of the electric field of the fundamental mode in two waveguides with dimensions 800×400 μm^2 (left) and 3000×400 μm^2 (right). 57
4.2 Effective index as a function of waveguide width variation for waveguides with different widths. 57
4.3 (a) derivative of effective index with respect to the waveguide width. (b) effective index versus width. 58
4.4 Effective index, (a), and group index (b) of SiN waveguide with transverse dimensions 3×0.4 μm^2 58
4.5 Comparison between an Euler bend versus a radial bend. 60
4.6 Transmission (%) versus wavelength and Euler bend effective radius for two waveguide widths equal to $W = 3$ μm and $W = 5$ μm. 60
4.7 Comparison between Euler bend and radial bend. Left plot : Transmission of the TE$_0$ and the total transmission in an Euler bend with $R_{eff} = 102$ μm. Right plot : Transmission of the TE$_0$ and the total transmission in a radial bend with $R = 102$ μm. 61
4.8 (a) Schematics of 50/50 MMI. (b) Field profile of the magnitude of the electric field inside the MMI. (c) Transmission at the two output prots. 62
4.9 Transmission of a linear taper versus its length 63
4.10 Transmission of the fundamental mode and the total power in a PSO-optimized MMI with core width equal to 9μm. It could be observed that since the PSO-optimized geometry leads to the excitation of higher order modes, the transmission to the fundamental mode has significant amount of loss. 63
4.11 INTERCONNECT simulation results of the fabrication tolerant interleaver. . . 64
4.12 Layout of the proposed fabrication tolerant design. 65
4.13 Effect of variation in the width of the waveguide on the transmission of the (a) Design II and (b) Design I. Each transmission is associated to a waveguide width, which has been chosen randomly between -30 nm to +30 nm. 66

Liste des abréviations

AR	Autoaggressive
ARMA	Autoaggressive Moving-Average
BOX	Buried Oxide
CMOS	Complementary Metal-Oxide Semiconductor
CW-WDM-MSA	Continuous-Wave Wavelength Division Multiplexing Multi-Source Agreement
DC	Directional Coupler
DIBL	Drain-Induced Barrier Lowering
DOF	Degree of Freedom
DWDM	Dense Wavelength Devision Multiplexing
FDE	Finite Difference Eigenmode
FIR	Finite Impulse Response
FOM	Figure of Merit
FSR	Free Spectral Range
FWM	Four Wave Mixing
GAA	Gate All Around
FET	Field Effect Transistor
GPU	Graphical Processing Unit
HBM	High Bandwidth Memory
HPC	High Performance Computing
IM-DD	Intensity Modulation Direct Detection
IOT	Internet of Things
IIR	Infinite Impulse Response
LIDAR	Light Detection and Ranging
LPCVD	Low Pressure Chemical Vapor Deposition
MA	Moving Average
MRM	Microring Modulator
MMI	Multimode Interference
MZI	Mach-Zehnder Interferometer
OPA	Optical Phased Array
OSNR	Optical Signal to Noise Ratio
OPL	Optical Path Length

PECVD	Plasma-Enhanced Chemical Vapor Deposition
PSO	Particle Swarm Optimization
RAMZI	Ring-Assisted Mach-Zehnder Interferometer
SiN	Silicon Nitride
SNR	Singal-to-noise Ratio
TE	Transverse Electric
TM	Transverse Magnetic
TPA	Two Photon Absorption
TOC	Thermo-Optic Coefficient
WDM	Wavelength Division Multiplexing

Liste des symboles

ϵ	Electrical permitivity
λ_0	Center wavelength
L_{DC}	Length of a directional coupler
κ	Power cross coupling coefficient
C	Amplitude cross coupling coefficient
S	Amplitude through coupling coefficient
L_{3dB}	3dB-coupler length
k_ν	Lateral wavenumber in MMI core
β_ν	Propagation constant in MMI core
W_M	MMI core width
L_π	MMI's beat length
ν	MMI core mode number
$W_{e\nu}$	Effective width of mode ν in MMI core
ψ_ν	MMI core eigenmodes
c_ν	Eigenmode expansion coefficients inside MMI core
x	Spacial direction
y	Spacial direction
z	Spacial direction
ω	Angular frequency
t	Time
n_{eff}	Waveguide effective index
n_g	Waveguide group index
$H_{Straight}$	Transfer function of straight waveguide
c	Speed of light in vacuum
τ	Propagation delay
L	Waveguide length
τ_0	Fundamental delay
$\mathfrak{z}$	z-transform variable
Z	Modified z-transform variable
ν_i^n	Velocity vector in particle swarm optimization algorithm
P_i^n	Position vector in particle swarm optimization algorithm
Pb_i^n	Best cognitive position in particle swarm optimization algorithm
g_i^n	Best collective position in particle swarm optimization algorithm

Δ_L	Imbalance length
ϕ_t	Phase shift due to thermal phase shifter
η	Waveguide loss
τ_r	Ring time delay
ϕ_r	Phase shift of all-pass system
τ_r	Time delay of lower arm in RAMZI
l_{ring}	Theoretical ring length for flat-top response
l_{ring}^{mod}	Modified ring length for flat-top response
R_r	Ring's radius
$\mathcal{S}$	Auxiliary variable
$\mathcal{F}$	Auxiliary variable
λ_{agg}	Reduced channel spacing in multi-fsr scheme
λ_{ch}	channel spacing in single-fsr scheme
N_{ch}	Number of channels in a single-fsr scheme
L_{int}	Number of interleaver stages in a link
$\Delta\lambda_{comb}$	Wavelength spacing between the lines of a comb source
ϕ_0	Global phase of MMI
l_{ϕ_0}	Equivalent waveguide section for global phase estimation
l_1	Length of the upper arm of MZI tunable coupler
l_2	length of the lower arm of the MZI tunable coupler
R_{eff}	Effective radius of an Euler bend
R_0	Constant of Euler bend
s	Path length
W_a	Access waveguide width
L_t	Taper length
n_{eff}^0	Effective index of the first mode in MMI core

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Introduction

Demand for more computation capacity

The digital world is experiencing rapid growth, mainly due to advanced applications such as artificial intelligence, the Internet-of-Things, high-performance computing, and social media. According to a study [1], it is predicted that in 2027, artificial intelligence data centers will use between 85 to 134 Terawatt-hours (Twh) per year, equal to the power consumed by the entire country of Sweden or Argentina.

The growth of AI and other computationally intensive applications depends on the scalable growth of computational capacity, which requires ever faster and smaller transistors. However, due to the wave nature of electrons [2] and complex manufacturability, transistors cannot reduce in size continuously, imposing a barrier on scalability. Therefore, the primary solution is to have more compute chips, e.g., GPUs, and connect them to increase the computation capacity. The "top500" organization [3] continuously lists the 500 most powerful supercomputers globally. The first rank supercomputer, according to this ranking system, boasts 8,699,904 CPU and GPU cores [4]. In order to fully exploit the immense computational power nested in these supercomputers, the cores must be able to communicate data as fast as the computation speed. In addition, data communication must consume a low amount of power.

The bottleneck

Gordon Moore, the founder of Intel, predicted in one of his articles that the number of transistors doubles every year in an integrated circuit. This prediction became a law in the electronics community known as Moore's law. However, in 2015, Gordon Moore himself admitted the impending end of Moore's law [5]. The first reason for Moore's law to fall is that to double the number of transistors, there is no other solution but to make the transistors smaller. Aside from the extreme manufacturing costs of tiny transistors, there is another barrier that is purely physical : when the transistor dimensions shrink in size, quantum effects, which disrupt classical transistor behavior, start to emerge [6; 2]. For example, when the transistor's channel shrinks in size, electrons tunnel through the gate oxide, which leads to current leakage [7]. This phenomenon is known as drain-induced barrier lowering (DIBL), one of the short-channel ef-

fects in transistors. As a result, a current might pass through the channel while the transistor is in its off state. Between 2010 and 2020, the bandwidth capacity of switch integrated circuits increased from 0.64 to 25.6 Tb/s [8]. In the same period of time, the process node decreased from 40 nm to 7 nm. It should be emphasized that even though the transistors can not reduce in size after a certain point, the compute bandwidth can still be improved by connecting them to form a network of computational nodes. However, the problem is the connections, or, more technically, the interconnects, which struggle to scale up with the ultra-high data rates required to make a network between fast compute nodes. Conventionally, copper-based interconnects are the primary method to communicate data on a chip or off the chip using Ethernet cables. The issue with conventional electrical interconnects is that they suffer from signal attenuation and distortion at high bitrates due to the skin effect and RC limitations.

Silicon photonics interconnects as a solution to interconnect bottleneck

Silicon photonics offers a solution to enable interconnects that exhibit ultra-fast and energy-efficient features. In 1984, Goodman et al. [9] proposed such silicon-based interconnects for the first time. These interconnects can be used between and within microelectronic chips [10]. In addition to the scalable speed silicon photonics offer, they are compatible with the complementary metal-oxide semiconductor (CMOS) process, which makes them economically suitable. Moreover, silicon photonics allows for the monolithic integration of photonics and electronics on a single chip.

What makes silicon photonics even more advantageous is that it is possible to associate different input data streams to different wavelengths and send them all together simultaneously. This method, called wavelength division multiplexing (WDM), increases the transceiver capacity by the number of wavelengths used. WDM technologies are categorized into two main groups : coarse wavelength division multiplexing (CWDM) and dense wavelength division multiplexing (DWDM). As the name suggests, the difference between the two is the spacing between frequency lines. However, in recent standards, the nomenclature is based on channel spacing rather than adjectives coarse or dense. For example, in Continuous-Wave Wavelength Division Multiplexing Multi-Source Agreement (CW-WDM MSA) [11], which is a standard group for WDM systems in O-band, there are DWDM-9nm, DWDM-18 nm and DWDM-36 nm wavelength grids standards. From a theoretical point of view, when the number of wavelength carriers increases, so does the link capacity. The primary question that arises when dealing with WDM systems is that what limits us from adding more wavelengths into the wavelength grid to even more increase the capacity. One major limitation in increasing the number of channels is complexity. For example, since today's most DWDM transceivers rely on frequency combs for the light source, increasing the number of channels put a heavy bur-

den on the frequency comb design and performance. Other limitations include cost, footprint, and power consumption. Besides the aforementioned limitations, inter-channel crosstalk due to modulation or multiplexing/demultiplexing is another reason why arbitrarily increasing the number of channels or wavelengths in a WDM system is not an optimal solution. We will discuss this limiting factor in more detail in section 2.3. The inter-channel crosstalk is directly related to the channel spacing [12; 13]. To decrease the impact of the crosstalk, one has to somehow separate the wavelength channels into two or more different paths, for example, odd-numbered and even-numbered channels (odd and even refers to the index of the input wavelength carriers where a number between zero to the maximum number of wavelength channels is associated to them). This strategy will automatically increase the channel spacing. This separation functionality can be realized using a photonic circuit component named interleaved.

Interleavers in WDM systems

Optical interleavers are passive optical devices that typically have one input port and two output ports. The output ports are complementary to each other in terms of frequency response, meaning the passband of the one output is the stopband of the other one. Interleavers are usually used in the begining section of an optical interconnect, on the transmitter side right after the laser source or on the receiver side. When the input multi-wavelength light comes from the laser to the interleaver, some of the wavelengths are sent to the first output port and other wavelengths are routed to the second output port. Interleavers can remedy the channel crosstalk by separating the wavelength channels into different routes (for example, odd and even number wavelength channels), increasing the wavelength spacing and reducing the crosstalk. In this text, a detailed working principle of interleavers has been provided, as well as its design considerations. Interleavers are realized using different methods such as cascaded Mach-Zehnder interferometers (MZI) or ring-assisted MZI (RAMZI) [14]. Take the example of a cascaded MZI interleaver. It consists of several MZIs connected serially. On top of each MZI, there are thermal phase shifters that act as post-fabrication fine-tuning. Furthermore, usually two or more stages of interleavers are used when the number of input wavelengths are increased [15]. In addition, the device has a relatively large footprint. The amount of loss in passive circuits such as MZI interleavers originate mainly from three sources : waveguide loss (due to sidewall roughness scattering and material loss), coupler loss, and bend loss. The loss increases when two or more interleavers are cascaded to increase the channel spacing. For instance, in a typical cascaded MZI interleaver, there are three MZIs connected to each other [16; 17], each one consisting of two input/output couplers, several 90-degree bends and hundreds of micrometers of waveguide length. If, for example, two of these interleaving stages exist, then the insertion loss will be twice as a single interleaver stage. Besides loss, another concern about the interleaver design is the phase error in the circuit associated with fabri-

cation errors. Since the operation principle of these devices is based on the interference of light, any phase error leads to deviation from the nominal design. The optical path length of the waveguide arms in the MZI interleavers are designed to realize a frequency response with accurate pass-bands for the output ports of the interleaver. Any fabrication error leads to a certain value of phase error. Since the fabrication errors, such as alternation in the width or thickness of waveguides, are inevitable [18; 19], the best way to tackle this issue is to implement thermal phase shifters on top of the waveguides in each MZI structure in the device [20]. By tuning the amount of current (thus, heat) in these micro-heaters, the refractive index of the waveguide changes slightly, providing an opportunity to compete against the phase error introduced by the fabrication errors. The phase tuning, however, does not come at no cost. The most important drawback of thermal phase tuning is power consumption. Typically, there are several phase shifters in one stage of interleaver [21]. For a π phase shift, a reasonably feasible value for a microheater's power consumption is around 15 mW per π phase shift [22; 23; 24; 25; 26; 27]. Assuming this amount of power consumption, on an approximate level, near 100 mW of power is consumed for a two stage interleaver with three MZIs in each stage. This could put a burden on realization of low energy per bit requirement in the future parallel petabits per second interconnects. Another challenge that arises due to microheaters is the complexity of measurement. When the number of microheaters increases, the test time will also increase accordingly, leading to higher test costs, especially in the volume production in the industry. Besides, the total footprint of the device will enlarge due to the need to put at least one electrical bond pad on the chip to apply DC voltage to turn the heaters on.

Realization of highly efficient interleavers

As discussed, future highly scalable petabits per second transceivers rely on the multi-wavelength parallelism, hence needing several (depending on the number of wavelength channels) interleaver stages. The efficiency of the interleaver hugely depends on the design parameters and also on the material platform that the interleaver is realized based on. A highly efficient interleaver, or any other photonic passive filters, is an interleaver that has 1) small footprint, 2) wideband operation, 3) low insertion loss, 4) high power handling, 5) high tolerance to fabrication tolerance. To address these requirements, in general, there are two approaches. The first one is to optimize the design using optimization methods. For example, there are several multimode interference (MMI) couplers used in the cascaded MZI-based interleavers. These MMIs have a certain amount of insertion loss, which contribute to the overall insertion loss of the interleaver. If the insertion loss of the MMI couplers can be reduced by using novel MMI design methods (for example using angled MMI structure [28]) then the whole interleaver can have less insertion loss. The second option to improve the efficiency of interleavers is to use an intrinsically lower-loss material platform rather than regular silicon in SOI. For instance, silicon nitride (SiN) is one of those materials that are preferred in passive optical filters because of its superior low-loss nature. In this text, we will design SiN-based interleavers. We

describe how SiN as an alternative material to silicon can partly contribute to the realization of a low-loss interleaver. The role of specific design methods to reduce the loss has also been discussed. It should be noted that SiN can be integrated into the SOI platform without adding much complexity because SiN is compatible with SOI. Numerous examples of SiN integration into silicon photonics platforms have been documented [29; 30; 31; 32; 33], highlighting SiN's appealing features for passive photonic devices.

Material-based improved efficiency

Compared to silicon, SiN has a lower bandgap (around 5 e.V). This lowers the intrinsic material absorption in telecommunication bands such as O-band and C-band. The propagation loss in a straight silicon strip waveguide at C-band is approximately 1.5 dB/cm. If we consider a typical cascaded MZI interleaver, for two stages of a silicon interleaver we can assume the total optical path length to be around 2 mm [34], resulting in 0.3 dB loss. If we use SiN interleaver, on the other hand, since its typical propagation loss is around 0.8 dB/cm, we will have a total of 0.16 dB loss due to waveguide propagation loss. Besides the material loss, silicon nitride waveguides are more immune to the sidewall scattering loss. Due to manufacturing imperfections, there is roughness at the sidewalls of the waveguides. These rough features act as nano-scatters that scatter the optical field in the waveguide, resulting in loss. The amount of scattered light due to roughness at the sidewalls is related to the index contrast of the waveguide core and its cladding [35]. While in the silicon platform, the waveguides' core index is about 3.5 and that of cladding (silica) is nearly 1.45, the SiN index is about 2, hence reduced index contrast in SiN waveguides. Note that even a waveguide sidewall roughness of only several nanometers can cause scattering loss equal to approximately 1 dB/cm [35]. Besides the nanometer scale feature size of the sidewall roughness in waveguides, which has a random behavior with a small correlation length, the thickness and width of the waveguide also vary with a much longer correlation length. For example, considering an MZI structure with waveguides designed to have 450 nm width, the waveguides that make this MZI structure can deviate in width (and also thickness) in the order of tens of nanometers. This sort of fabrication variation will cause phase errors in interferometric circuits such as MZIs and interleavers. As a rule of thumb, 1 nm waveguide width variation can lead to 1 nm shift in the spectral behavior of an MZI [35]. The low index contrast of the SiN waveguides can also reduce these phase errors because the effective index of these waveguides are less sensitive to variations in width and thickness [35; 19]. However, the downside of low index contrast is a larger footprint, mainly because of bend structures. Simply put, the optical mode in low index contrast waveguides is less confined in the mode, and as a result, more power will scatter in a sharply bend waveguide. Another benefit of SiN platform is its low sensitivity to thermal fluctuations. The thermo-optic coefficient of the SiN is $2.45 \times 10^{-5} K^{-1}$, which is about one order of magnitude smaller than that of Si which is $1.8 \times 10^{-4} K^{-1}$ [36; 37; 38]. The thermal fluctuations lead to phase error that could be compensated using an additional micro heater to control the overall temperature

of the device, which in turn consumes electrical power. In other words, the SiN reduces the optical loss directly by reducing material and scattering loss and the electrical loss indirectly by reducing the post-fabrication thermal tuning.

Another benefit of SiN is its high power handling capacity. Choosing advanced modulation formats such as 64-QAM increases the spectral efficiency of the link. However, higher-order modulation formats require increased optical power to maintain the signal-to-noise ratio (SNR). In other words, when the constellation points in a high-dimensional modulation format increase in number, in order to have a reliable SNR one must increase the distance between those constellation points, which could be realized either by reducing the loss of different components in the circuit or by increasing the level of the input optical power.

The first solution, i.e., reducing the component loss, seems too challenging, especially when dealing with complex coherent systems with numerous components such as splitters, filters, interleavers, I/O couplers, several modulators, phase shifters, etc. Therefore, increasing the optical power is the most straightforward solution. Increasing the input power in silicon photonics circuits causes unwanted behavior. Silicon has a strong two-photon absorption (TPA) coefficient [39; 40] in telecommunication bands. TPA is a nonlinear process that leads to the generation of free carriers. On the other hand, generated free carriers in silicon cause loss, and more importantly, increase the temperature. If the input power is increased, it can also lead to the physical damage of silicon components. The amplitude of TPA coefficient is directly related to the band gap of the material. Since SiN has larger bandgap, its TPA coefficient is significantly smaller than silicon, hence increasing the power handling of integrated circuits based on SiN.

Design-based improved efficiency

In the previous subsection, we described briefly how SiN can replace Si in a circuit such as interleavers to improve the overall power consumption and power handling, two important metric of such devices. Now, we explain what strategies other than alternative material platforms can improve the metrics. These strategies include 1) using infinite impulse response (IIR) filters instead of a finite impulse response (FIR). Note that a cascaded MZI structure is a FIR filter [43], 2) using wide waveguide widths in waveguide designs, and 3) improving the performance of the couplers in the circuit. In the following, these three strategies are explained.

In general, any filter implementation relies on FIR or IIR architectures. In FIR filters, a finite set of feed-forward arms are used to create a certain frequency response. A simple MZI is an FIR filter, which constitutes two feed-forward arms (two arms of the MZI) that interfere with each other and make the well-known sinusoidal type of transfer function. In order to make a more sophisticated frequency response, one must use more MZIs in a cascaded (higher order filters) topology. Even though these filters are well-established in academia and in the

market, they suffer from a large footprint. In particular, to create a multi-stage interleaver using a cascaded MZI structure, the footprint becomes very large, which leads to a smaller bandwidth density of the link. Another downside of large-footprint interleavers is that they will be more susceptible to fabrication variations and hence, phase errors. These phase errors must be compensated by applying thermo-optic phase shifters to tune the response, which will consume power. On the other hand, filters could be realized using a feedback mechanism to provide a pole in the Z-domain transfer function, making an IIR filter. In optical domain, this feedback mechanism could be realized using a resonator in the circuit. For example, if there is a ring resonator on top of one of the arms of the MZI, the ring will act as a feedback mechanism, and hence, an IIR filter would be realized. In contrast to FIR filters, the resonator's optical path length in IIR filters is used as a delay line not just one time but infinitely many times [44]. Due to this feedback mechanism, the IIR filters are much smaller in size compared to FIR ones. When a ring is used on one arm of the MZI it is called ring-assisted MZI (RAMZI). The drawback of RAMZI structures is the high optical intensity inside the ring. As in any resonator system, the light coupled into the ring experiences intensity buildup as with every round trip. As a result, the light intensity increases significantly in the ring, enabling nonlinear effects [45]. However, as discussed earlier, the SiN platform has a higher tolerance than Si ring resonators to handle high amount of optical power. The rigorous study of power handling of SiN RAMZI interleavers is beyond the scope of this manuscript, and to the best of our knowledge at the time of writing this text, there is no such analysis in the literature.

Another design strategy to improve the performance of the interleavers is to make them more tolerant to fabrication errors. Fabrication variations are inevitable due to insufficient accuracy of the tools in foundaries used for processes such as lithography exposure dose, chemical mechanical polishing, etching, and etc [18]. Fabrication errors will eventually lead to geometry deviation of components such as deviation in the width and thickness of waveguides. These deviations, in turn, change the effective index of waveguides, leading to phase error. However, it has been shown [46] that choosing a wider waveguide width will decrease the sensitivity of the effective index with respect to the variations in waveguide width. Hence, we will design an interleaver based on wide waveguide width.

Finally, as the third strategy to improve the insertion loss of the interleavers, we will use inversely designed MMI couplers with low loss and wide bandwidth behavior to be used in the interleaver circuit. Note that the insertion loss and bandwidth of the interleaver hugely depend on that of the couplers used in it.

Practical considerations

As for any integrated photonic filters, there are some important practical considerations. In this section, we briefly discuss some of these challenges.

Firstly, the interleaver must be compatible with CMOS foundries in order to make it financially viable. The interleaver proposed in this work is based on the SiN platform, which is CMOS compatible. The advantage of being CMOS compatible is that the platform will be much more efficient in terms of cost because the foundry process and infrastructures are already established and mature. It will also enable the co-integration of photonics and electronics on the same chip. This integration is critical for applications in future datacenters and in particular for co-packaged optics where electronic and photonic integrated circuits are located in the same chip. There are several prototyping/industrial foundaries such as Applied Nanotools and Advanced Micro Foundry. With their multi-project-wafer service, their SiN platform can be available at reasonable costs. Since the interleaver is considered a passive device, it could be fabricated in a passive-only run by the foundry. In the passive-only run, as the name suggests, only passive waveguides as well as metal layers are available. Hence, it will significantly reduce the number of required masks and fabrication cycles. In addition, it will reduce the cost and lead time of the fabrication, making it ideal for rapid prototyping. Another important consideration is the scalability. This is a challenge for high capacity links where data rates beyond petabits per second are required. In these links, several stages of interleavers are used to further increase the wavelength spacing between the input frequency channels. In Section 2.3, this concept is discussed in depth. The main challenge of cascading several stages of interleavers is the insertion loss of each stage. If four stages are used, then in the transmitter we have eight times the insertion loss of one interleaver. The reason we have eight times the insertion loss (instead of four) is that in the transmitter there is one interleaver (de-interleaver to be precise) before the modulation stage and one interleaver after it. In order to make the device more scalable, the insertion loss of each stage should be minimized as much as possible. Another consideration regarding scalability is about thermal tuning. As we will see later, there are three thermal tuners used in each of the interleavers. Increasing the number of interleaver stages significantly increases the power consumption due to these thermal tuners. Power consumption is one of the most important issues for future data centers, as discussed in the beginning of this section. In addition to power consumption, another issue related to thermal tuners is the number of bondpads needed to apply DC voltage. For each thermal tuner, there must be two bondpads. Even though the ground of different thermal tuners could be shared, in a multi-stage interleaved link, there will rise many different issues related to complex routing, packaging cost and complexity due to increased number of bondpads. Besides, the total footprint of the device will increase significantly. The increased footprint is in particular important for SiN-based interleavers, becuase of the fact that SiN filters already consume large footprint due to SiN's low index contrast which leads to enlarged bend radius.

Motivation

SiN-based integrated photonic components and devices have played a key role in systems that require low-loss behavior. While a variety of SiN devices have already been successfully implemented, there is no report of SiN interleaver. The motivation of this work is to design and analyze a SiN interleaver that not only boasts the low-loss property of SiN, but also it features compactness in terms of footprint, wide in regards of bandwidth, and high tolerance to fabrication errors.

Objective

The final objective is to realize a high-performance petabits per second link with scalable capacity. In particular, we aim to implement a hybrid Si/SiN platform to benefit from the advantages of both platforms simultaneously. The current study aims to shed light on one of the components of this transceiver, namely, interleaver. A transceiver of this kind will be essential in the coming decades to keep up with the demands of intensive digital workloads.

Thesis organization

The organization of the thesis is as follows. In **Chapter 1**, necessary information about the SiN platform and its advantages and disadvantages over the silicon platform has been provided. In addition, the working principles of components that are used in our interleaver design is investigated. In this regard, we start with the modal analysis of the waveguide. Then directional couplers and MMIs are described shortly. After that, other components such as I/O couplers (edge coupler and grating coupler) and Euler bends are briefly discussed.

Chapter 2 is dedicated to the theories behind interleavers. In this chapter, we will describe how an interleaver works using z-transform formalism. In order to comprehend the principles, we first take the example of a simple MZI structure and we will show how this device leads to significant performance deterioration. After that, we analyze the RAMZI structure and we will demonstrate how the addition of a ring on one arm of an MZI can significantly boost the performance of the interleaver. Additionally, mathematical requirements for a maximally flat RAMZI have been provided. In section 2.11 of this chapter, we go through an in-depth discussion on how interleavers can increase link capacity. In this regard, we first analyze conventional single-FSR links, and after showing its shortcomings, we will understand how a multi-FSR link (enabled by interleavers) could remedy the capacity scaling limitation.

In **chapter 3**, a wideband interleaver has been designed in an step-by-step manner. Firstly, the components used in the circuit are designed and optimized such as MMIs. Then, we extract the compact model of this components to run a circuit level simulation and extract the key figures of merit of the device.

In **chapter 4**, a fabrication tolerant interleaver has been designed. We will take almost the same steps in chapter 3, except that in the interleaver designed in this chapter, we will use wide waveguides. Using wide waveguides allows the existence of higher-order waveguide modes. Hence, in order to prevent coupling to higher order modes, which leads to loss, we will use Euler bends. The reason for using a wide waveguide width in this design is to reduce the overlap between the optical mode and the edges of the waveguide, which will lead to less fabrication variation as the width of the waveguide is prone to those variations. At the end of this section, we will compare the fabrication tolerance of this design to that of chapter 3 using a simulated corner analysis.

Finally, the results of the two designs are concluded and future pathways for SiN interleaver design are provided.

Chapitre 1

Fundamentals of Silicon Nitride devices and principles

In this chapter, we start with a short overview of SiN material platform and its unique advantages. Following this, we proceed with discussing the sub-components that will be used in the interleaver circuit e.g., waveguides, MMIs, Euler bends, and other essential elements.

1.1 Silicon nitride material platform

SiN is an ideal CMOS-compatible material platform for passive photonic integrated circuits. Firstly, SiN exhibits a minimal amount of material loss, thanks to its large bandgap [48]. It is therefore used in high Q-factor ring resonator systems used in comb generation [49]. In addition to its low material loss, because of its low index contrast, sidewall roughness has less impact on the SiN waveguide loss, compared with Si waveguides. Secondly, SiN has a smaller thermo-optic coefficient (TOC) than silicon by approximately an order of magnitude [37]. Low TOC enables designing of circuits with small sensitivity to thermal fluctuations. Its low TOC is especially valued in interferometric devices such as interleavers where phase errors due to thermal fluctuations can alter the interference behavior, leading to shift of the spectrum. Moreover, SiN has a high Kerr coefficient, making it ideal for nonlinear photonic applications. Furthermore, it has a low TPA coefficient, which makes it tolerant to high optical powers, increasing the power handling of the devices [50].

SiN is specifically advantageous in interferometric optical filters. As discussed previously, today's modern scalable optical transceivers rely on parallelism on wavelength, i.e., WDM systems. In WDM-based optical tranceivers, an interferometric optical filter as a mux/demux is usually responsible for routing or separating different wavelengths. This interferometric filter could be realized with different architectures such as RAMZI interleaver or cascaded MZI filter. What is common in all the conventional interferometric circuits is their low tolerance to

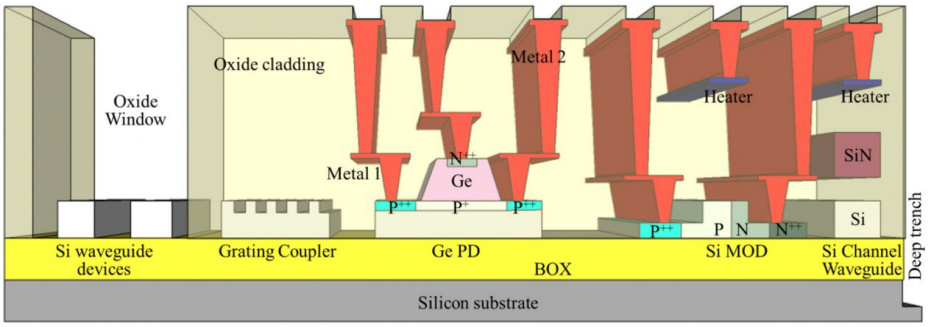


FIGURE 1.1 – Cross section of different layers in AMF fabrication process (adopted from [56]).

TABLE 1.1 – Comparing different material properties of Si and SiN in telecommunication O-band.

Parameter	**Si**	**SiN**
Refractive index ($1.31\mu m$)	3.5 [51]	1.98 [47]
Bandgap (eV)	1.14 [52]	$4.55 \sim 5.3$ [48]
TOC (K^{-1})	1.8×10^{-4} [38]	2.45×10^{-5} [37]
TPA coefficient (cm/GW) at 1310 nm	1.16 [40]	~ 0 [40]
Waveguide loss (dB/cm) at 1310 nm	~ 3.5 [41]	< 1 [42]

fabrication variations. SiN can improve this tolerance by offering low variation to temperature (because of its low TOC) and low phase error due to waveguide variations (due to its low index contrast). Hence, SiN can be especially advantageous in these sorts of devices that are sensitive to phase errors. In addition, as mentioned earlier, WDM filters can exhibit loss, if designed poorly. Therefore, the low-loss behavior of the SiN platform can contribute to the overall low-loss feature of the whole filter. It must be noted that the drawback of SiN as a material platform is that the device footprint becomes much larger than Si devices. The reason for this enlarged footprint is the large bends used in them. Since the index contrast is low, in order to reduce the radiation loss in SiN bends, its radius must increase, which leads to larger footprints. Attempts have been made to reduce the bend radius using novel bend designs [53; 54; 55].

Fig. 1.1 shows the cross-section of a multi-project wafer (MPW) process by Advanced Micro Foundry (AMF) [56]. This process is used as a reference in our designs. As shown in Fig. 1.1, SiN layer is deposited on top of Si layer. After the patterning of Si layer, SiO_2 is deposited and then it will be polished (planarization) using a process called chemical mechanical polishing (CMP) to make a flat surface prepared for deposition of SiN [57]. After CMP, SiN is deposited using either low-pressure chemical vapor deposition (LPCVD) or plasma-enhanced chemical vapor deposition (PECVD).

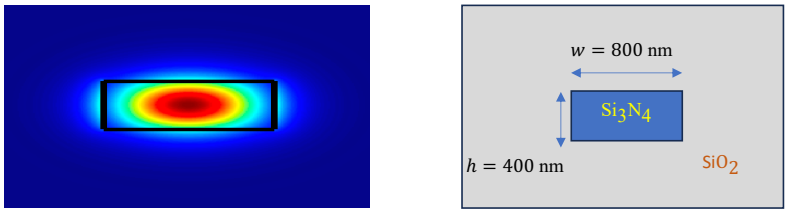


FIGURE 1.2 – Waveguide dimensions of a standard SiN waveguide in O-band (right) and mode profile of the first TE mode (left).

1.2 Silicon nitride passive devices

In this section, key components that will be used in an interleaver are analyzed. First we discuss SiN waveguiding analysis including its dispersion and single mode operation. Then, the working principles of two main coupling methods, i.e., MMI and directional couplers (DC), are discussed. Next, we focus on the Euler bend designs. As discussed later, we will use these Euler bends in our fabrication tolerant interleaver designs (chapter 4), where a wide waveguide bend with the least amount of intermodal coupling is required. We start with a quick overview of waveguides and their characteristics in the O-band spectral window.

1.2.1 Waveguides

Waveguides are essential elements in every photonic integrated circuit. In order to properly design a filter, one must first have enough information on the waveguides that will be used in that filter. For instance, and particularly relevant to interleavers, the group index of the waveguide (as we will see later) greatly impacts the frequency response of the interleaver. In addition, the waveguide's width must be chosen to meet the single-mode requirement. In addition, we need the compact model of the waveguide in order to perform a circuit-level simulation of the whole interleaver. As we will see in later chapters, a wideband response of the waveguide, which includes the dispersion information of it, will be obtained and used in circuit simulations.

Strip SiN waveguides

Fig. 1.2 shows a standard SiN waveguide with SiN index around 2 and cladding (SiO_2) index near 1.44 at $\lambda_0 = 1310$ nm. Notice that the thickness of this waveguide is chosen as 400 nm, which is within the range of SiN thickness in silicon photonic foundries, such as AMF [56]. This waveguide geometry is called "strip" waveguide in literature. Note that this waveguide dimension meets the single mode condition, as the only transverse electric guided mode are the fundamental TE_0 and TM_0 modes, which is shown in Fig. 1.3.

Dispersion and group index

Fig. 1.4 shows the dispersive characteristics of SiN waveguide. The effective index varies significantly in the O-band frequency window, but the group index variation is minimal. We will see later that in the interleaver design, we neglect the frequency dependence of the group index.

1.2.2 Directional coupler

In the interleaver designs that we propose in chapters 4 and 5, we will use MMI couplers because compared to DCs, they are less wavelength dependent and less sensitive to the variations in the width of the fabricated waveguides [58]. In addition, MMIs are more compact. Before we calculate the size of the DC for our application and compare it to that of an MMI coupler, a brief introduction to DCs are presented in the following.

A schematic of a DC is shown in Fig. 1.4. As shown, two waveguides are placed nearby each other so that the evanescent field coupling takes place. The coupling could be best described by the supermode concept (discussed in-depth in [59]). An excellent analogy is two pendulums connected by a spring. As shown in Fig 1.6. there are two symmetric and anti-symmetric motion modes that could be regarded as the eigenmodes of the two-pendulum system. Any motion could be expanded as a linear superposition of these two motions, same as the light in the coupled waveguides where every field distribution in those waveguides could be expressed as a linear combination of the two symmetric and anti-symmetric modes. For a more mathematically rigorous analysis of eigenmode expansion, the reader is referred to chapter 7 of [60].

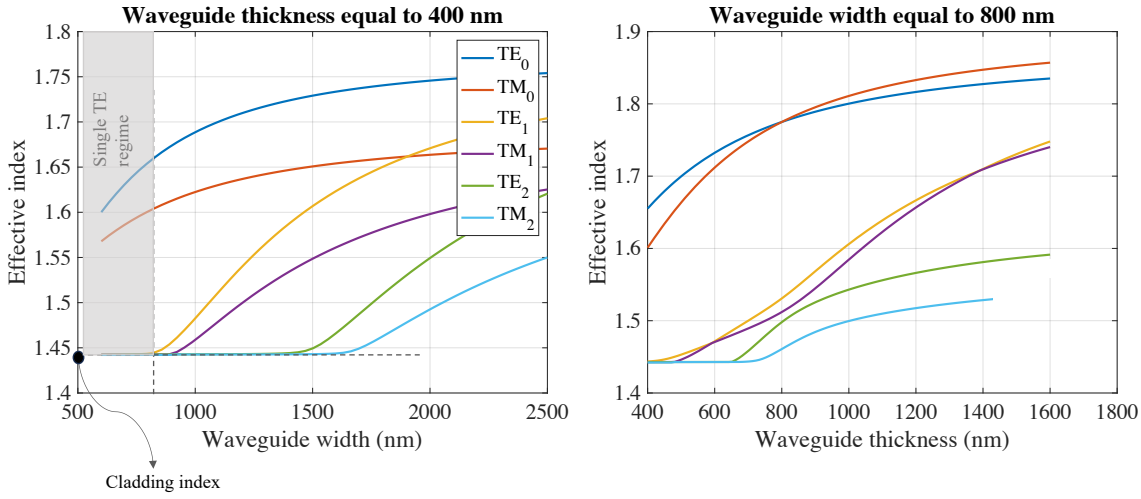


FIGURE 1.3 – Variation of the effective index versus waveguide width (left) and versus waveguide thickness (right).

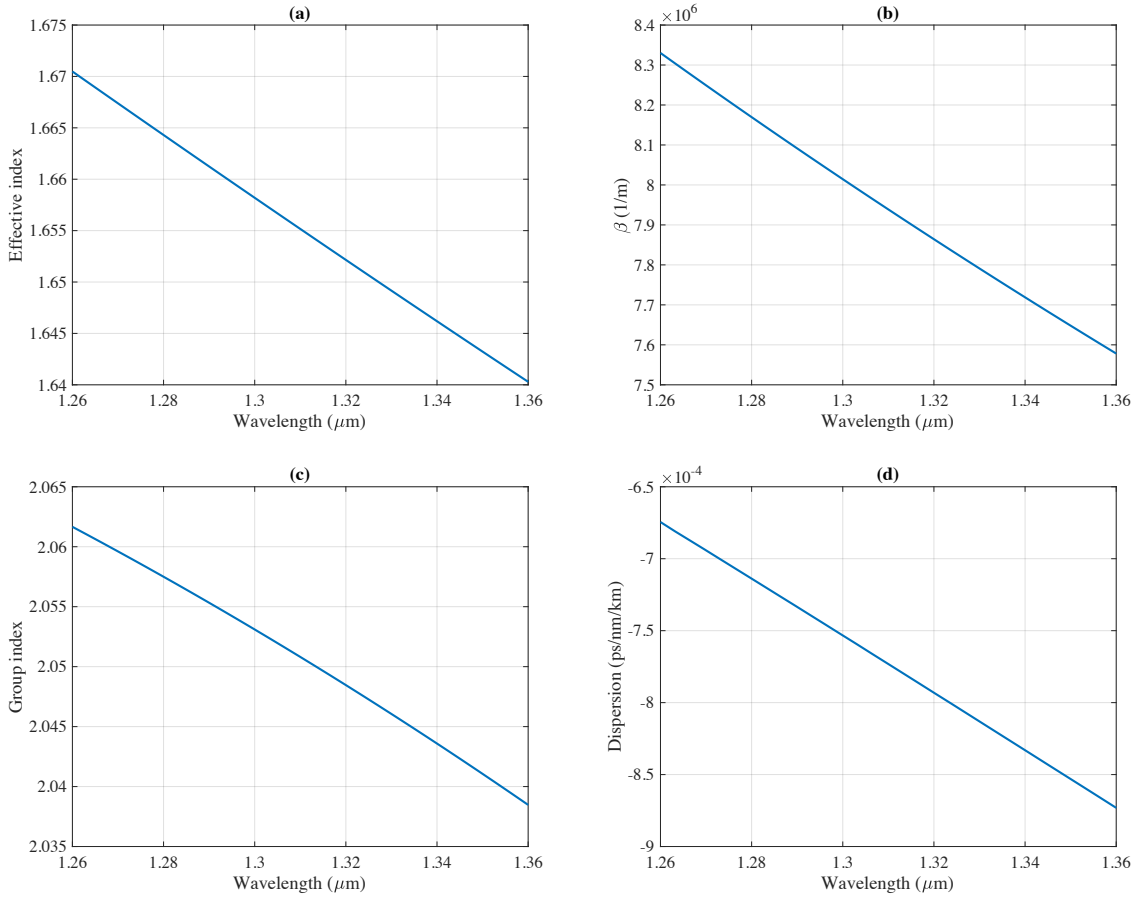


FIGURE 1.4 – Wavelength dependence of (a) effective index, (b) propagation constant, (c) group index, and (d) dispersion coefficient in the O-band corresponding to the waveguide dimensions indicated in Fig. 1.2.

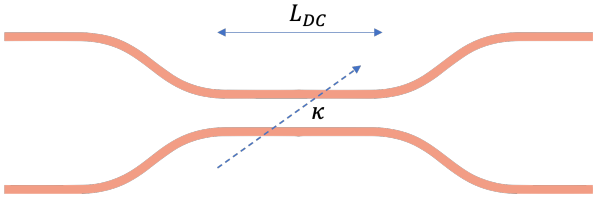


FIGURE 1.5 – Schematics of a directional coupler with field coupling coefficient κ

The field transfer matrix of a DC could be written as [84]

$$\begin{bmatrix} E_3 \\ E_4 \end{bmatrix} = \begin{bmatrix} \cos\psi & -\mathrm{i}\sin\psi \\ -\mathrm{i}\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \end{bmatrix} \tag{1.1}$$

where ψ is $\psi = \frac{\pi \Delta n_{eff}}{\lambda} L_{\mathrm{DC}}$ and Δn_{eff} is the difference between the effective index of the two supermodes (shown as the asymmetric and symmetric modes in Fig. 1.6). The coupling power could be chosen by varying the coupling length (for a fixed gap between the two waveguides).

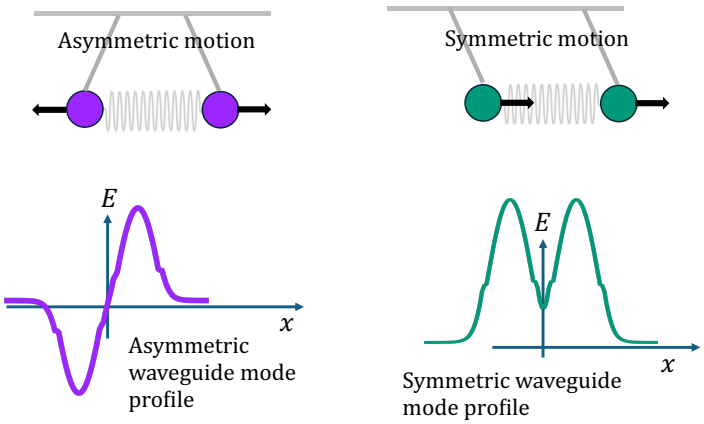


FIGURE 1.6 – Supermodes in a coupled two-state system such as two coupled waveguides or two connected pendulums by a spring.

For an arbitrary coupling coefficient κ, we have

$$\mathbf{H}_{\mathrm{DC}} = \begin{bmatrix} C & -\mathrm{i}S \\ -\mathrm{i}S & C \end{bmatrix} \tag{1.2}$$

where

$$\begin{aligned} C &= \cos\psi = \sqrt{1-\kappa} \\ -\mathrm{i}S &= -\mathrm{i}\sin\psi = -\mathrm{i}\sqrt{\kappa} \end{aligned} \tag{1.3}$$

In chapter 5, we will propose an architecture for interleaver that utilizes wide waveguides (3 μm $\times$ 0.4 μm). For this particular waveguide dimension and choosing the gap equal to 300 nm, which is the minimum possible spacing foundaries such as AMF [56], Δn_{eff} is calculated as $\Delta n_{eff} = 6.6 \times 10^{-4}$. For the calculation the effective index, we use commercially available Ansys Lumerical MODE Solutions. In case of a 3-dB DC, we have $\kappa = 0.5$. Therefore, at the center frequency of $f = 229$ THz (corresponding to center wavelength 1.31 μm), $L_{3dB} = 496.21\mu$m, which is impractically long. The main problem with this long DC, even if we neglect the large footprint that it consumes in the die, is that in general the bigger a device the higher the chance to fabrication variations. Notice that if the gap between the two waveguides could be smaller, then a more compact DC would be designed. However, considering AMF as foundry, 300 nm gap is the minimum, as mentioned earlier. Using Eq. 1.2 and Eq. 1.3 and doing some algebra, we can relate the input and output powers of a DC as follows.

$$\kappa = \frac{P_{\mathrm{cross}}}{P_0} = \sin^2(\psi), \tag{1.4}$$

where P_0 is the input optical power and P_{cross} is the power coupled to the cross port.

1.2.3 Multimode interference devices

The theory of self-imaging in a multimode waveguide was first introduced by Bryngdahl [61] and it was explained more rigorously by Ulrich [62]. The main idea behind self-imaging is as

follows. Self-imaging is a phenomenon that happens inside waveguides that support multiple modes, and it refers to the fact that the input light profile is reproduced at multiple distances along the propagation direction. MMI devices have two components. The core component is the center multimode waveguide and a couple of access waveguides based on the number of the inputs/outputs of the device (Fig. 1.7). The core waveguide (center waveguide) supports many modes, as it's usually much wider than the access waveguides. An MMI with M input access waveguides (located on the left side) and N output access waveguides is called an M×N MMI coupler. Typically, the access waveguides are designed to support only one mode.

The rigorous mathematical analysis of self-imaging could be found in primary works by Soldano et al [64] or Bachmann et. al [63]. Here, we pinpoint only those expressions and equations that will be used when designing a 50/50 MMI in chapters 3 and 4. We will understand how to design a 50/50 splitting ratio MMI and more importantly, we will highlight the global phase of MMI (discussed in depth in Chapter 3), which plays a key role in designing interleavers based that includes a tunable coupler structure to couple the light from the MZI to the ring. We begin our discussion with the geometrical and mode analysis of MMI, then we highlight the beat length and imaging condition (imaging length). Finally, the scattering matrix form of the MMI as well as its global phase is expressed.

In Fig. 1.8, MMI core with an index of n_r and cladding index of n_c is shown. The width of the core waveguide is W_M and the propagation direction is along z. Fig. 1.8 shows the amplitude of the modes inside the core waveguide. As shown, the MMI core can support several modes because of its wide cross section.

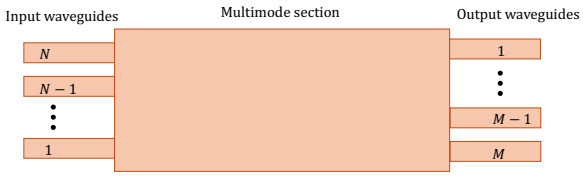


FIGURE 1.7 – Schematic of a generic N-by-M MMI coupler.

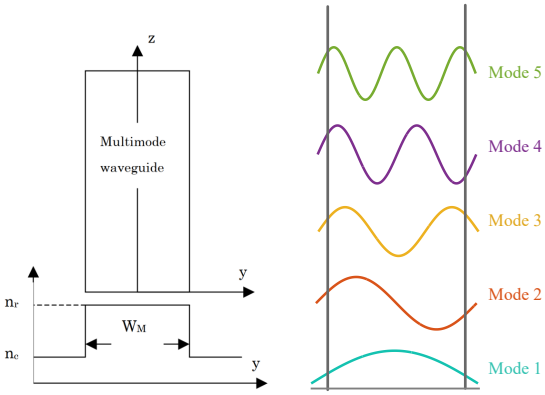


FIGURE 1.8 – Index profile and mode distribution in the core of MMI.

The MMI core (multimode section shown in Fig. 1.7) supports many modes because of its wide width. We indicate these modes by an integer ν. Indicating the longitudinal wavenumber of core modes by β_ν, it is shown in [64] that β_ν can be expressed as

$$\beta_v = n_r k_0 - \frac{(v+1)^2 \pi \lambda_0}{4 n_r W_e^2}, \tag{1.5}$$

where $k_0 = 2\pi/\lambda_0$ is the vacuum wavenumber at wavelength λ_0, and W_e is the effective width of the modes inside the core. A physically meaningful parameter, L_π, is defined as

$$L_\pi \equiv \frac{\pi}{\beta_0 - \beta_1}. \tag{1.6}$$

L_π is important when designing an MMI because as we will see later, the first step to design an MMI is to calculate its L_π. Note that L_π has a physical meaning. It means, for the two lowest order modes it takes a distance equal to L_π for the modes to constructively interfere and make a periodic pattern. Using this definition, the mode spacing becomes

$$(\beta_0 - \beta_v) \approx \frac{v(v+2)\pi}{3L_\pi}. \tag{1.7}$$

Since the modes in the MMI core are orthonormal [64], any field distribution inside the MMI core could be written as an algebraic sum of the eigenmodes of the MMI core. If we consider the input field as $\Psi(y,0)$, then it could be written as $\sum_v c_v \psi_v(y)$, where c_ν are coefficients that could be obtained from the orthonormal property of the guided modes inside the MMI core. Notice that $\psi_v(y)$ are the amplitude of MMI core guided modes and each of these eigenmodes propagate inside the core, indicated by a phase term equal to $\exp\left(j\left(\omega t - \beta_v z\right)\right)$. Assuming a harmonic time dependence, and using Eq. 1.7, it is shown step-by-step in [64] that the field distribution at a certain distance $z = L$ in the MMI core could be written as

$$\Psi(y,L) = \sum_{v=0}^{m-1} c_v \psi_v(y) \exp\left(j \frac{v(v+2)\pi}{3L_\pi} L\right). \tag{1.8}$$

Using this equation, it can be shown that at $L = p(3L_\pi)/2$, where p is an odd integer, we would have a duplicated image of the input filed, realizing a 50/50 splitting ratio MMI.

By using $L = p(3L_\pi)/2$ an plugging it into Eq. 1.8 and performing some algebraic manipulations, we will have

$$\Psi\left(y, \frac{p}{2} 3L_\pi\right) = \frac{1 + (-j)^p}{2} \Psi(y,0) + \frac{1 - (-j)^p}{2} \Psi(-y,0) \quad . \tag{1.9}$$

Notice that the first duplicated image occurs at $z = 3L_\pi/2$, and the second one at $z = 9L_\pi/2$, corresponding to $p = 1$ and $p = 3$, respectively. However, when 50/50 splitting application is

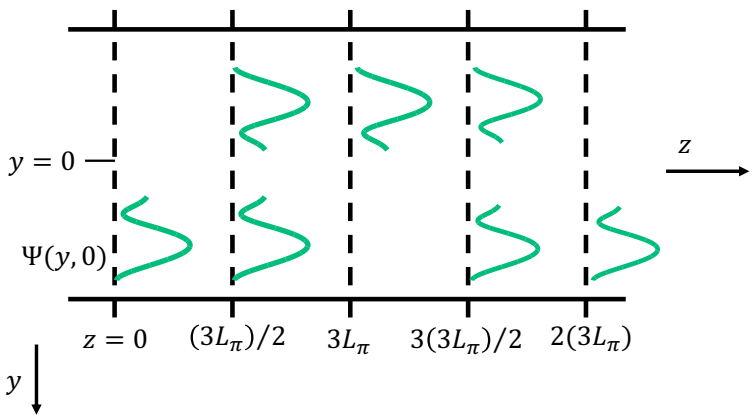


FIGURE 1.9 – Schematics of field propagation inside MMI core. At distances equal to odd multiples of $3L_\pi/2$ two-fold or duplicate image of the input is formed. At distances equal to even multiples of $3L_\pi/2$, we hve single images.

considered, MMI length is always chosen to be $3L_\pi/2$ to have a more compact device. This is shown in Fig. 1.9.

A MATLAB script for calculating the electrical field at the output of the MMI has been provided in code listing 3 in appendix A. In this script, the input field is first decomposed into the core modes (ψ_ν), which have been assumed to have a sinusoidal behavior inside the core, and then after calculating the propagation constant of the modes and summing the propagated fields, the overall field inside the core at every point is calculated.

It is quite insightful to derive a scattering matrix representation of a 50/50 splitting ratio MMI (see section 3.4, where the scattering matrix of 50/50 MMI is used). We start from Eq. 1.10 and consider $p = 1$, which corresponds to the first double image (see Fig. 1.9) and can be written as :

$$\Psi\left(y, \frac{3}{2}L_\pi\right) = \frac{1-j}{2}\Psi(y,0) + \frac{1+j}{2}\Psi(-y,0) \quad , \tag{1.10}$$

which shows that the field at the 50/50 length, i.e. $3L_\pi/2$, is a linear combination of the input fields. To obtain the scattering matrix, we need another expression for $\Psi\left(-y, \frac{3}{2}L_\pi\right)$, which is equal to $\frac{1-j}{2}\Psi(y,0) + \frac{1+j}{2}\Psi(-y,0)$. Hence, we can form the following scattering matrix :

$$\begin{pmatrix} \Psi\left(y, \frac{3}{2}L_\pi\right) \\ \Psi\left(-y, \frac{3}{2}L_\pi\right) \end{pmatrix} = \begin{pmatrix} \frac{1-j}{2} & \frac{1+j}{2} \\ \frac{1+j}{2} & \frac{1-j}{2} \end{pmatrix} \begin{pmatrix} \Psi(y,0) \\ \Psi(-y,0) \end{pmatrix}. \tag{1.11}$$

Note that the above matrix could be written as

$$S = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-j\frac{\pi}{4}} & e^{j\frac{\pi}{4}} \\ e^{j\frac{\pi}{4}} & e^{-j\frac{\pi}{4}} \end{pmatrix}. \tag{1.12}$$

A generalized form of the above scattering matrix is

$$S = \frac{j e^{j\varphi_o}}{\sqrt{2}} \begin{pmatrix} 1 & j \\ j & 1 \end{pmatrix}, \tag{1.13}$$

where $e^{j\varphi_o}$ is know as the global phase term, and in ref [63], it is proven to be equal to

$$\varphi_o = -\beta_o L_N^{M=1} - \frac{3\pi}{4} \tag{1.14}$$

where β_0 is the propagation constant of the first mode in the core of the MMI and L_N is the length of the MMI to have N images at the output. Notice that for 2-by-2 MMI, $N = 2$. Also note that L_N is related to the beat length as $L_N = 3L_\pi/N$. As we will discuss in chapter 3, *the global phase term in Eq. 1.14 should be taken into consideration when designing a RAMZI interleaver.*

1.2.4 Euler bends

Since SiN waveguide index contrast is relatively low, a waveguide bend is more prone to radiation loss compared to silicon platform. Hence, the most straightforward solution is to increase the radius of the bend [66]. However, this comes at the cost of an increased footprint. Another approach is using a different bend profile, such as Euler bends. In Euler bends, the bend's curvature increases gradually along the length of the bend. This adiabatic increase in curvature will prevent loss due to radiation.

However, the main reason we will use the Euler bend in our design is not to mitigate radiation loss. As we will see in Chapter 4, the waveguides used in the interleaver have a width equal to 3 μm. This waveguide can support higher-order modes. The existence of a higher order mode in a waveguide bend can lead to a relatively high amount of loss as the power of the fundamental mode could be coupled to those higher order modes along the waveguide bend [67]. In an Euler bend, because its curvature increases linearly along the bend (we will see that in chapter 4), less power, compared to radial bend, is coupled from the fundamental mode to the higher order modes. This behavior is depicted in Fig. 1.10. This figure compares the transmission behavior of a radial bend and an Euler bend. The waveguide width and bend radius are kept the same for both bends. The fundamental mode (TE_0) was injected as the input, and transmission to the first four TE modes was obtained. In Fig. 1.10 (a) and (b), the transmission of the input to these four modes is depicted. As it can be observed, in an Euler bend, most of the input power is transferred to the output TE_0 mode. In other words, the power is preserved in the fundamental mode, and only a tiny amount of power is leaked into the higher-order modes.

In contrast, for the radial mode, we can see that much more power (compared to that of Euler bend) has been transferred to the higher-order modes. Another way to look at this is by calculating the overlap between the field profile of the fundamental mode and the output field. In Fig. 1.10 (d), it could be seen that the field profile at the output port of the Euler bend, compared to the radial bend, has more overlap with the field profile of the fundamental mode.

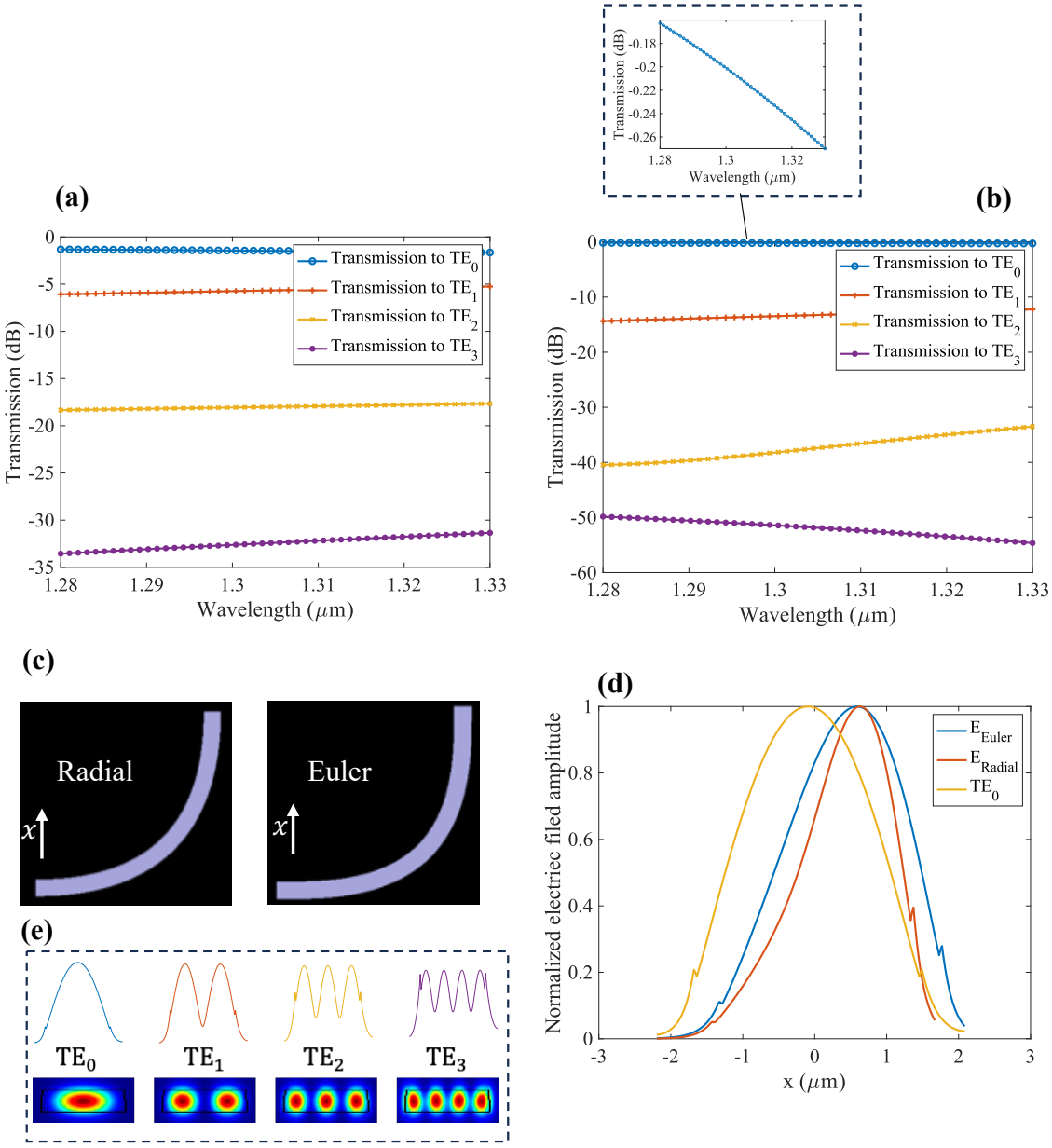


FIGURE 1.10 – Comparative analysis of loss due to the transmission to higher order modes in an Euler bend versus a radial bend. (a) Transmission of the fundamental mode at the input to the first four modes at the output of a 90° radial bend. (b) Transmission of the fundamental mode at the input to the first four modes at the output of a 90° Euler bend. As shown, the transmission to TE_0 mode in an Euler bend is much larger compared to radial bend. (c) 2-D profile of a radial and en Euler bend. As could be observed, the curvature of the radial bend is constant while the curvature of the Euler bend increases linearly along the bend. (d) Electric field profile of the output fields at the output of the radial and the Euler bends and the electric field profile of the fundamental mode. As shown, the field profile at the output port of the Euler bend has more overlap with that of the fundamental mode. (e) Field profile of the first 4 modes inside a waveguide.

In chapter 4, we will use Euler bends to realize fabrication tolerant interleavers with wide waveguide widths. In Appendix A, Listing 1, a MATLAB script for generating an Euler bend with arbitrary parameters has been provided.

1.2.5 Y-splitters

Another device that is used in our design is Y-splitter. A Y-splitter is a three port device with one input and two outputs and it directs 50% of the input power to the first output port and the rest to the other output port. Unlike MMIs, output ports are at the same phase with each other (notice to the MMI's output phase difference in Eq. 1.10). This feature will allow us to use less amount of power by excluding the need for additional phase shift on the ring inside the interleaver circuit (this is discussed in chapter 4 in detail).

1.2.6 Input/Output couplers

Input/Output couplers are essential components in every integrated circuit where the light comes from an external source such as a laser or a fiber. There are two main methods to couple the light into the chip : edge coupling and vertical coupling. Vertical couplers are commonly realized using a grating coupler. In this scheme, the fiber is placed above the chip at a certain angle [68; 69; 70; 71; 72; 73]. Grating couplers are ideal for rapid testing because compared to edge coupler they are easier to align. Additionally, since they are compact in size, it allows more devices be placed on a certain area. There are two main drawbacks for grating couplers. Firstly, they exhibit a relatively strong wavelength dependency, and secondly the coupling efficiency is lower than the edge couplers [74]. On the other hand, edge couplers have a larger footprint and are more difficult for coupling alignment [75; 77; 78; 79; 80; 81; 82; 83]. However, they excel in coupling efficiency, broad bandwidth and polarization independence. Edge couplers are less commonly used for rapid testing and they are usually used in product manufacturing.

The most simple design for an edge coupler is an inverse taper structure, which has become a standard solution for both silicon and SiN circuits. The inverse taper edge couplers typically exhibit wide bandwidth covering over 100 nm. One challenge in these structure is the leakage of mode from the inverse taper to the silicon substrate. The tip of the inverse taper should ideally be small enough so that the mode at the tip be expanded to a diameter of approximately 10 μm for optimal coupling with SMF-28 fiber. At this point we have discussed essential components used for realization of interleavers, which include MMIs, waveguides, I/O couplers. We emphasized more on the theory behind MMIs because we will see later how this component can have a significant impact on the overall design performance. In the next chapter, the we will focus on the mathematical principles of RAMZI interleavers and also we will understand what advantages do we get from these interleavers in a WDM-based optical transceivers.

Chapitre 2

Principles of interleaving using ring-assisted Mach-Zehnder interferometer

In this chapter, we will discuss the mathematical principles of interleaving structures. We start with a simple MZI structure and explain why such structures lead to a significant bandwidth reduction. After understanding the limitation of a simple MZI structure as an interleaver, we will explain how a RAMZI structure could remedy those shortcomings by introducing a flat top response. For the flat top behavior of the filter, we analyze different conditions on the length of the ring, the MZI's differential length, and the coupling coefficient from the top arm of the MZI to the ring. After discussing the mathematical foundations of the interleaver, we will consider using the interleaver in a transceiver link and explain on how the interleavers could lead to the bandwidth scaling of those links.

2.1 MZI interleavers

We start our discussion using a simple MZI interleaver. The fundamental configuration of an interleaver based on a MZI is illustrated in Fig. 2.1, and its power transfer function is presented in Fig. 2.2. The MZI can be viewed as a sequence of three components : the input directional coupler, a differential delay section, and the output coupler. Although MZIs are well-established and straightforward to analyze, we will start our discussion with them. After mathematically observing the drawback of a simple MZI interleaver, it will be more evident why a RAMZI can improve performance.

Following the methodology introduced by Madsen and Zhao [84], this section will start with an overview of the analytical tool for examining filters using the Z-transform and its connection to the conventional transfer function

In the analysis provided in the following sections, we follow the notations from [84]. Z-transform is used for simplifying the notations. Notice that Z-transform is used frequently in digital filter design and analysis. Simply put, the Z-transform formalism is adopted for optical filter design because optical filters have discrete time delays. For instance, a MZI is consisted of two time delays associated to the two arms with different lengths. Also notice that in all of the following analysis, without losing generality, we assumed lossless waveguides.

2.1.1 z-transform and transfer function

The transfer function of a straight waveguide with length L could be written as [84] :

$$H_{\text{straight}}\left(f\right) = \mathrm{e}^{-\mathrm{i}k_0 n_{eff} L} \tag{2.1}$$

where $k_0 = 2\pi f/c$, and n_{eff} is the waveguide's effective index. This transfer function can be written also as a function of delay inside the waveguide as

$$H_{\text{delay}}\left(f\right) = \mathrm{e}^{-\mathrm{i}2\pi f\tau}, \tag{2.2}$$

where $\tau = Ln_{eff}/c$. In the lattice filters and RAMZI structures (to be explored subsequently), various delays exist, all of which are integer multiples of a fundamental delay denoted as τ_0. It is therefore logical to introduce a normalized frequency

$$f' = f\tau_0 \tag{2.3}$$

enabling us to express Eq. 2.2 as

$$H_{\text{delay}}\left(f'\right) = \mathrm{e}^{-\mathrm{i}2\pi f'} \tag{2.4}$$

By defining a new complex valued variable $\mathscr{z}$ as

$$\mathscr{z} = \mathrm{e}^{\mathrm{i}2\pi f'}, \tag{2.5}$$

Eq. 2.4 becomes

$$H_{\text{delay}}\left(\mathscr{z}\right) = \mathscr{z}^{-1}. \tag{2.6}$$

If a filter has zeros only in its transfer function, then it belongs to finite impulse response (FIR) or moving average (MA) filters. On the other hand, if it has at least one pole in its transfer function, it is categorized as infinite impulse response (IIR) filters. If IIR filters have only poles then it is an autoregressive filter (AR) and if it consists of both zeros and poles, then it belongs to autoregressive moving average (ARMA) filters. [84].

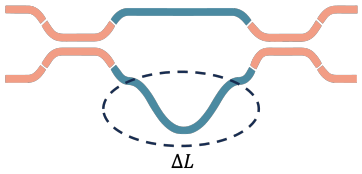


FIGURE 2.1 – A simple MZI structure consisted of two directional couplers and one imbalanced section ΔL

2.1.2 Imbalanced section

The imbalanced section is formed by two waveguides of different lengths ($L1$ and $L2$). In practice, these imbalanced structures comes with a thermo-optic phase shifter on top of one of the waveguides. In order to account for that phase shifter in our analysis we include ϕ_t, which represents the amount of phase shift introduced by a phase shifter, in one of the branches as follow [85]

$$\mathbf{H}_{\text{delay}} = \begin{bmatrix} \mathrm{e}^{-\mathrm{i}k_0 n_{eff}(f) L_1} \mathrm{e}^{-\mathrm{i}\varphi_t} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}k_0 n_{eff}(f) L_2} \end{bmatrix} \tag{2.7}$$

where τ is the differential delay and is given by

$$\tau = \frac{1}{\Delta f_{\text{FSR}}} = \frac{(L_1 - L_2)\, n_g}{c} = \frac{\Delta L n_g}{c} \tag{2.8}$$

where ΔL is the imbalance length and n_g is defined as [85]

$$n_g = n_{eff} + f \frac{\partial n_{eff}}{\partial f} \tag{2.9}$$

Note that the reason that the differential delay system is represented in a matrix form is that it has two inputs and two outputs, such as an MZI with couplers at the input and output sections. If we factor the phase shift introduced in second path, we would have the transfer matrix as

$$\mathbf{H}_{\text{delay}} = \mathrm{e}^{-\mathrm{i}\beta(f) L_2} \begin{bmatrix} \mathrm{e}^{-\mathrm{i}2\pi f \tau} \mathrm{e}^{-\mathrm{i}\varphi_t} & 0 \\ 0 & 1 \end{bmatrix} \tag{2.10}$$

Note that the bold notation for $\mathbf{H}_{\text{delay}}$ indicates that it is written as a matrix form. By introducing a new variable $Z^{-1} = \mathrm{e}^{-\mathrm{i}\varphi_t} \mathscr{z}^{-1}$ we arrive at

$$\mathbf{H}_{\text{delay}} \cong \begin{bmatrix} Z^{-1} & 0 \\ 0 & 1 \end{bmatrix} \tag{2.11}$$

2.1.3 MZI based interleaver

Since the imbalanced MZI is consisted of one imbalance length section between two directional couplers, the transfer matrix describing such device could be written as

$$\mathbf{H}_{\text{MZI}} = \mathbf{H}_{\text{DC}_2} \mathbf{H}_{\text{delay}}\, \mathbf{H}_{\text{DC}_1} \tag{2.12}$$

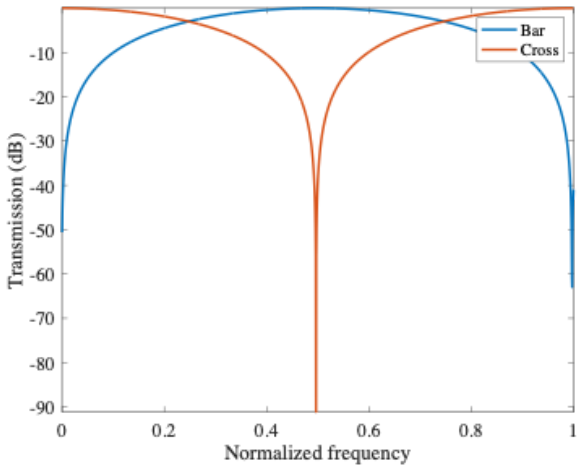


FIGURE 2.2 – Amplitude response of an imbalanced MZI at the cross and bar ports

Using the transfer matrices that was derived for delay section (Eq. 2.11) and for DCs, the MZI's transfer matrix is written as [85]

$$\begin{aligned}\mathbf{H}_{\mathrm{MZI}} &= \begin{bmatrix} H_{11}(\mathrm{z}) & H_{12}(\mathrm{z}) \\ H_{21}(\mathrm{z}) & H_{22}(\mathrm{z}) \end{bmatrix} = \begin{bmatrix} A(\mathrm{z}) & B^R(\mathrm{z}) \\ B(\mathrm{z}) & A^R(\mathrm{z}) \end{bmatrix} = \\ &= \begin{bmatrix} -S_1S_2 + C_1C_2Z^{-1} & -\mathrm{i}\left(C_1S_2 + S_1C_2Z^{-1}\right) \\ -\mathrm{i}\left(S_1C_2 + C_1S_2Z^{-1}\right) & C_1C_2 - S_1S_2Z^{-1} \end{bmatrix}\end{aligned} \tag{2.13}$$

If the input/output couplers have similar coupling coefficient then we have

$$\begin{aligned}\mathbf{H}_{\mathrm{MZI}} &= \begin{bmatrix} -S^2 + C^2Z^{-1} & -\mathrm{i}SC\left(1 + Z^{-1}\right) \\ -\mathrm{i}SC\left(1 + Z^{-1}\right) & C^2 - S^2Z^{-1} \end{bmatrix} = \\ &= \begin{bmatrix} -S^2Z^{-1}\left(Z - \frac{C^2}{S^2}\right) & -\mathrm{i}SCZ^{-1}(Z+1) \\ -\mathrm{i}SCZ^{-1}(Z+1) & -C^2Z^{-1}\left(Z - \frac{S^2}{C^2}\right). \end{bmatrix}\end{aligned} \tag{2.14}$$

Using $Z^{-1} = \mathrm{e}^{-\mathrm{i}\varphi_t}\mathrm{z}^{-1}$, together with Eq. 2.5, Eq. 2.3, and Eq. 2.14, and assuming $C = S = \sqrt{1/2}$ (50/50 coupling ratio), we have the frequency response (power transfer) at the bar port [85]

$$|H_{11}(f)|^2 = |H_{22}(f)|^2 = \sin^2\left(\frac{2\pi f\tau}{2} + \frac{\varphi_t}{2}\right) \tag{2.15}$$

and

$$|H_{12}(f)|^2 = |H_{21}(f)|^2 = \eta_{\Delta L}\cos^2\left(\frac{2\pi f\tau}{2} + \frac{\varphi_t}{2}\right) \tag{2.16}$$

for the cross port of the MZI.

As seen from the last two equations, the output power from MZI has a sinusoidal pattern. This is actually the drawback of MZI as an interleaver because the stopband is significantly narrow,

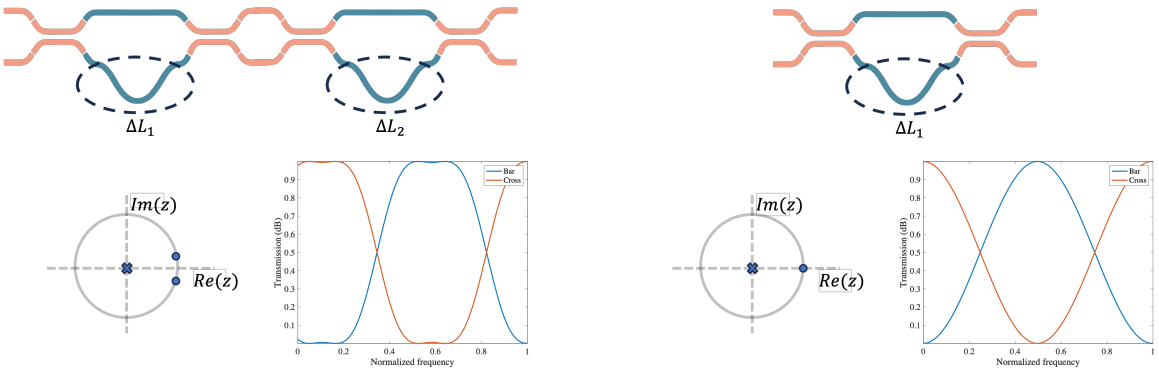


FIGURE 2.3 – Comparison between the response of single stage and two-stage MZI interleaver. Cascading more interleavers leads to creation of more zeros, which could potentially flatten the response if the coefficients are chosen properly.

making most of the bandwidth ineffective. Therefore, we need a more rectangular shape for the pass band of the filter.

Notice that ideal rectangular frequency response means that in time domain we need a sinc-shape pattern that is extended infinitely in time domain, which is physically unattainable because of causality. However, there are methods to approximately achieve the flat-top rectangular shape for the frequency response. One example of these methods is by simply cascading MZI structures with varying imbalance length. In Fig. two stages of MZIs have concatenated to form a filter with a more flat response. The reason, as shown in Fig. 2.3, is that cascading two MZIs lead to creation of an extra zero. Now, having two zeros (its position could be changed by proper choice of coupling coefficients of DCs) they are placed around the unit circle such that the transmission becomes more flat near the pass band of cross or bar port. Adding more MZI stages will introduce more and more zeros and hence a more flat response over the pass band can be achieved. However, adding more stages consumes huge amount of space area on the chip and also since each stage introduce a certain amount of insertion loss, the overall multi stage circuit could be prone to high amount of loss.

2.2 Ring-assisted MZI interleavers

Apart from cascaded MZI structure, an alternative approach for designing a flat-top interleaver is by integrating a ring resonator within an asymmetric MZI, as depicted in Fig. 2.4 [86; 87; 88]. The inclusion of the ring introduces a frequency-dependent nonlinear phase shift in one arm, while preserving a unity amplitude response. Before delving into the analysis of this interleaver, we will examine the frequency response of a single ring resonator. In Fig. 2.5, the ring resonator is depicted alongside a waveguide coupled to it. A fraction of the light propagating through the waveguide is coupled into the ring. Following one roundtrip, a portion of that light is re-coupled into the straight waveguide, while the remainder proceeds for a second roundtrip. This process continues until it reaches to steady state. Upon defining $z_r = e^{i2\pi f \tau_r}$ where τ_r is

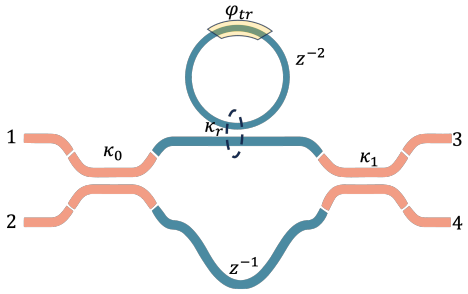


FIGURE 2.4 – Schematics of a RAMZI interleaver.

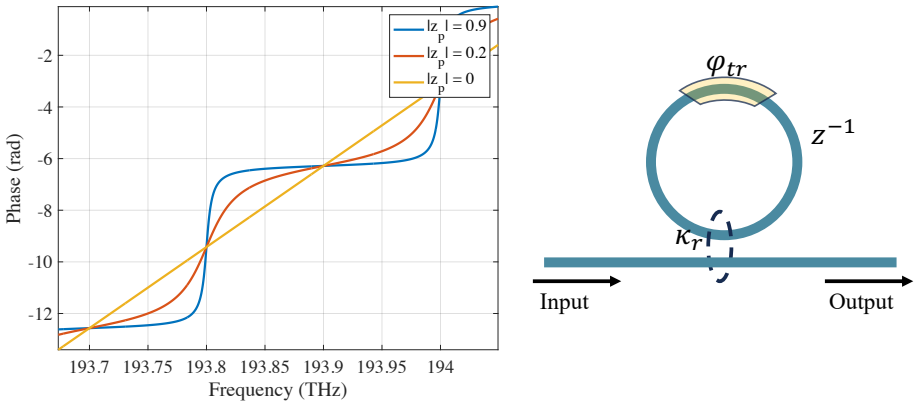


FIGURE 2.5 – An all-pass system. The phase response in this figure indicates the phase difference at the output relative to the input.

the time delay inside the ring, the transfer function of the system could be written as [84]

$$H_r(\mathfrak{z}_r) = \frac{\mathrm{e}^{-\mathrm{i}\varphi_{tr}}\left(c_r \mathrm{e}^{+\mathrm{i}\varphi_{tr}} - \mathfrak{z}_r^{-1}\right)}{1 - c_r \mathrm{e}^{-\mathrm{i}\varphi_{tr}}\mathfrak{z}_r^{-1}} \tag{2.17}$$

where $c_r = \sqrt{1-\kappa_r}$ and φ_{tr} is the extra phase shift on the ring (see Fig. 2.5). Note that without loss of generality, we assume lossless scenario here. If we take the phase of the above transfer function we have

$$\varphi_r(f) = \tan^{-1}\left[\frac{\left(1-c_r^2\right)\sin\left(2\pi f\tau_r + \varphi_{tr}\right)}{2c_r - \left(1+c_r^2\right)\cos\left(2\pi f\tau_r + \varphi_{tr}\right)}\right]. \tag{2.18}$$

The phase response is illustrated in Fig. 2.5 for various values of the modulus of the pole location, $|\mathfrak{z}_p|$, determined in the lossless scenario from Eq. 2.17 as $|\mathfrak{z}_p| = c_r$. The extreme case $|\mathfrak{z}_p| = c_r = 0$ corresponds to $\kappa = 1$, indicating that all incident light couples from the input into the ring, completes precisely one roundtrip, and then fully recouples to the straight waveguide. This scenario is analogous to a single waveguide, which has a linear phase response.

2.2.1 Maximally flat condition

By properly choosing c_r, the frequency response of the interleaver becomes maximally flat at the passband regions. In order to achieve a flat pass-band response, it is necessary to have a constant phase difference between the upper and the lower arm of the MZI in the RAMZI structure throughout the passband region is required. One approach to achieve this is by aligning the slope of $\varphi_r(f)$ with the group delay of the lower arm, denoted as τ_t. This can be expressed as :

$$\tau_r\left(f_c\right) = -\left.\frac{d\varphi_r(f)}{df}\right|_{f=f_c} = \tau_t \tag{2.19}$$

where f_c is the passband center frequency. From Eq. 2.18, $-d\varphi_r(f)/df$ could be written as

$$\tau_r(f) = \frac{\left(1-{c_r}^2\right)}{1+{c_r}^2-2c_r\cos\left(2\pi f\tau_r+\varphi_{tr}\right)}\tau_r. \tag{2.20}$$

The ring's group delay at anti-resonance is

$$\tau_r(f)|_{2\pi f\tau_r+\varphi_{tr}=(2m+1)\pi} = \left(\frac{1-c_r}{1+c_r}\right)\tau_r. \tag{2.21}$$

Upon equating the group delays of the ring and the lower arm of the MZI, we have

$$\tau_t = \left(\frac{1-c_r}{1+c_r}\right)\tau_r \tag{2.22}$$

The solution for the pole magnitude for $\tau_t = \tau_r/2$ is $c_r = 1/3$, hence, the coupling ratio for flat-top response is $\kappa_r = 8/9 \approx 0.89$. This means, with every other parameter well designed, in order to have the maximally flat-top condition, we must set the coupler so that 89% of the power couples from the waveguide into the ring.

The nonlinear phase shift of the ring (Eq. 2.18) can be employed within the MZI as depicted in Fig. 2.4. As shown in Fig. 2.6, in order to achieve a flat-top response, the time delay of the ring must be twice the time delay of the imbalance section of the MZI. The bar and cross transfer functions of the ring-assisted MZI is given by

$$\begin{aligned} |H_{11}(f)|^2 &= \sin^2\left(\frac{\Delta\varphi(f)}{2}\right) \\ |H_{12}(f)|^2 &= \cos^2\left(\frac{\Delta\varphi(f)}{2}\right) \end{aligned} \tag{2.23}$$

where DCs of the MZI have are both 3dB coupler and $\Delta\varphi(f) = \varphi_r(f) - 2\pi f\tau_t$, representing the difference between the phase of the ring-path and the phase of the through-path arm of the MZI.

The normalized bar transfer function $|H_{11}(f)|$ is null when $\Delta\varphi(f) = 2m\pi$, where m is an integer, and it equals to 1 for $\Delta\varphi(f) = (2m+1)\pi$. Achieving passband flattening needs tuning the ring to be in anti-phase ($\varphi_{tr} = \pi$) at the point of maximum transfer for the MZI. Note

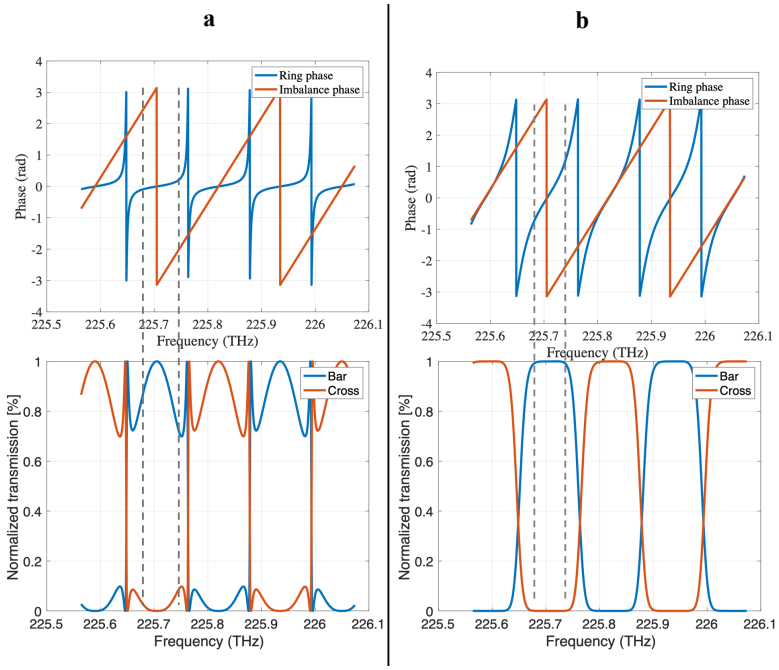


FIGURE 2.6 – RAMZI interleaver response for a) $|\kappa_r|^2 = 0.29$ and b) $|\kappa_r|^2 = 0.89$. In (b), dashed lines specify the region where the phase difference between the upper and lower arm of the MZI have π phase difference, leading to maximum transmission in one port and null transmission in the other port. In (a), the ϕ phase difference is not satisfied, hence leading to a non flat-top amplitude response. Note how the value of κ determines the region where π phase difference holds in the phase response. A MATLAB script has been provided in Appendix A code listing 4 that generates these plots.

that this extra phase shift in the ring is due to the π phase shift introduced by the input DC. If, however, the input DC is replaced by a phase preserving power splitter, such as a Y-branch, then there is no need for this final phase shift in the ring. This is in fact an attractive idea when it comes to power consumption enhancement of the device because when a DC is used for the input coupler, then the extra phase shift must be introduced using a thermo-optic phase shifter, which consumes a certain amount of power.

Fig. 2.6 (b) illustrates that the intensity transfer of the filter is now dictated by the phase difference between the two channels. The centers of the pass-band and stop-band align at a phase difference of $m\pi$. The coupling coefficient has been calculated such that near the center of the pass- and stop-bands, the phase slope of the ring equals that of the differential imbalance length. This results in a constant zero (or π) phase difference between the branches over a substantial fraction of these bands. In contrast, as shown in Fig. 2.6 (a), when an unsuitable value for coupling and length parameters are chosen, it will result in unequal phase slope between the ring's phase and imbalance length's phase, which subsequently leads to a response other than a desired flattop one.

2.3 Anatomy of optical interconnects

There are numerous types of optical interconnects based on the data rate, transmission distance (reach), form factor, etc. In standard IEEE 802.3, different types of interconnects are introduced with their specifications, architectures, and abbreviated names. For instance, 400G-DR4 refers to an interconnect with four single wavelength channels and each channel transmits a PAM-4 signal with a 100 Gbits/s data rate. As another example, in 400G-FR4 transceivers there are four parallel wavelength channels all operating at 100 Gbps speed with PAM-4 signaling. Such interconnects do not meet the scalable beyond-Terabit/s requirement set by today's and future's modern AI and HPC data center networks.

As discussed earlier, a WDM-based transceiver can be deployed to increase the data rate. Therefore, there are other architectures that are inherently different with DR4 in regard that they are designed to use the parallelism in wavelength. Nevertheless, no matter what architecture is being considered and for which application, in general, all the optical transceivers have three main sections, including : 1) light source, 2) multiplexing/demultiplexing, 3) modulation, and 4) detection.

Wavelength parallelization in WDM systems is a key technology enabling scalable transceivers. The number of input frequencies in a WDM link determines the type of input laser source to be used. When the number of input frequencies increases (for example in 32 wavelength grid standard by CW-WDM MSA), using array of single wavelength lasers (usually distributed feedback lasers, or DFB) is not an optimum choice. Firstly, using 32 (or higher) CW lasers will significantly increase the footprint of the overall transceiver as well as cost and complexity. Because of fabrication variations, those large number of individual lasers might not have the same frequency distance between the neighbor frequencies. Hence, wavelength uniformity reduces when the number of lasers increases.

If the laser sources are integrated on the chip, thermal management is another issue [89]. Additionally, if these arrays of CW lasers are deployed externally, the packaging and alignment of the laser array to the chip would become increasingly challenging. Semiconductor mode-locked laser (MLL) is another option as a source for WDM transmitters. The benefit of MLL over array of DFB lasers is that there is an inherent mode-locking in the generation of frequencies, hence reducing the need for controlling the wavelength spacing. MLLs are, however, restricted in the gain bandwidth, leading to a limited number of generated combs and, therefore, not ideal for scalable beyond-Tb/s interconnects [90; 91].

In DWDM optical links, frequency combs are considered as a scalable solution for the multi-wavelength light source [92; 93]. Frequency combs are compact, power efficient, and can produce hundreds of wavelengths. In addition, the generated wavelengths are inherently phase-locked, and therefore, the wavelength spacing is uniform, eliminating the need for wavelength control and considering channel guard bands [94]. These features have made them a promising

choice in DWDM optical interconnects [95; 96; 97].

There exist different kinds of optical frequency combs. In a general way, optical frequency combs can be categorized into five different types, i.e., mode-locked lasers, gain-switched lasers, $\chi^{(3)}$ combs, $\chi^{(2)}$ combs, and electro-optic combs. The mechanism and principles behind each type of comb generation is different and the reader is referred to [98] for an overview. Kerr frequency combs, which are a subgroup of $\chi^{(3)}$ combs have been successfully demonstrated in massively parallel high capacity DWDM optical interconnects [15]. These comb sources are based on ultra-high-quality (Q) ring resonators and have applications in not only optical communications but also in LiDAR, optical clocks, quantum optics, and etc [99; 100; 101; 102].

Kerr combs are operated using a single-frequency CW input pump. When the CW pump is injected into the cavity (ring resonator or disk resonator, for example) and after a certain threshold in the intensity of the light in the cavity, a nonlinear process called four-wave mixing (FWM) happens by which new frequencies are generated. Since the nonlinear interaction has low intensity, a high Q factor cavity is needed to provide the intensity build-up. SiN Kerr combs are ideal because of the SiN's low loss (required for high-Q ring resonator) and high third-order nonlinearity, two main features needed for a high-performance Kerr comb. The newly generated frequencies will again undergo another FWM process to generate new frequencies and this process will continue until a wideband comb is generated. Note that the newly generated frequencies must satisfy the resonance condition of the ring, hence a careful dispersion engineering must be performed. Kerr combs can be realized in both normal and anomolous dispersion regimes. In case of normal dispersion, SiN combs can offer more than 30 % pump-to-comb conversion efficiency [103].

After the laser source, the next building block in optical interconnects is de-multiplexing block. Note that de-multiplexing is required only when there is a parallel transmission scheme such as WDM or polarization division multiplexing (PDM) and space division multiplexing (SDM). The role of multiplexer/de-multiplexer is to direct each frequency line into its corresponding modulator or directing modulated signals to the channel fiber. In a transceiver architecture, de-multiplexers are also used in the receiver side, where different tones from the fiber need to be directed to a photodiode. Note that if an array of CW lasers with different frequencies are used, we still have wavelength parallelism, however, in this case no de-multiplexer is needed because single tones are naturally directed to modulators (see Fig 2.7. (b)). De-multiplexers are frequency-selective passive filters that can be realized in different configurations such as cascaded MZI, arrayed waveguide gratings, or RAMZI interleavers.

After the generation of WDM signals, different frequency tones are directed to a modulator. Microring modulators (MRMs) have several advantages over other types of modulators. They exhibit extremely low power consumption, very tight footprint, and high bandwidth [104; 105; 106; 107; 108]. In addition, MRMs are inherently frequency selective and by engineering the

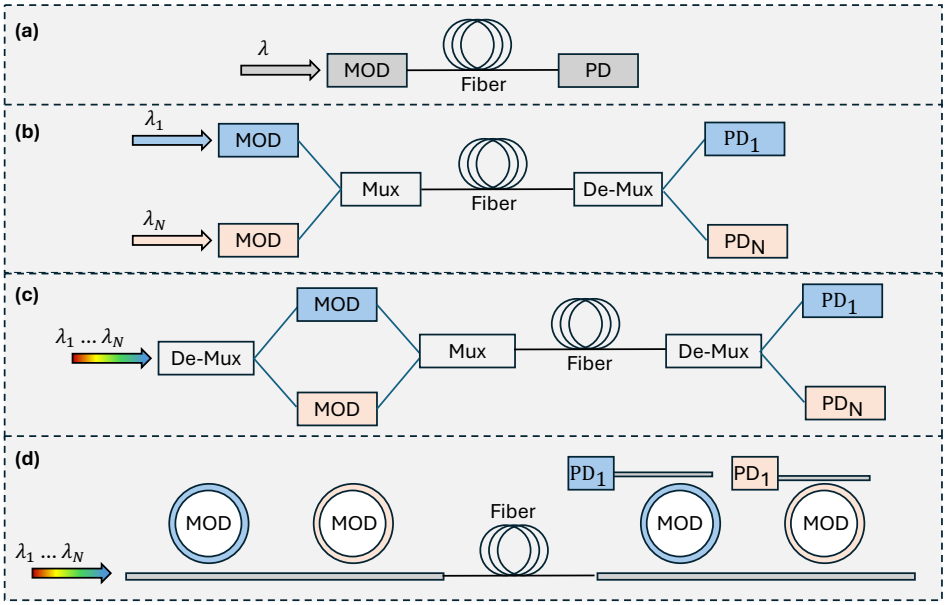


FIGURE 2.7 – Schematics of different optical link architectures. (a) a simple point-to-point link using a single frequency CW laser input source. (b) A WDM link using array of single frequency lasers as input. (c) A WDM link using a comb source. (d) A WDM link where MRMs perform the modulation and mux/demux funtionality at the same time. MOD : modulator. PD : photodetector.

length of the ring, we can select which single tone from the comb source get modulated by which modulator in the modulation bus. Therefore, the need for a multiplexer will be relaxed (Fig. 2.7 (d)).

Regarding the interconnect architectures, it is also noteworthy to mention that in today's short reach data center interconnects, where low latency and low power consumption are of high priority, the intensity modulation direct detection (IM-DD) scheme is preferred to coherent transmission. Coherent links are more complex, have higher power consumption due to extensive digital signal processing, and need a local oscillator at the receiver side [15].

The most straightforward architecture of an optical transmitter is shown in Fig. 2.7 (a) and (b). As shown in this figure, it is comprised of a comb source and then each generated tone gets directed to a single modulator. This scheme has two drawbacks : 1) the large footprint from the routing of different tones to distinct buses, which makes it almost impossible for large Pb/sec scale interconnects with many frequency channels, and 2) the insertion loss introduced by couplers used for separating and then collecting all the single tones is significant. The only major difference between the architectures in Fig. 2.7 (b) and (c) is that the light source in the latter is of comb type, eliminating the need for implementing DFB arrays. MRRs are inherently frequency selective. In other words, when several ring resonators are placed on the same waveguide bus, then only one frequency from the comb source will be coupled into a ring resonator that has been judiciously engineered to trap that frequency inside. For instance, if

four comb lines are generated from the source, by choosing right ring length, it is possible to direct each of those tones to their associated ring resonator, without using any separate de-multiplexer circuit. This multi-ring-on-a-bus or single-FSR configuration (Fig. 2.7 (d)), in contrast to the one shown in Fig. 2.7 (c) has the advantage of small footprint because of removed mux/demux circuits.

However, the limitation of this architecture is that only a certain number of channels can be placed inside the FSR of the ring. See Fig. 2.9 for a schematic view of FSR and channel spacings. Notice that if the number of channels in one FSR increases, the channels will be placed more compactly near each other in the frequency domain. While increasing the number of channels is possible by reducing the channel spacing, previous studies have demonstrated that channel spacings below 100 GHz result in significant crosstalk penalties [109; 13].

A naïve solution to increase the number of MRMs on a single bus is to increase the FSR of the MRMs by decreasing the radius of the rings. Ring's FSR, up to the first order dispersion approximation, could be written as [44]

$$\mathrm{FSR} = \frac{\lambda^2}{n_g(2\pi R_r)}, \tag{2.24}$$

where n_g is the group index of the waveguide, R_r is the radius of the ring, and λ is the center wavelength. However, when the ring's radius becomes smaller, so does the length of the phase shifter (pn junction), reducing modulation efficiency [110]. Furthermore, when the ring's radius becomes smaller, the radiation loss due to tight bending increases significantly [111]. Therefore, the architecture shown in Fig. 2.7(d) still cannot comply with the Pbts/sec scalable designs with many input comb lines.

The solution is to have a multi-FSR configuration. The multi-FSR link structure could be realized using an optical *interleaver* [112; 113]. The idea of using an interleaver before doing the modulation is to somehow pack the input frequency combs into separate groups and then put each pack in a single modulation bus in a multi-FSR format. Two main configurations in optical interconnects that use interleavers exist : 1) band interleaving and 2) even-odd interleavering. A more detailed analysis of these two interleaving schemes is provided in [112], however, we highlight the essential concepts that are relevant to our work.

2.3.1 Band interleaving

The schematic of band interleaving is shown in Fig. 2.8 (b). The interleaver, in this case, has a flat-top (ideal case) frequency response with a period equal to the bandwidth of the comb. Since the interleaver has two frequency-complementary outputs (the stop band of one output is the passband of the other), the input frequency combs will be packed into two groups. Then, they will be directed to two modulation buses. Note that, based on the number of the input combs, there can be additional stages of band interleavers to further split the combs and put

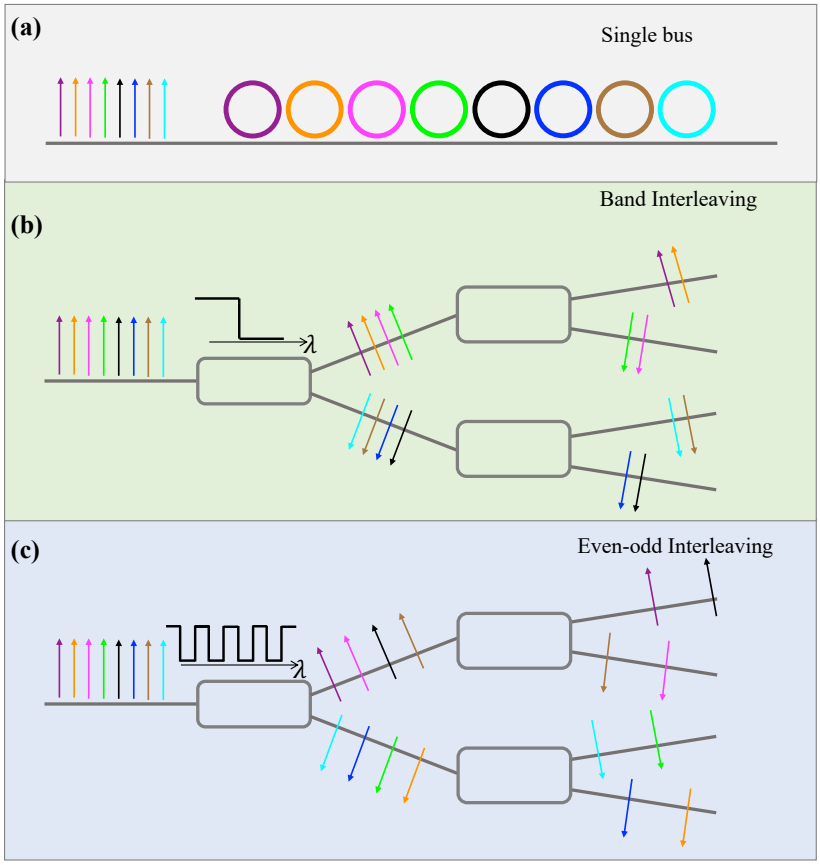


FIGURE 2.8 – Three architectures of interconnect transmitters using MRMs. a) single bus transmitter. b) band interleaving configuration where input frequency combs are segmented into separate groups. In this configuration, frequency spacing between adjacent channels remains the same as the frequency spacing of the input comb lines. c) even-odd interleaving where the frequency spacing is increases and segmented as well.

them into comb groups with smaller number of combs. Remember that the reason for this strategy is that a limited number of combs can be put into the FSR of the ring with a fixed channel spacing. After the band interleaving, channel spacing will remain the same, however, the number of combs in each group will reduce. In order to determine the number of band interleaver stages, one must ensure that the bandwidth of the final stage of the interleavers does not exceed the resonators' FSR. The main challenge in this scheme is the realization of compact and efficient band interleaver filters with sharp roll-off. Roll-off is a figure-of-merit in filter design that specifies how fast a passband state transits to the stop band and vice versa and has a unit of dB/nm. Additionally, the channel spacing remains intact in the band interleaving structure. We will see in the following sections that other interleaving strategies (i.e., even-odd interleaving) can increase the channel spacing, which has a significant impact on inter-channel crosstalk.

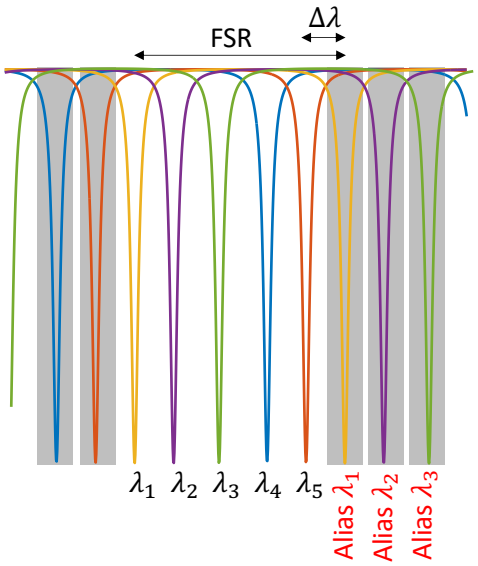


FIGURE 2.9 – Channel arrangement of a single-FSR link architecture.

2.3.2 Even-odd interleaving

An alternative approach to band interleaving is even-odd interleaving. As the name suggests, even-odd interleaving subdivides the input combs into two groups of combs with even and odd indices. Subsequently, these two groups of odd and even combs will be directed toward separate modulation buses. It is crucial to notice (as shown in Fig. 2.8 (c)), the channel spacing in each modulation bus is increased (doubled in case of using one stage of interleaver). Because the periodicity of the even-odd interleaver is higher than the band interleaver, it is easier to implement the desired frequency response with sharp roll-off [114].

In the even-odd interleaving scheme, one can configure the link architectures in different ways. For example, the number of interleaving stages, the ring's FSR, number of modulators on a single bus, and etc. A detailed analysis of even-odd interleaving link configuration, also known as multi-FSR link architecture, has been reported in the supplementary information of ref [15]. Fig. 2.10 shows the schematics of a multi-FSR link architecture. This figure shows two stages of even-odd interleavers, leading to four modulating buses.

To better understand multi-FSR link architecture we will consider a simple example. In this example, we assume that the input comb laser has 20 tones and the spacing between each comb line is $\Delta\lambda_{comb}$ = 200 GHz (1.14 nm) and the FSR of MRMs is equal to 5.7 nm ($\sim$ 1 THz). With this choice of ring FSR and comb spacing, a maximum of 5 channels could be positioned on a single channel modulation bus. If we assume a 50 GHz bandwidth for each modulator, then an aggregate capacity of equal to 250 Gbps could be achieved using on-off keying (OOK). Now we want to show that *how much enhancement in aggregate capacity can we get by using multi-FSR architecture enabled by 2 stages of interleavers*. An illustration of

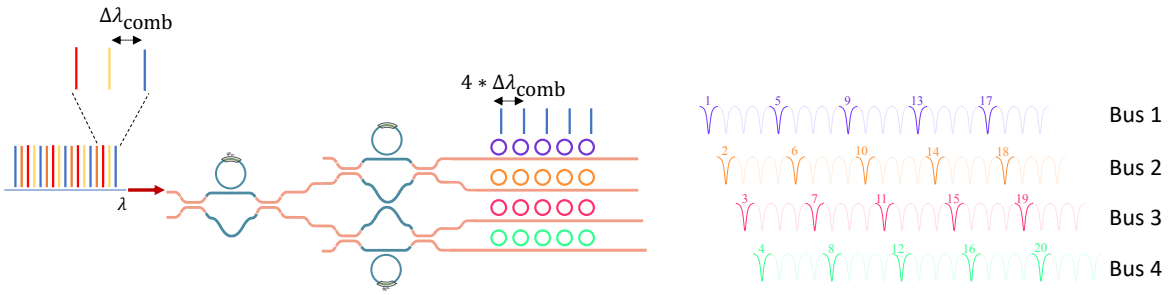


FIGURE 2.10 – An example of a multi-FSR link with two stages of interleavers, enabling a four-fold increase in comb line spacing and channel capacity.

this link configuration is depicted in Fig. 2.10. As shown in this figure, 5 channels could be positioned on each bus, and since we have 2 levels of interleavers there exist 4 modulation buses. Note that the channel spacing on each modulation bus increases by a factor of 4. Also, since the number of channels on each bus is the same as a single-FSR link, the channel capacity also increases by a factor of 4 (number of modulation buses). Therefore, in this example, with two stages of interleavers in the input, the aggregate capacity increases from 250 Gbps to 1 Tbps.

We conclude our analytical discussion about interleavers at this point and extra points and remarks about it will be covered in the next chapters.

Chapitre 3

Designing a wideband SiN interleaver

In this chapter, we aim to design a SiN wideband and maximally flat low-loss interleaver. The design flow is shown in Fig. 3.1. The whole design process is split into two sections : device simulations and circuit simulations. The outputs from device simulations (i.e. compact models) are fed into the circuit simulations. We will use commercially available Ansys Lumerical packages such as MODE, FDTD, and INTERCONNECT. Device simulations will provide waveguide compact model and the S-parameters for the MMI and Y-splitter. In addition, the ring's length for a maximally flat interleaver could be calculated from the group index, which is obtained from solving the mode properties of the waveguide. Using the waveguide compact model, S-parameters of the designed MMI, and calculated ring length, we then perform a circuit simulation in INTERCONNECT and extract and analyze different interleaver metrics such as extinction ratio, bandwidth, and etc. We will discuss each section in Fig. 3.1 starting from the requirement section in the following.

3.1 Requirements

The first step for our design is to specify the interleaver's specifications. In this design, the frequency spacing of the input (considering a frequency comb in the input stage) is considered 100 GHz, which is standard in highly scalable interconnects [15]. The center wavelength of the filter is $\lambda_0 = 1310$ nm, and the device will be designed to operate in the O-band window. We follow the standard 400×800 nm dimensions for the waveguides used in the circuit. A sidewall angle equal to 83° is considered. As shown in Fig. 1.3, this waveguide dimension remains in the single mode condition. Using this dimension and considering the bulk index of SiN ($n \simeq 1.98$ at $\lambda_0 = 1310$ nm), the effective index and the group index for this waveguide are equal to $n_{eff} = 1.647$ and $n_g = 2.05$. The field profile of the calculated mode for the described waveguide is shown in Fig. 3.2. In this figure, the intensity of the electric field is depicted in transverse dimensions. As shown, the electric field almost vanishes at the boundary of the mode solver.

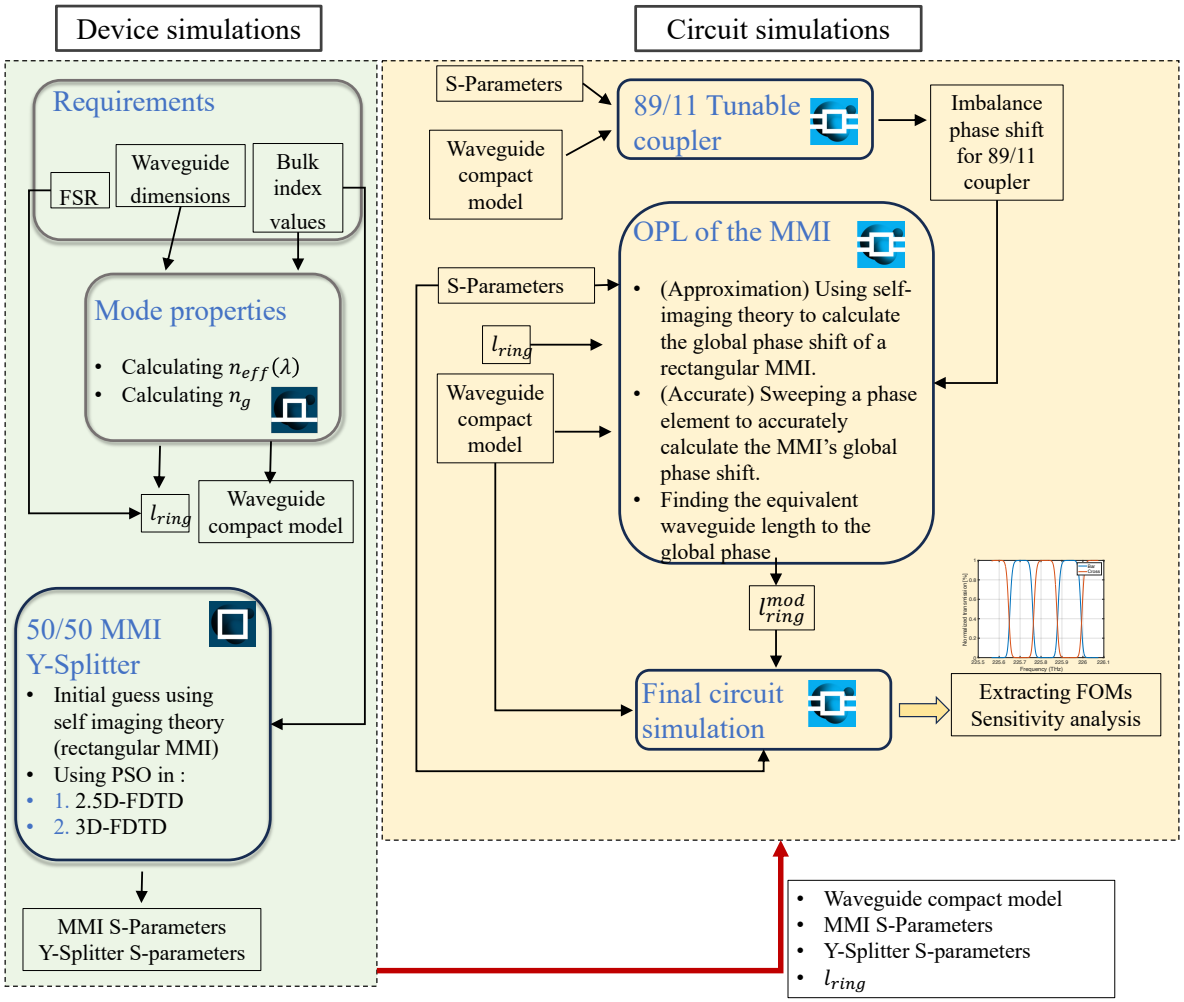


FIGURE 3.1 – A summary of the design flow. The design is performed in two steps, namely, device simulation section and circuit simulations. The results of the device simulation (indicated by red arrow) are fed into the circuit simulation. In the device simulation, we will design the components used in the circuit simulation and obtain compact models that could be used in the circuit simulations. In this chapter, we will discuss each section of this flow. As shown in the subset of the boxes in this figure, the simulations are mainly performed in commercially available Lumerical MODE, FDTD, and INTERCONNECT packages.

Notice that the interleaver is going to be designed for TE mode. In order to accurately solve the mode, a sufficient number of mesh cells across the transverse coordinates must be set. In this regard, a mode effective index convergence analysis versus the number of mesh in each transverse directions must be performed. As seen in Fig. 3.3, the effective index converges to the value of 1.6473 by setting 1500 number of mesh cells in each direction while having a finite difference eigenmode (FDE) solver region with $8 \times 5 \mu m^2$ dimensions. Note that the FDE solver is integrated into Lumerical MODE solutions.

We use the calculated group index to obtain the length of the ring as $l_{ring} = \frac{2c}{n_g FSR} = 1462.402 \mu m$ and the length of the differential section at the lower arm of the MZI would be $l_{ring}/2 = 731.201 \mu m$.

3.2 Inverse designed MMI

As shown in Fig. 2.4, there is a coupler between the upper arm of the MZI and the ring. According to the analysis in Chapter 3, the amount of power that must be coupled into the ring to achieve a flat-top spectral response, according to the analysis in section 2.2.1, is equal to 89% of the total power in the upper arm. There are two main methods to implement such power coupling. We can either use a single coupling device, such as a DC or an MMI with a 0.89/0.11 power splitting ratio, or a tunable coupler (imbalanced MZI) with 3-dB couplers. Using the latter coupling method is more practical because the splitting ratio could be tuned using integrated heaters. Hence, it reduces the fabrication error. Notice that the input/output 3-dB tunable coupler in our design are 50/50 MMIs. The insertion loss of this tunable coupler depends mainly on the performance of the input/output MMIs. Since MMIs have relatively wide band transmission bandwidth, they are used in the input and output couplers in the interleaver as well (Fig. 3.4). Hence, there are 4 MMIs in this circuit. Notice that this circuit is only a first order interleaver. Increasing the order of the interleaver by adding more rings

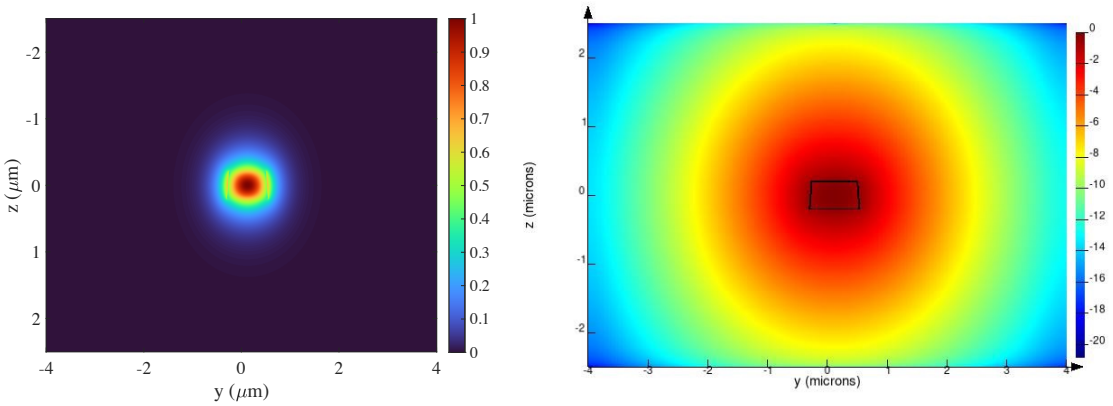


FIGURE 3.2 – Calculated mode profile of the waveguide used in the interleaver design in linear (left) and logarithmic (right) scales.

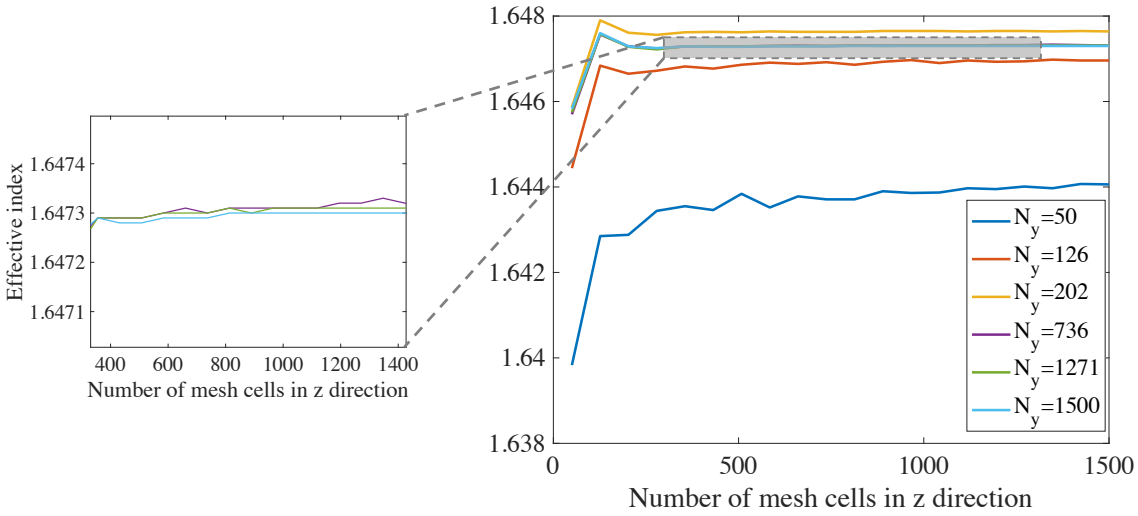


FIGURE 3.3 – Mode convergence versus number of mesh cells in each transverse coordinates. Dimensions of the FDE solver is set to $8 \times 5\mu m^2$

on both sides of the MZI, would improve some metrics, such as roll-off factor or cross-talk (we will discuss these metrics at the end of the chapter). However, enhancement comes at the cost of increased insertion loss because the device would become larger and will have more MMIs, more bends, and longer waveguide sections. As shown in Fig. 3.4, by adding one ring to the circuit, two MMIs and two thermal phase shifters (one on the ring and one for the tunable coupler) must be added to the circuit, which is the main source of power consumption. Therefore, in order to have a low-loss interleaver, we should be able to reduce the insertion loss of the MMIs in the circuit. The first solution to have a MMI with low insertion loss is to use a wide core. However, increasing the width of the MMI's core, leads to exponential increase in the length of the MMI. As a result, the footprint of the entire circuit will increase substantially. The beat length calculated in Eq. 1.7, could be written as a function of effective mode width in core, W_e, as [63]

$$L_\pi = \frac{\pi}{\beta_0 - \beta_1} \approx \frac{4 n_r W_e^2}{3\lambda_0} \tag{3.1}$$

According to Eq. 3.1, by increasing the core width by a factor of m, the beat length and hence the 3-dB length would increase by m^2. To simultaneously have small footprint and low insertion loss, we should use another paradigm for designing an MMI. One widely used method is to have a varying core width instead of a rectangular core. However, this design could be optimized numerically by sweeping all core widths along the device, which is computationally expensive. Notice that when the geometry of the device becomes complicated, there is no analytical solution. Hence, the method of direct design (working with analytical solutions) could not be exploited. Inverse design, also referred to as the inverse problem, represents a set of innovative research methodologies gaining prominence in the fields of natural sciences and engineering. In general terms, an inverse design problem involves determining the specific configuration of a

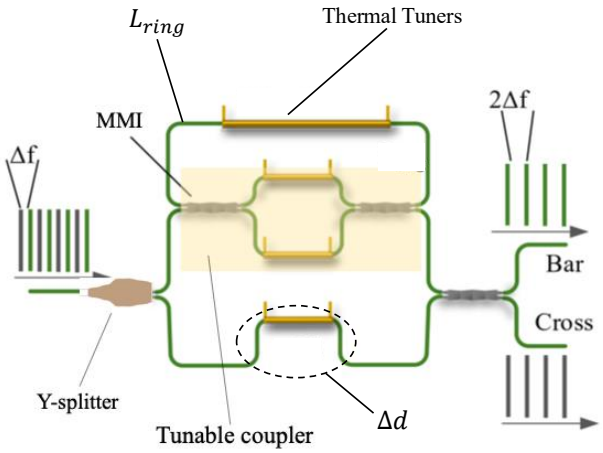


FIGURE 3.4 – A schematics of the interleaver under design. Δd is the extra path length compared with the top arm. L_{ring} is the length of the ring's waveguide. The shaded yellow region indicates the tunable coupler.

material, geometry, or process through a targeted search activity [115; 116; 117; 118]. While the techniques employed for this search may vary, the distinguishing feature of the inverse approach lies in formulating the functional requirements for a given design as an optimization problem. In essence, the inverse approach entails a systematic and automated parameter search until a design solution is discovered that optimally fulfills the specified objectives. This structured exploration of the design space allows for a comprehensive analysis of trade-offs among various performance aspects. Consequently, this approach not only enhances our understanding of the previously unexplored relationships between properties and their effects but also propels the development of novel solutions that may have remained undiscovered until now. In order to design the MMI we use Lumerical's integrated particle swarm optimization (PSO) utility.

PSO is an bio-inspired (collective intelligence of birds or fishes) optimization algorithm. PSO enables finding the global maxima (or minima) without the knowledge of the gradient of the cost function. It relies on the movement of particles within the parameter space, which are influenced by their own experiences and the experiences of neighboring particles. Numerous nanophotonic applications utilize this algorithm for optimization problems [119; 120; 121; 122; 123]. The basic operation of PSO involves, in the first step, generating particles at random positions (a set of random vectors with a dimension equal to the number of parameters). Then, the position of these initially generated particles are updated using a velocity vector at each generation. Subsequently, after the first update, a new set of particle positions as well as new velocities are generated. This iteration continues until a convergence is reached. The crucial point the that the velocity vector is determined by the personal (cognitive) best solution and

TABLE 3.1 – Geometrical parameter range for the optimization of the 50/50 MMI.

Parameter	Min	Max
w1 (μm)	2	3
w2 (μm)	2.6	4.3
w3 (μm)	0.8	1.3
l1 (μm)	1	4
l2 (μm)	5	8
l3 (μm)	5	7

collective (global) best solutions. For a rigorous description of PSO algorithm, the reader is referred to [124]. Nevertheless, a brief overview is presented here. Let's denote the particles by a vector P_i^n, where i and n indicate particle number and iteration number, respectively. Notice that the dimension of P_i^n is determined by the number of the parameters in the parameter space of the given optimization problem. After generating particle positions, the next step is to update these positions using a velocity vector as $P_i^{n+1} = P_i^n + \nu_i^{n+1}$. The velocity vector itself should be initialized. Often, the velocity vector is initially set to zero. The velocity vector is then updated using the following formula :

$$\nu_i^{n+1} = \omega\nu_i^n + c_1\eta_1(Pb_i^n - P_i^n) + c_2\eta_2(g_i^n - P_i^n), \tag{3.2}$$

where c_1 and c_2 denote cognitive and social factors, respectively. The inertial weight is denoted by ω, and η_1 and η_2 are random numbers ranging between 0 and 1. Pb_i^n and g_i^n are the cognitive and collective best solutions and are determined using the FOM.

Lumerical's PSO implementation employs default values for c_1 and c_2 and ω, which have demonstrated effective convergence in numerous test optimization problems for photonic design.

As discussed above, the first step is to define an FOM. In our design, since we want to have a 50/50 splitting ratio MMI and at the same time decrease the insertion loss, we define the FOM as below.

$$FOM = \sum (0.5 - y_1(\lambda))^2 + \sum (0.5 - y_2(\lambda))^2 + \sum (y_1(\lambda) - y_2(\lambda))^2 . \tag{3.3}$$

where $y_1(\lambda)$ and $y_2(\lambda)$ correspond to the normalized optical power at the through port and cross port of the coupler. Notice that this FOM ensures the wideband behavior of the device, as the summation is over wavelength λ. The second step is to define the degrees of freedom in the core geometry. A typical approach is to have varying core widths. We have used the values listed in Table. 3.1. The parameters in Table. 3.1 are shown in Fig. 3.5.

Next, we have to define the number of generations and the size of the generation. We set generation size equal to 45 and the number of generation to 20. By running the optimization, the best value for the FOM is 1.193. This FOM was obtained by the final values of the DOFs as shown in Table. 3.2. See that the W4 and W5 parameters (shown in Fig. 3.5) are fixed and not considered as DOF in optimization.

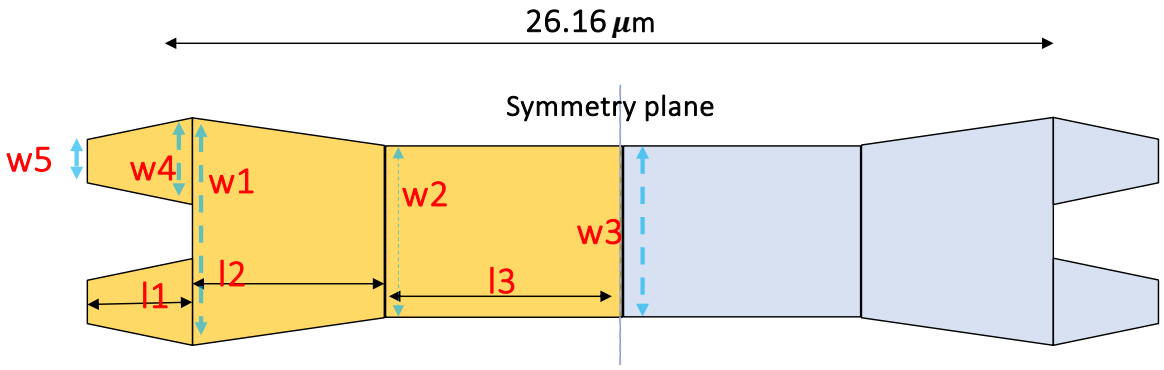


FIGURE 3.5 – Illustration of defined DOF for MMI PSO optimization.

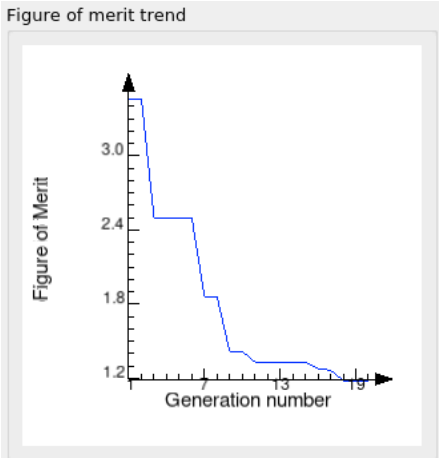


FIGURE 3.6 – Convergence of FOM after 20 generations.

TABLE 3.2 – Final values for the DOFs to achieve minimum FOM.

Parameter	Value
w1	2.78617 μm
w2	3.50796 μm
w3	1.26294 μm
l1	4 μm
l2	8 μm
l3	7 μm

Notice that the reported PSO optimization was performed in Lumerical 2.5D-FDTD, which is much faster compared to 3D-FDTD. The next step is to use the final parameter values obtained by PSO to run a 3D-FDTD simulation and extract the S-parameters to use in circuit level simulations using INTERCONNECT. The frequency response of the MMI in the O-band is shown in Fig. 3.7. Note that better performance from the optimized MMI could be achieved by increasing the widths and length of the MMI. However, we use the one presented here because of its compact footprint. As shown in Fig. 3.5, the core length is only 26.16 μm .

The next step is to implement a tunable coupler with 89/11 splitting ratio to couple the light

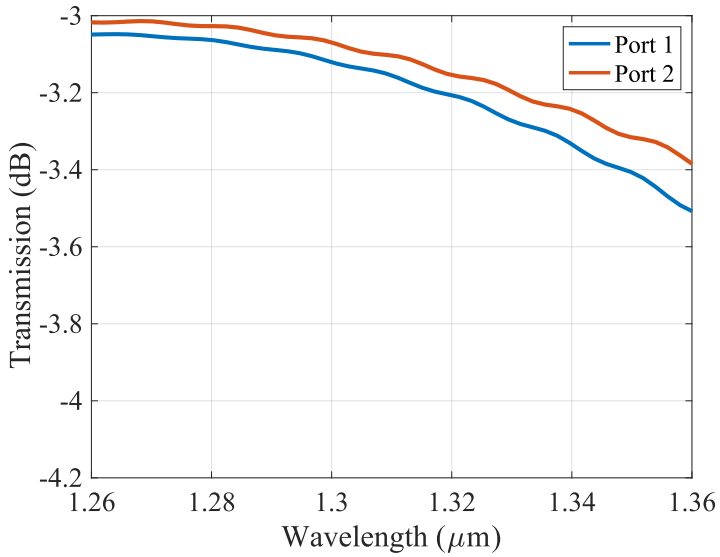


FIGURE 3.7 – Power transmission of the PSO-optimized MMI.

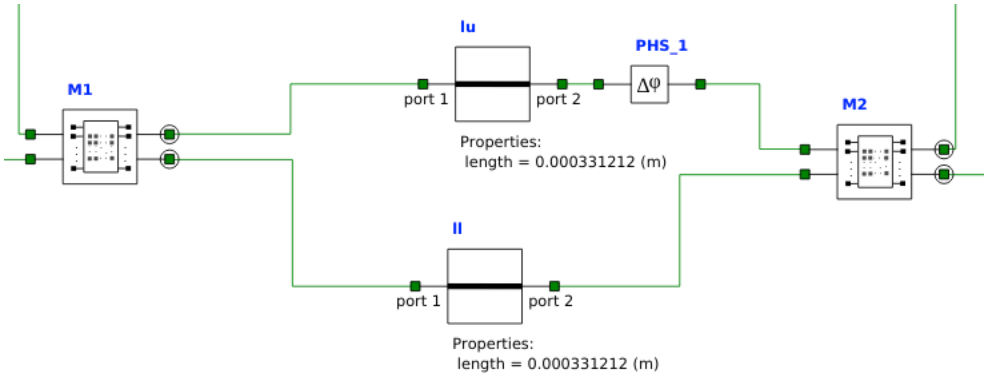


FIGURE 3.8 – Schematics of 89/11 splitting ratio tunable coupler.

from the upper arm of the MZI to the ring. We use the PSO-optimized MMIs to implement this tunable coupler. The objective is to find the required phase shift on the one arm of the MZI tunable coupler. In other words, as written in Eq. 2.15 and Eq. 2.16, for certain amount of ϕ_t, the ratio between transmission at the two ports, i.e., $|H_{12}(f)|^2 / (|H_{11}(f)|^2 + |H_{12}(f)|^2)$ becomes 0.89. A code listing in Appendix A section A.4 has been provided which calculates the power at the two ports of the tunable coupler for different values of ϕ_t.

In order to determin ϕ_t, we import the S-parameters of the inverse designed MMIs in INTERCONNECT and form the tunable coupler circuit and use a phase shift element to sweep its value and find the phase shift required for the 89/11 splitting ratio for the tunable coupler. A schematic of the tunable coupler in INTERCONNECT is shown in Fig. 3.8.

As shown in Fig. 3.8, there are two MMIs (M1 and M2) and two arms of the MZI in between has same length. Note that M1 and M2 are S-parameter components, which means the S-parameters from a FDTD simulation has been imported to them. On the top arm of the MZI

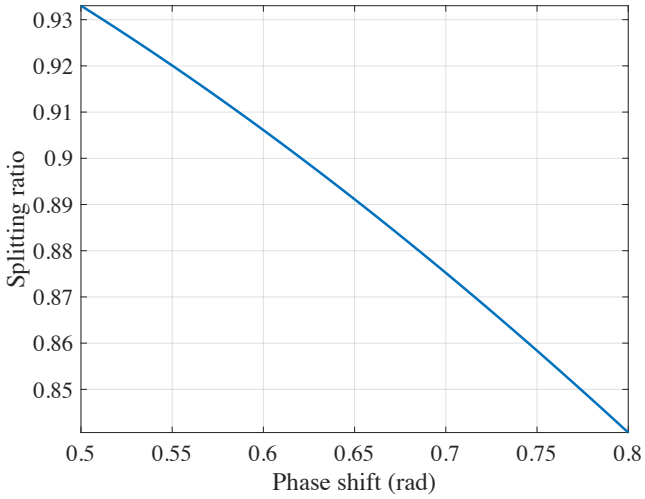


FIGURE 3.9 – Coupling ratio of the tunable coupler versus phase shift on the top arm of the MZI.

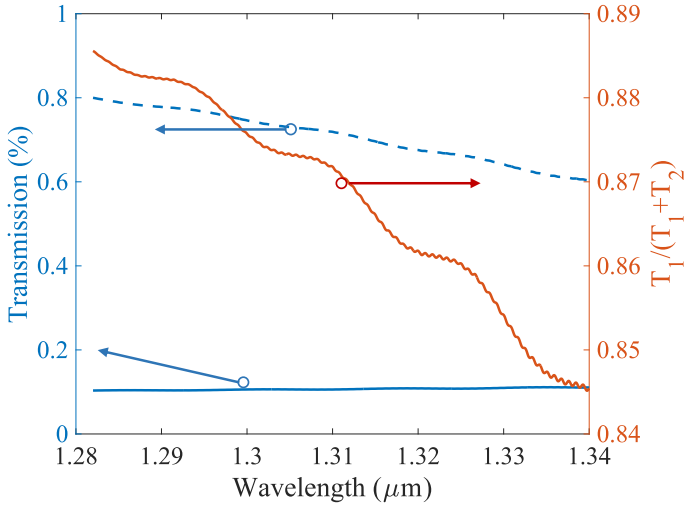


FIGURE 3.10 – Transmission of the tunable coupler with $\phi_t = 0.6484$ rad at the two output ports (left axis) and the power ratio (right axis). As shown, the ratio is close to theoretical 89/11 splitting.

there is a phase shift element. This phase shift could be realized using a thermal phase shifter. Nevertheless, we sweep the value of this phase shift element from $-\pi$ to π. As shown in Fig. 3.9, the amount of phase shift to obtain a 89/11 splitting ratio is 0.648 radians. Using this amount of phase shift, the transmission of the tunable coupler is depicted in Fig. 3.10.

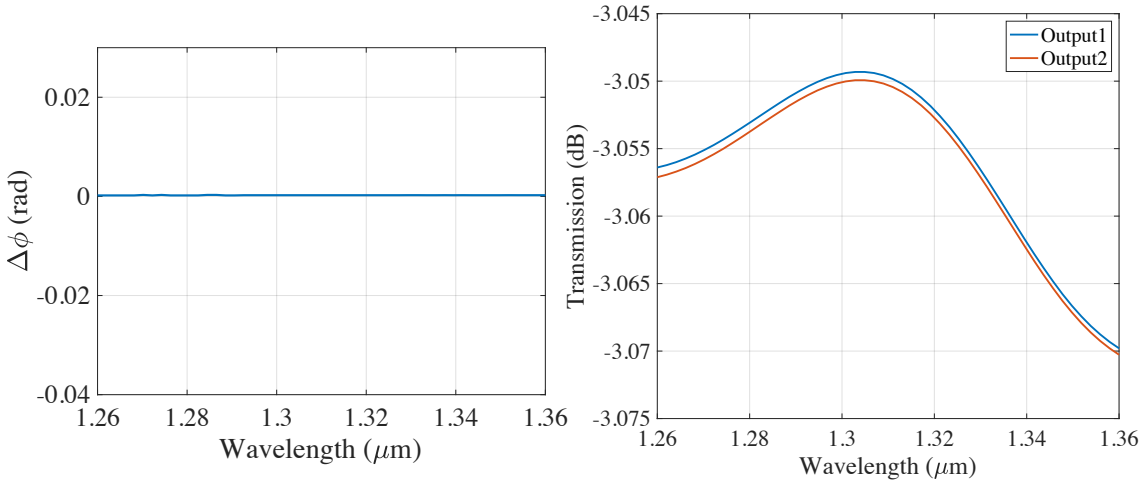


FIGURE 3.11 – Power transmission of the PSO-optimized Y-Splitter at the two output ports (right). Phase difference in radians at the two output ports(left).

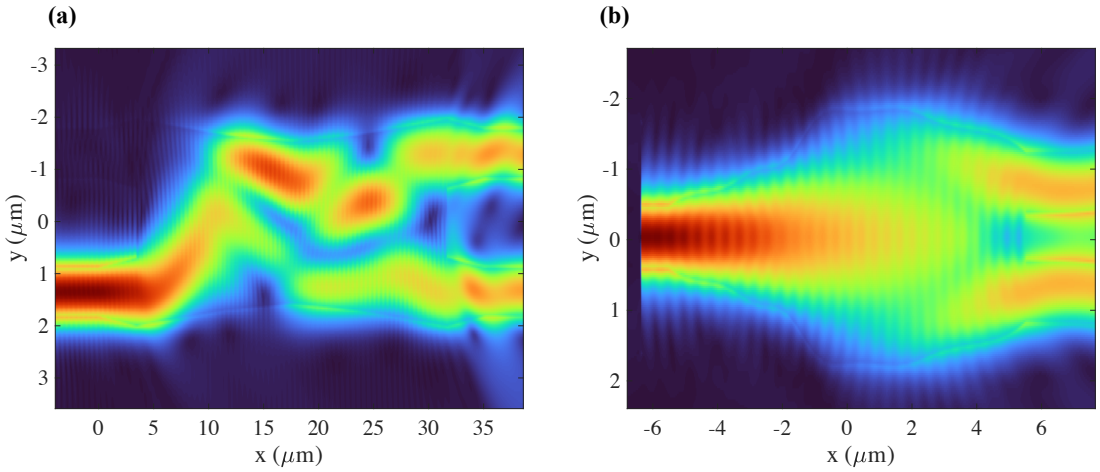


FIGURE 3.12 – Magnitude of the electric field of the (a) MMI and (b) Y-splitter.

3.3 Designing a compact Y-splitter

Notice that an additional π-phase shift must be applied on the ring to compensate for the π phase shift at the output ports of the input MMI in the interleaver, which could be realized by using a thermal phase shifter for example. However, in order to eliminate this extra phase shifting we can use a Y-splitter insetad of an MMI at the input section. Another benefit of using Y-splitter is that its footprint is smaller than MMI. For designing the Y-splitter we use the PSO optimization again to optimize the geometry for having 50/50 splitting ratio. We have used variation points along the Y-splitter core as shown in Fig. 3.13 and the final values after optimization have been listed in Table. 3.3. The 3D-FDTD spectral response of the device has also been reported in Fig. 3.11. Fig. 3.12 shows the field profile (magnitude of the electric field) for the optimized Y-splitter (b) and the MMI (a).

TABLE 3.3 – Optimized values for different widths of the Y-splitter. Values are in μm units.

Parameter	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Value	1.131	1.309	1.74	2	3.102	3.28	3.3	3.3	3	2.814	2.654	2.2

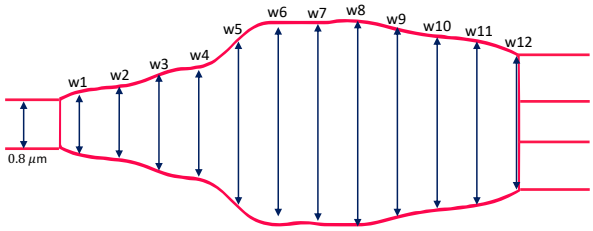


FIGURE 3.13 – Illustration of DOFs defined for PSO optimization of the Y-splitter.

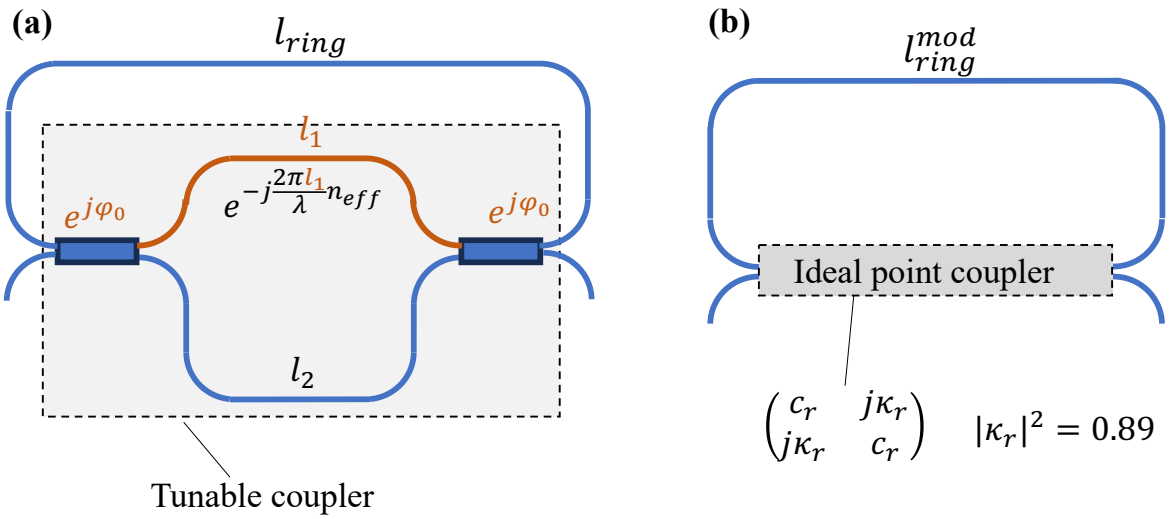


FIGURE 3.14 – (a) Coupling into the ring using a tunable coupler. (b) the equivalent circuit of (a) using an ideal point coupler instead of a tunable coupler.

3.4 Calculating the OPL of the tunable coupler

At this point, we have every component needed to form the interleaver circuit in INTERCONNECT. The important point in designing an interleaver (or any other passive filter structure with infinite impulse response) is to considering the optical path length (OPL) of the tunable coupler. A tunable coupler, as shown in Fig. 3.14 (a), is consisted of an MZI between two MMI couplers. Hence, the OPL of the tunable coupler is the sum of the OPLs of the two MMIs and those of the MZI arms.

Consider Fig. 3.14 (a). In this figure, a schematics of the upper section of a RAMZI interleaver (the section where the ring is connected to the upper arm of the MZI) is depicted. Remember that in the section 2.2, it was assumed that the ring is coupled to the MZI's waveguide using an **ideal point coupler**. *In other words, the transfer function in Eq. 2.17 was derived based*

on the fact that the coupler is in fact an ideal point coupler. Based on this assumption, we derived the ring length that guarantees the maximally flat response. However, in practice there is no ideal point coupler. In a point coupler, first of all, there is no insertion loss. Secondly, and more relevant to our discussion in this section, the coupling mechanism happens in a one point without any propagation of the light. For the coupler in our design, as discussed earlier, we chose a tunable coupler, as shown in Fig. 3.14(a). The question now is that how should we modify the l_{ring} (and subsequently the differential length of the MZI) now that the coupler is no longer a point coupler ? In other words, given the l_{ring} and the components used in the point coupler (l_1 in Fig. 3.14 and the structure of the designed MMI), how can we obtain l_{ring}^{mod} (shown in Fig. 3.14(b)). Notice that the ring length that we consider in practice either for fabrication or in a final circuit simulation is l_{ring}^{mod}. In summary, we should calculate the OPL of the tunable coupler and modify the ring's length based on it. The first step towards calculation of l_{ring}^{mod} is to obtain the global phase response of the MMI at the output ports. In the two following subsections, we discuss in depth on that.

3.4.1 Global phase response of a rectangular MMI

For a rectangular MMI, self-imaging theory predicts the amount of global phase introduced by the MMI. However, the MMIs that we used in our design are not rectangular. Hence, there is not any analytical solution, up to our knowledge, for the phase response of a varying width MMI. On the other hand the width variations along the core of our designed MMI is not large. Hence, we first get an estimate of the global phase of the MMI by assuming that it is a rectangular MMI, and then using numerical tools, which we discuss soon, we will obtain a more accurate estimation of the global phase of the MMI. In order to approximate the global phase of the PSO-optimized MMI, we assume it is rectangular MMI with core width equal to the largest core section of the original MMI. Then, we will use Eq. 1.14 to calculate the global phase of the rectangular equivalent MMI of the PSO-optimized MMI.

The widest section of the designed MMI is 3.507 μm (see Table. 3.2). Hence we assume a rectangular MMI with core width equal to 3.507 μm and we calculate β_0 and L_N for this MMI core. The effective index of the first mode is calculated and is equal to 1.7618. Therefore, the propagation constant at the center wavelength of 1310 nm is equal to $\beta_0 = 8.45 \times 10^6$ rad/m. The beat length for this MMI is equal to 24.7 μm and therefore the 3-dB length is 37.04 μm. Notice the reduction from 37. 04 μm of the rectangular MMI to 26.16 μm of the PSO-optimized MMI. According to 1.14, the general phase of the rectangular MMI is equal to ϕ_0 =315.41 rad. We will use this value of ϕ_0 to modify the ring's length. We define l_{ϕ_0} as a waveguide length that introduces ϕ_0 phase shift. We will use this definition in the following analysis.

3.4.2 Modification of the ring's length considering the MMI's global phase

Referring to Fig. 3.14(a), it could be observed that the upper arm of the tunable coupler as well as the two MMIs contribute to the length of the ring. In other words, the phase shift introduce by the ring is not only due to the waveguide section with length l_{ring}, but also due to the phase shift of the coupler's length l_1 and the global phase shift of the two MMIs. Therefore l_{ring}^{mod} could be approximately written as

$$l_{ring}^{mod} \approx l_{ring} - l_1 - 2l_{\phi_0}. \tag{3.4}$$

As could be seen from the equation above, l_{ring}^{mod} is *approximately* equal to $l_{ring} - l_1 - 2l_{\phi_0}$. The reason for this approximation lies in the fact that l_{ϕ_0} represents the equivalent waveguide length that introduces a phase shift equal to ϕ_0, the global phase shift of a *rectangular* MMI. However, in our design, the MMI core is not a rectangular. Nevertheless, as mentioned earlier, the core variation along the PSO-optimized MMI is small. Therefore, l_{ϕ_0} is a good approximation. However, in order to prevent further post fabrication tuning, we aim to compute the exact global phase shift of the PSO-optimized MMI. Before discussing the method to obtain the exact global phase shift of the MMI, we show how to obtain l_{ϕ_0} from ϕ_0. The transfer function of a lossless straight waveguide with length L is

$$\mathrm{H(f)} = e^{-i\left(\frac{2\pi f_0}{c} n_{\mathrm{eff}} + \frac{2\pi}{c} n_g (\mathrm{f}-\mathrm{f_0})\right)\cdot \mathrm{L}} \tag{3.5}$$

where f_0 is the center frequency. Notice that the higher order dispersion terms have been neglected. Using this equation we set $H(f) = e^{-j\phi_0}$ and solve for L. Note that we set n_g and n_{eff} to those of the waveguide that is used in the ring. Then, $L = l_{\phi_0} = 39.92\mu m$. Obviously, this length is not equal to the core length of the MMI. In order to match the FSR of the ring and MZI to generate a flat top response, the same amount of length that was substracted from l_{ring}, i.e., $l_1 - 2l_{\phi_0}$ must be added to the differential length of the MMI on the lower arm.

Having l_{ϕ_0}, we can perform a fine tuning length sweep in INTERCONNECT to find the exact phase shift of the tunable coupler. In this regard, we form the entire interleaver circuit in INTERCONNECT, and then sweep the ring length from $l_{\phi_0} - l_{trial}$ to $l_{\phi_0} + l_{trial}$ in equal steps, and for each step we overlap the transmission response with that of an ideal interleaver. In the step that this overlap maximizes, we conclude that the global phase shift of the MMI has been accurately calculated. Note that in our design we chose $l_{trial} = 2$ μm. Other values for l_{trial} could be chosen based on the geometry of the device. In the circuit that was used in INTERCONNECT, we assume 331.2 μm length for the arms of the tunable coupler (l_1 and l_2 in Fig. 3.14) and also 5 μm for the waveguide tapers connecting to the MMI. After running this length sweep, the corrective length was calculated as 1.2 μm. Therefore, the the global pahse shift of the PSO-optimized MMI is equal to $l_{\phi_0} - 1.2\mu m = 37.72\mu m$. Therefore, the equivalent length of the tunable coupler $(l_1 + l_2 + 2(l_{\phi_0} - 1.2\mu m))$is equl to 426.64 μm. In

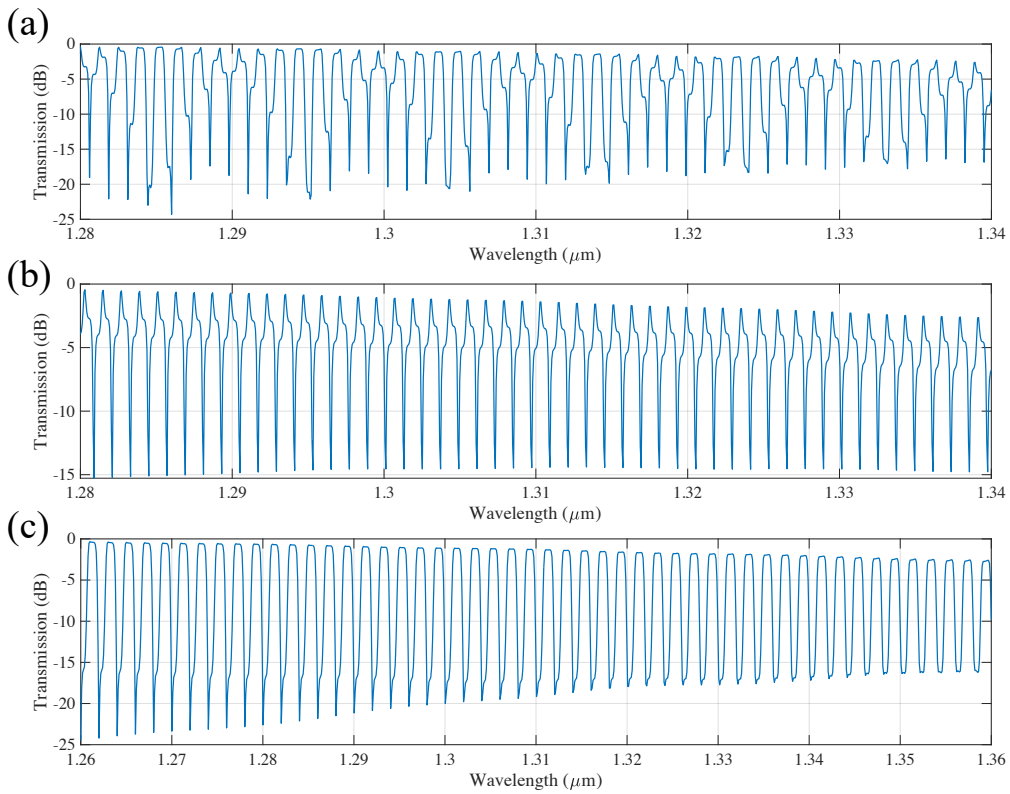


FIGURE 3.15 – Transmission of the interleaver at one of the output ports when (a) OPL of the tunable coupler is not considered, (b) OPL considered assuming a rectangular core MMI, and (c) a final length sweep has been performed in INTERCONNECT to correct for the rectangular assumption of the MMI.

section 3.1, assuming the coupler to the ring is a point coupler, the ring length was calculated as $l_{ring} = \frac{2c}{n_g FSR} = 1462.402$ μm. Therefore, for the modified ring length, we have $l_{ring}^{mod} = 1035.762$ μm. After modifying the length of the ring, a final circuit simulation was performed. Fig. 3.15 shows the results. As could be seen from Fig. 3.15 (a), the response is far from flat top when the tunable coupler's phase shift is not substracted from the ring. In Fig. 3.15 (b), the calculated tunable phase shift (assuming rectangular MMIs) has been considered. As a result the spectral response of the interleaver has become closer to an ideal flat top one. However, it is still not the response that is sought after. By running the final length sweep, as discussed earlier, and additional 1.2 μm should be subtracted from the ring's length. This will compensate for the miscalculation of the general phase in the MMI. After subtracting this value from the ring's length, the achieved transmission response is quite close to an ideal interleaver (Fig. 3.15 (c))

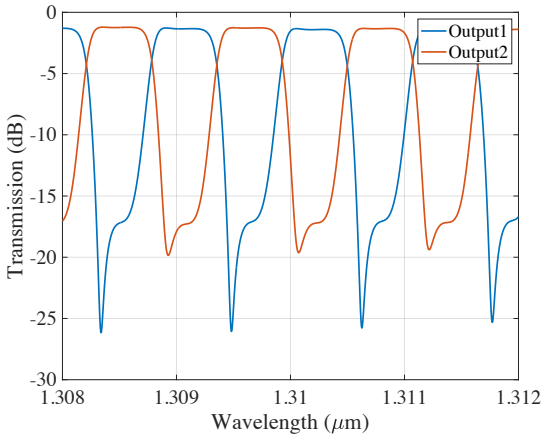


FIGURE 3.16 – Transmission of the interleaver at the two output ports.

3.5 Simulation results

Fig. 3.16 shows the transmission at the two output ports. From this figure, perfromance metrics of the device could be extracted in the center frequency. As shown in this figure, the extinction ratio of the interleaver which is defined as the difference in the transmission of the two output ports at the center of the band, is equal to 16 dB. The insertion loss of the interleaver at the center frequency is equal to 1.27 dB. The 3-dB bandwidth of the device (shown in Fig. 3.17) is 70 GHz. In order to calculate the bandwidth, simply we obtained the 3 dB bandwidth in terms of wavelength and converted it to frequency using $\Delta f = \frac{c}{\lambda^2}\Delta\lambda$ where c is the speed of light in vacuume and λ is the center wavelength at which the bandwidth is being calculated. The passband ripple amplitude is around 0.03 dB at the center of the band. Also, the roll-off factor is equal to 0.5 dB/GHz.

3.6 Layout for fabrication

The design reported here is under fabrication by Advanced NanoTools offering Ebeam technology for SiN platform. The layout of the device is shown in Fig. 3.18. As shown in this figure, the footprint of the device (without considering the DC bond pads shown by brown color in the figure) is about 600×800 μm^2. As could be observed in Fig. 3.18, there are 4 thermal phase shifters for post fabrication phase tuning : two of them on the tunable coupler, one on the ring, and another one on the lower arm of the MZI. In Fig. 3.18 there are two thermal isolating deep trench. These deep trench structures prevent thermal cross talk between two heaters. Different test structures was considered for characterization of components used in this design. Back-to-back structures was considered for the characterization of the edge couplers and imbalanced MZI structure for characterization of the thermal phase shifters. Also, cascaded MMIs and Y-splitter was considered to measure the loss of these components. Fig. 3.19 shows our previously fabricated transmitter which includes O-band SiN interleavers. As

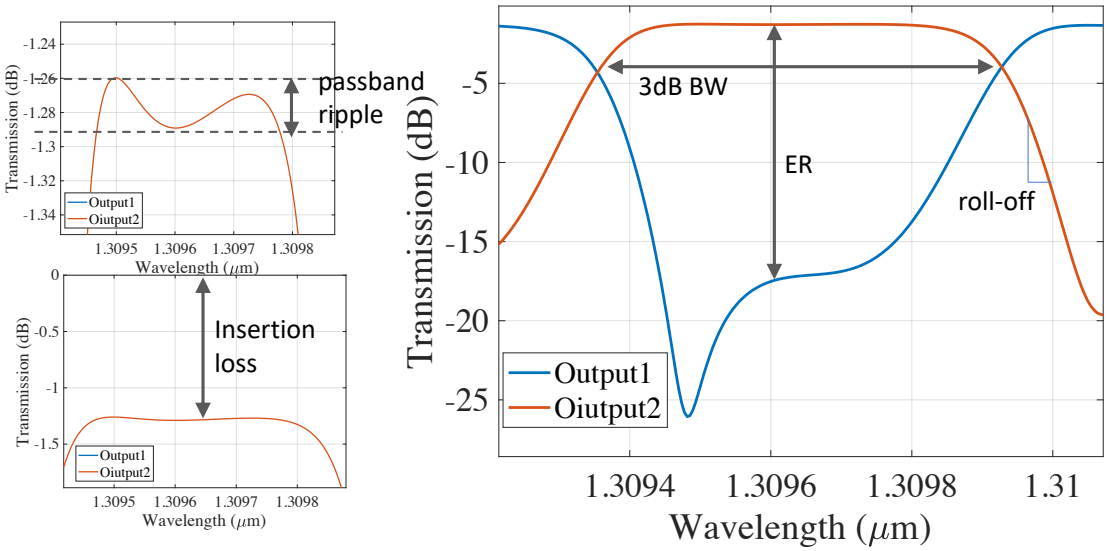


FIGURE 3.17 – Different performance metrics of an interleaver

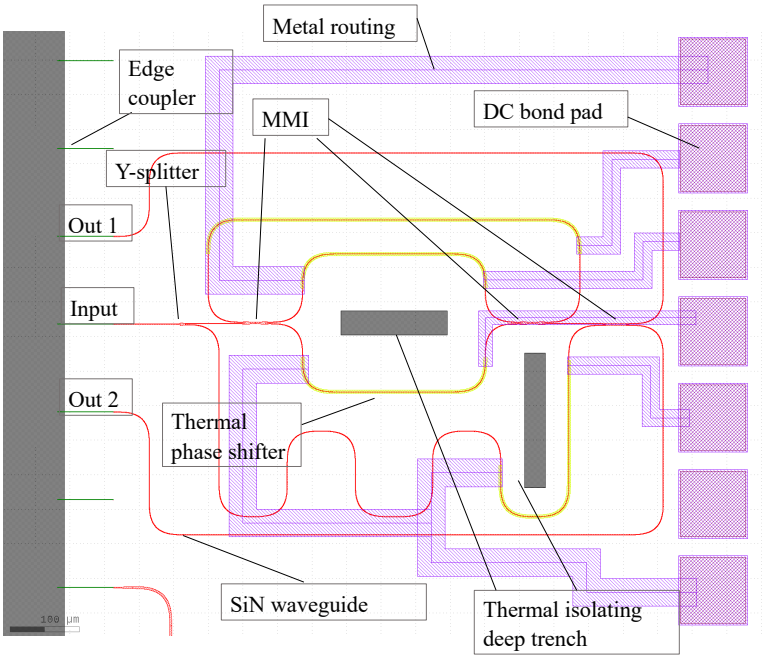


FIGURE 3.18 – Fabrication layout of the designed interleaver.

shown in this figure, the transmitter has one stage of interleaver and there are two modulation buses each containing 6 MRMs.

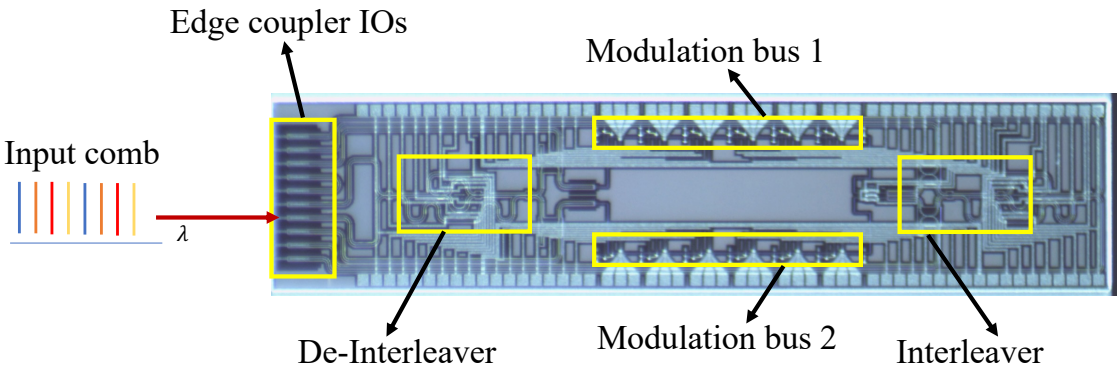


FIGURE 3.19 – Fabricated O-band transmitter with one stage of interleaver and two modulation buses each containing 6 MRMs.

Chapitre 4

Designing a Fabrication Tolerant Interleaver

In this chapter, we design an interleaver that is relatively more tolerant to fabrication errors. The origin of errors due to fabrication comes from the overlap of the optical field with the physical boundaries of devices, where the fabrication imperfections occur. For example, the width and the thickness of a waveguide could be varied along the chip due to multitude of factors during the fabrication process. One way to relatively reduce the adverse effects of fabrication errors is to somehow reduce the overlap between the optical field and the boundaries of the devices. In case of waveguides, from an elementary waveguide analysis it could be understood that when the transverse dimensions of a waveguide increases, the optical mode will have more overlap with the bulk of the waveguide rather than the boundaries. Therefore, if we keep the thickness the same (varying thickness is less practical in a standard fabrication process compared to varying the width), wider waveguides are less prone to fabrication errors. However, when the waveguide is wide it supports higher order modes. Since the operation of devices is mainly based on the fundamental TE_0 mode, any perturbation along the waveguide or even waveguide bends can enable transferring power from the fundamental mode to the higher order modes. In order to mitigate this issue we will use Euler bends to reduce the amount of power coupled to those modes. In the interleaver that was designed in the previous chapter, radial bends were used because the width of the waveguides in that design was 800 nm, for which higher order modes are in their cut-off regime in the O-band window.

Photolithographic errors can cause issues such as overexposure or underexposure of the photoresist, irregular line width, non-uniform etching, and the absence or misalignment of features. These errors, in turn, result in semiconductor components with dimensions that deviate from specifications and performance that falls outside of the designed dimensions. In Si platform, the fabrication errors can be compensated by post-fabrication tuning by utilizing an integrated heater that locally changes the effective index of the waveguide through thermo-optical

effect. Since the TOC of SiN is one order of magnitude lower than that of Si, post fabrication thermal tuning of SiN devices consumes a lot more power. Therefore, a crucial consideration in designing SiN devices is to ensure that they need the least amount of thermal tuning.

4.1 Wide SiN waveguides

As mentioned above, larger waveguides are less affected by fabrication errors. The reason for this lies in the fact that, when the dimensions of a waveguide increases, the light is less intense at the boundaries of the waveguide. The optical properties of the guided mode are related to the real and imaginary parts of the effective index.

The change in effective index is computed by the overlap between the optical electric field and the local change in the index caused by geometric variation that originates from the roughness of the sidewall. The effective index then could be calculated as [125]

$$n_{\text{eff}} = n_{\text{eff}_0} + \frac{dn_{\text{eff}}}{dn_{\text{SiN}}} \times \frac{\int_{\Omega \pm |\delta w_x|} E^*(x) \Delta n_{\text{SiN}}(x) E(x) dx}{\int_{\Omega \pm |\delta w_x|} E^*(x) E(x) dx}. \tag{4.1}$$

In this equation, Δn_{Si} a perturbation function that describes the dimension variation of the waveguide and is non-zero only at the sidewalls. As shown in Fig. 4.1, the boundaries of the ideal waveguide (without any fabrication variation) is defined by Ω. If the fabrication errors lead to the reduction of the waveguide dimensions, i.e., Ω gets shrunk, then the value of Δn_{Si} has a negative sign and is equal to the difference between the cladding index and the core index. On the other hand, if the fabrication errors lead to the enlargement of the waveguide dimensions, i.e., Ω gets enlarged, then the value of Δn_{Si} becomes positive and is equal to the difference between the core index and the cladding index. In Eq. 4.1, for the sake of simplicity, the other transverse coordinate, y, has been omitted. If we define δw_x the maximum amount of variation in the waveguide width (along with direction x) then the integration in Eq. 4.1 has values only in the δw_x neighborhood of Ω because Δn_{Si} is nonzero only when it is in the error zone, i.e., $\Omega \pm \delta w_x$.

It could be observed from this equation that if the electrical field intensity is small at the boundaries of the waveguide, then the overlap integral and therefore the change in effective index will be small too. Fig. 4.1 compares the magnitude of the electrical field for a single mode waveguide with dimensions 800×400 μm^2 and a multimode waveguide with dimensions 3000×400 μm^2. In both case, the modes are the fundamental TE mode. As shown in this figure, the magnitude of the electric field on the left and right boundaries (defined on Fig. 4.1) has decreases by a factor of 3 when increasing the width of the waveguide from 800 nm to 3000 nm.

As discussed above, when the waveguide width is increased, it is predicted that the rate of change of effective index versus waveguide width variation will decrease. This could be observed in Fig. 4.2. In this figure, the effective index for 7 different waveguides with different widths is

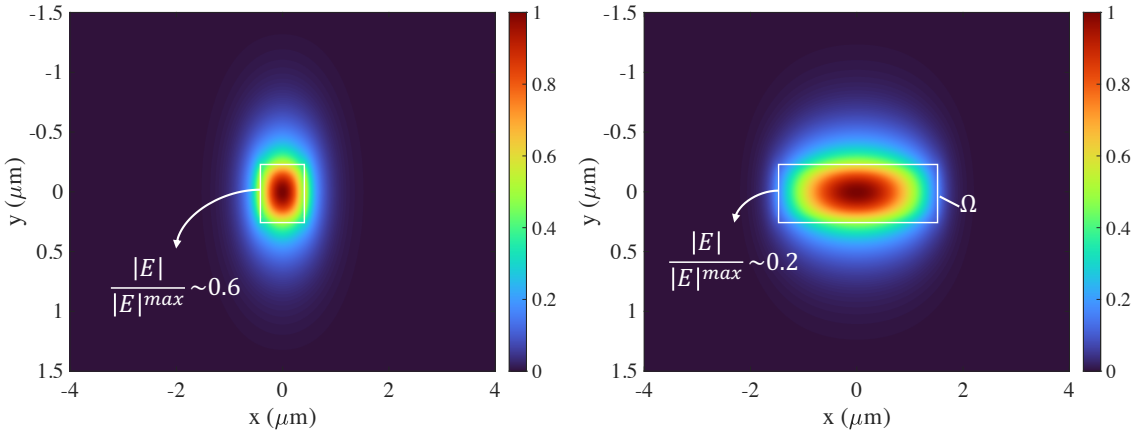


FIGURE 4.1 – Comparison of magnitude of the electric field of the fundamental mode in two waveguides with dimensions 800×400 μm^2 (left) and 3000×400 μm^2 (right).

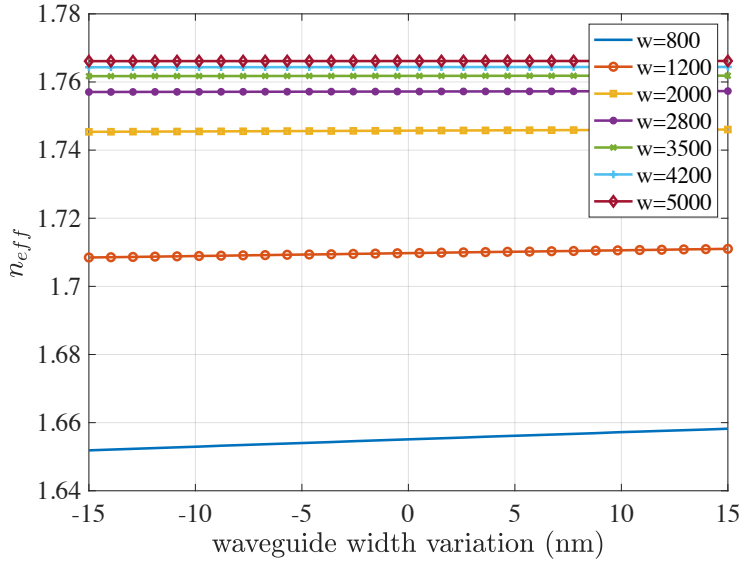


FIGURE 4.2 – Effective index as a function of waveguide width variation for waveguides with different widths.

plotted versus variation in the width of those waveguide. As could be seen, for the waveguide with minimum value of width ($W = 800$ nm) the slope of the curve is highest with respect to other waveguide widths.

For the interleaver that we are going to design, we choose $W = 3000$ nm. For this choice of width, a sidewall roughness of equal to 15 nm can only change the effective index by a value of 6×10^{-5}. Fig. 4.3 (a) shows the variation of effective index versus waveguide width. As shown in this figure, when the waveguide width increases, the variation of the effective index becomes less sensitive to the variation of the width of the waveguide. From this figure, it could be observed that for a single mode waveguide with 800 nm width, the variation in effective

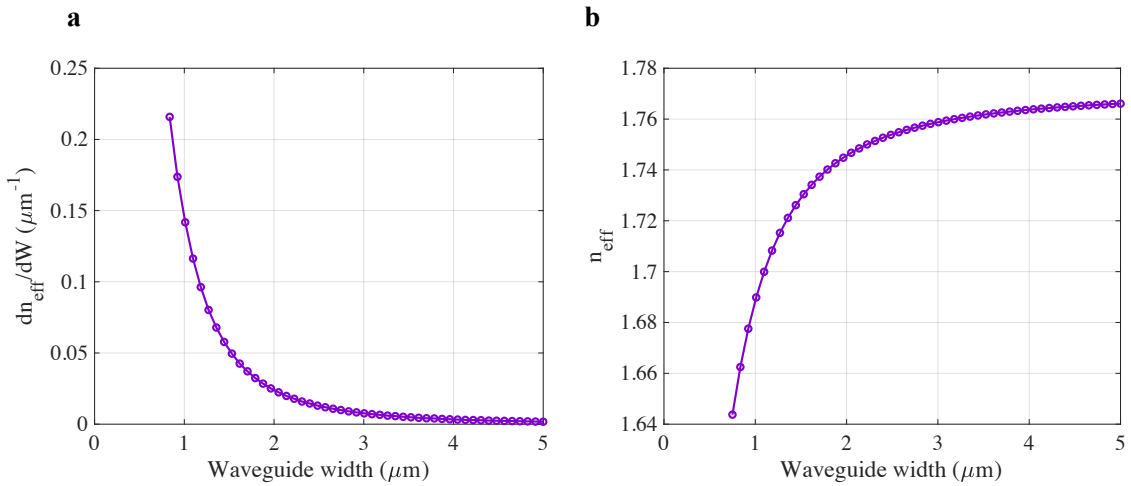


FIGURE 4.3 – (a) derivative of effective index with respect to the waveguide width. (b) effective index versus width.

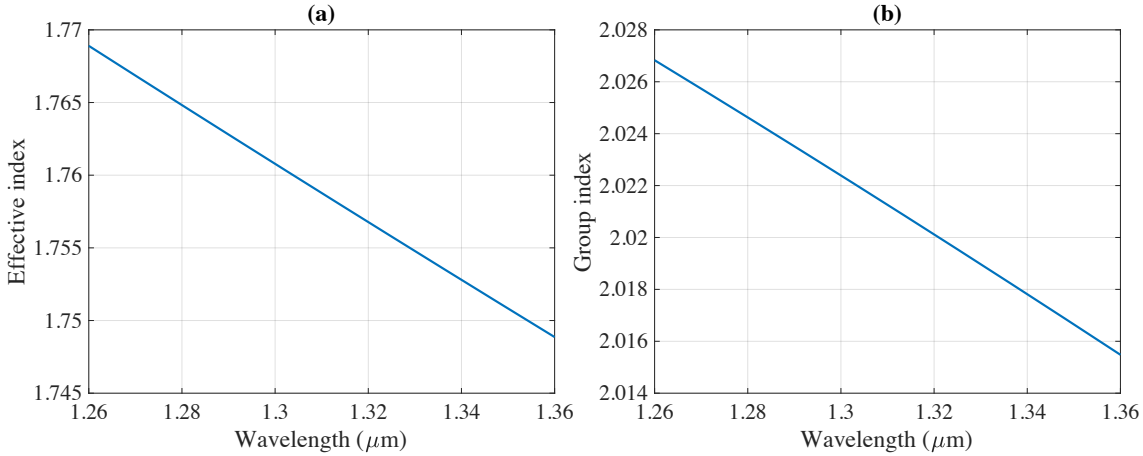


FIGURE 4.4 – Effective index, (a), and group index (b) of SiN waveguide with transverse dimensions 3×0.4 μm^2

index versus change in the width of the waveguide is less than 0.2 μm^{-1} and for a 3000 nm waveguide it is near 0.007 μm^{-1}, which demonstrates that the effective index sensitivity becomes very small for wide waveguides. Choosing larger widths comes at the cost of large footprint of the device. This increase in footprint, as we will see in this chapter, is associated with large bend radia and elongated MMIs.

4.2 Group index and length of the ring

In order to implement the interleaver using the 3μm waveguides, we need to define the length of the ring, which requires the group index of the waveguide. Fig. 4.4 shows the effective index and group index of the waveguide. According to Fig. 4.4, the waveguides group index n_g (at

$\lambda_0 = 1.31\mu$m) is equal to 2.0212. According to $FSR = \frac{2c}{l_{ring} n_g}$, for an FSR equal to 200 GHz, the length of the ring must be 1483.234 μm. Hence, the length of the differential section of the MZI, which is half the length of the ring is equal to 741.619 μm. Note that the same calculations for the general phase of the MMI (and hence the entire tunable coupler) should be applied here in order to obtain the modified value for the ring length. Specifically, we will see in section 4.4 that the l_{ϕ_0} is 222.12 μm. In this case, if we choose $l_1 = 100$ μm, then the modified ring length becomes $l_{ring}^{mod} = 938.99$ μm.

4.3 Euler bend

As discussed previously, we use Euler bend instead of a typical radial bend because in multimode waveguides Euler bends minimize coupling of energy from the fundamental mode to the higher order modes.

Using parametric equations, an Euler bend is describe by its coordinates as (In Appendix A, Listing 1, a MATLAB script for generating an Euler bend with arbitrary parameters has been provided.) [87]

$$\begin{aligned} x(s) &= \int_0^s \cos\left(\frac{t^2}{2R_0^2}\right) dt \\ y(s) &= \int_0^s \sin\left(\frac{t^2}{2R_0^2}\right) dt, \end{aligned} \tag{4.2}$$

where s is the path length and R_0 is the constant of the bend. According to above equation, the curvature could be derived as $\kappa(s) = 2s/R_0^2$. It is insightful to notice that when s is small (beginning section of the bend), the curvature is also small and as the path length increases, so does the curvature in a linear manner. This is shown in Fig. 4.5. Notice that in this figure, the red curve, which represents a radial bend, has a constant curvature but the curvature of the Euler bend increases linearly with the path length.

We compare Euler bend versus radial bend using 3D-FDTD in terms of transmission of the fundamental mode. For sake of comparison, the effective bend radius (R_{eff}) of an Euler bend is defined such that it has the same start and end coordinates as a radial bend with $R = R_{eff}$. Fig. 4.6 demonstrates this space exploration for two different waveguide widths, i.e. $W = 3\mu$m and $W = 5\mu$m. In this figure, the colorbar represents the transmission to the fundamental TE mode. As shown in this figure, in both cases when the effective bend increases, the transmission will increase as well. It could also be observed that choosing a wide waveguide leads to higher amount of loss for a fixed value of radius compared to a narrower waveguide. The reason lies in the fact that when the waveguide becomes wider, the number of modes that it can support will increase. Having more number of modes in a waveguide means that less power would be coupled to the fundamental mode.

Based on the obtained value shown in Fig. 4.6, we choose an Euler bend with $R_{eff} = 102$ μm.

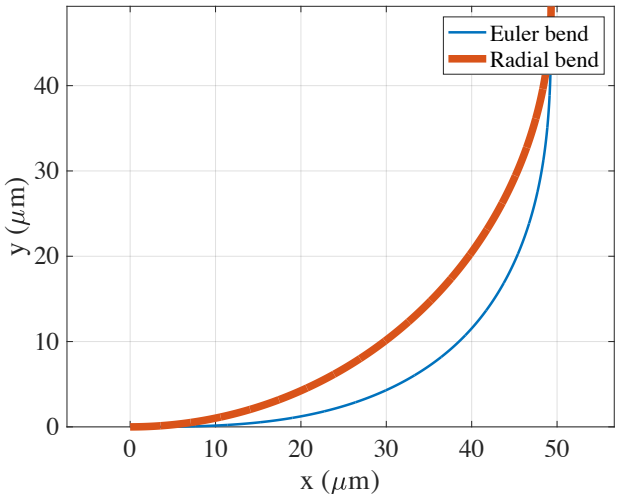


FIGURE 4.5 – Comparison between an Euler bend versus a radial bend.

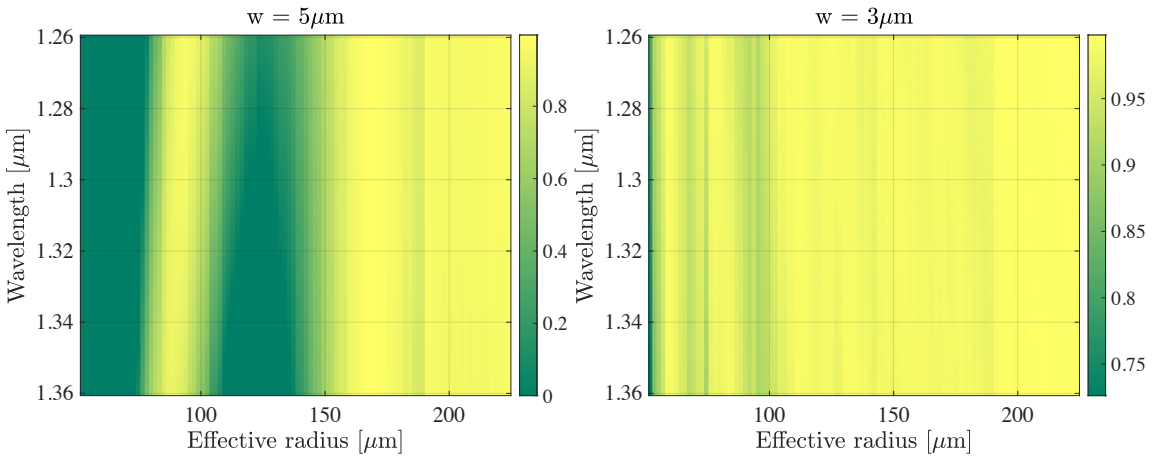


FIGURE 4.6 – Transmission (%) versus wavelength and Euler bend effective radius for two waveguide widths equal to $W = 3$ μm and $W = 5$ μm.

As shown in Fig. 4.7, the TE_0 transmission for this bend is between 0.01 and 0.03 for a 90° bend.

4.4 MMI design

The MMIs used in this design are rectangular. The reason that we do not use MMI structures such as those used in previous chapter is that since the core of MMI is not rectangular in PSO-optimized MMIs, the propagation constant will not remain straight. This will excite higher order modes at the output of the MMI. In fact, PSO-optimized still yields in a 50/50 splitting ratio, however 50% of power at each output port is an aggregate of the powers for all the modes inside the wide waveguide. This is shown in Fig. 4.10. As shown in this figure, the blue curve shows the total power transmission and the red curve shows the transmission for

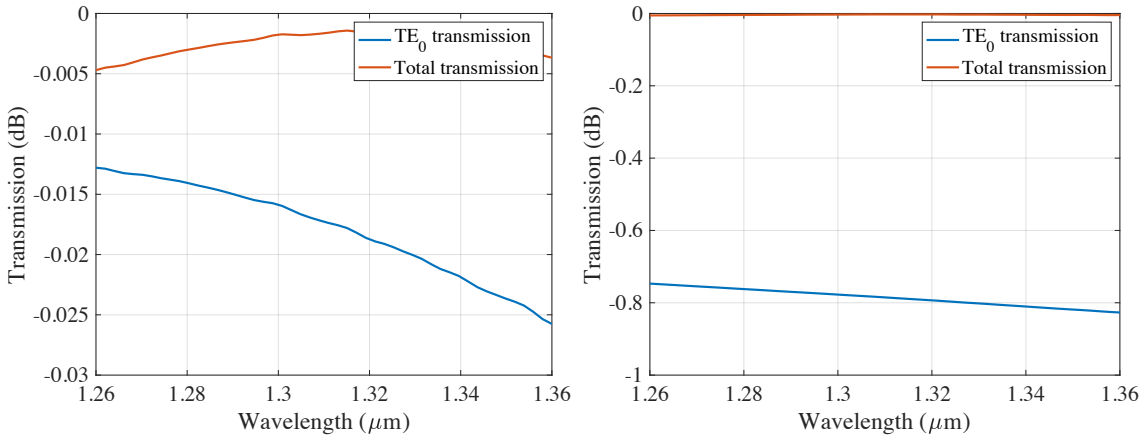


FIGURE 4.7 – Comparison between Euler bend and radial bend. Left plot : Transmission of the TE_0 and the total transmission in an Euler bend with $R_{eff} = 102$ μm. Right plot : Transmission of the TE_0 and the total transmission in a radial bend with $R = 102$ μm.

the fundamental mode only. The remaining amount of power is coupled to the higher order modes. The structure of the rectangular MMI used for this design is depicted in Fig. 4.8 (a). The field profile (magnitude of the electric field) is also depicted in Fig. 4.8 (b). As shown, the width of the core is set to $W = 9$ μm and the access waveguide aperture is $W_a = 3.9\mu$m. Calculating the first two modes' propagation constant and using Eq. 1.7, the length of the core to achieve a 50% splitting ration is $L = 220.541$ μm. For this MMI, the transmission at the two output ports is shown in Fig. 4.8(c). As shown in this figure, the power imbalance between the two output ports is zero at the center of the band and in the left and right most sides of the spectrum, the power imbalance is almost the same and equal to 0.08. Note that the 10 μm choice for the input/output tapers was based on a separate simulation. In that simulation, a taper with wide and narrow widths equal to $W = 9$ μm and $W_a = 3.9\mu$m was considered. Then, the length of the taper was swept from 5 μm to 30 μm. Fig. 4.9 shows the results of this length sweep. As shown, for $L_t = 10$ μm the insertion loss of the taper is near 0.05 dB. These results were obtained using 3D-FDTD simulations.

As discussed in previous section, when designing the interleaver we should calculate the general phase in the MMI. Again, we follow the same procedure as in the previous chapter. We start with Eq. 1.14. The effective index of the fundamental mode inside the core of the MMI is equal to $n^0_{eff} = 1.7691$. For the center wavelength of $\lambda_0 = 1.31\mu m$, the propagation constant for this mode is $\beta_0 = 8.4856 \times 10^6$. Therefore, the general phase inside the MMI is equal to 596.439π . The effective index of the waveguides with $3\mu m$ width used in this design is equal to 1.7588. For this waveguide, the general phase of 596.439π will be equal to a length of 222.12 μm. As could be seen, the length of waveguide to introduce the general phase of the MMI is slightly higher than the length of the MMI.

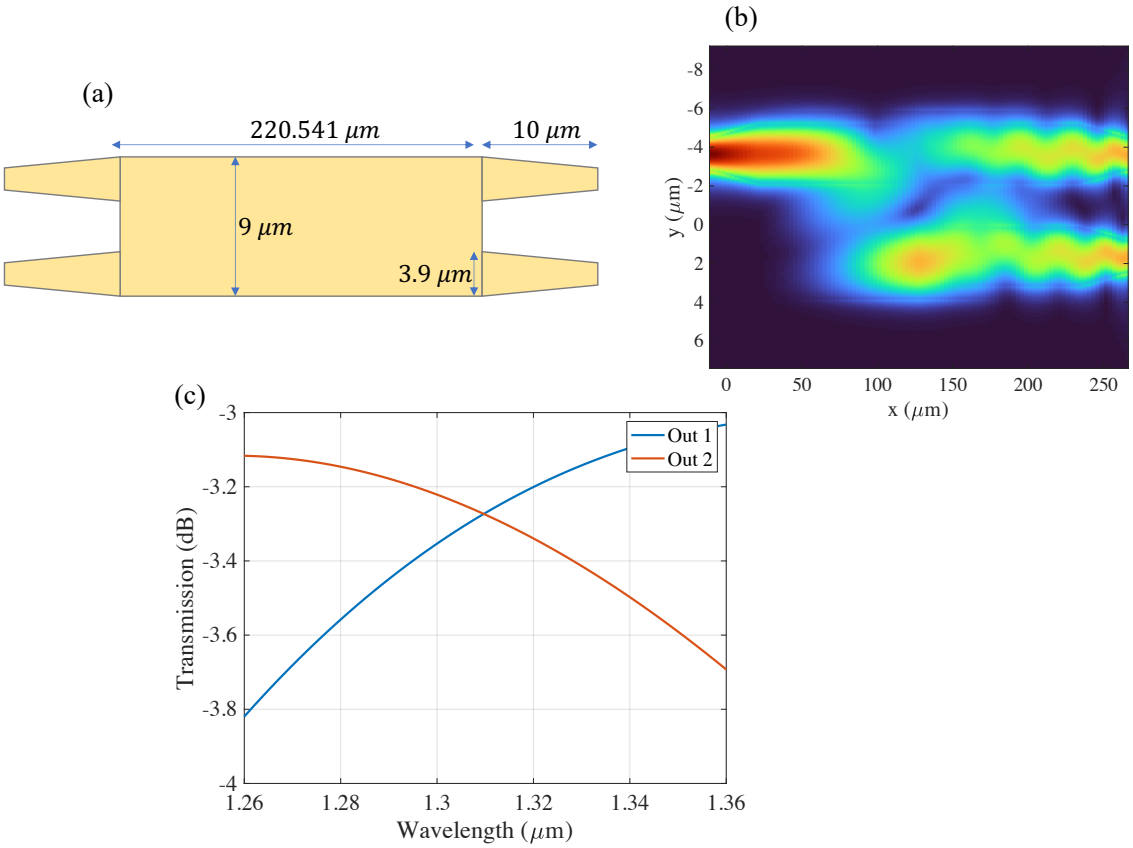


FIGURE 4.8 – (a) Schematics of 50/50 MMI. (b) Field profile of the magnitude of the electric field inside the MMI. (c) Transmission at the two output prots.

4.5 Simulation results

As shown in the design chart in Fig. 3.1, having the S-parameters of the components and obtaining the general phase of the tunable coupler, we can start a circuit level simulations. By running the circuit level simulations, the response of the interleaver is shown in Fig. 4.11. The performance metrics of the designed interleaver could be obtained from Fig. 4.11. According to this figure, the ER, passband bandwidth, IL, and passband ripples are, respectively, 13 dB, 2.5 dB, 0.012 dB, and 50 GHz.

The layout schematics of the designed device has been shown in Fig. 4.12.

4.6 Sensitivity analysis

In this section, we perform simulations to compare the interleaver designed in this chapter with that of the previous chapter in terms of tolerance to fabrication variations. In a fabrication process, numerous parameters are susceptible to variation. These variations could occur among wafers as well as among dies across a wafer.

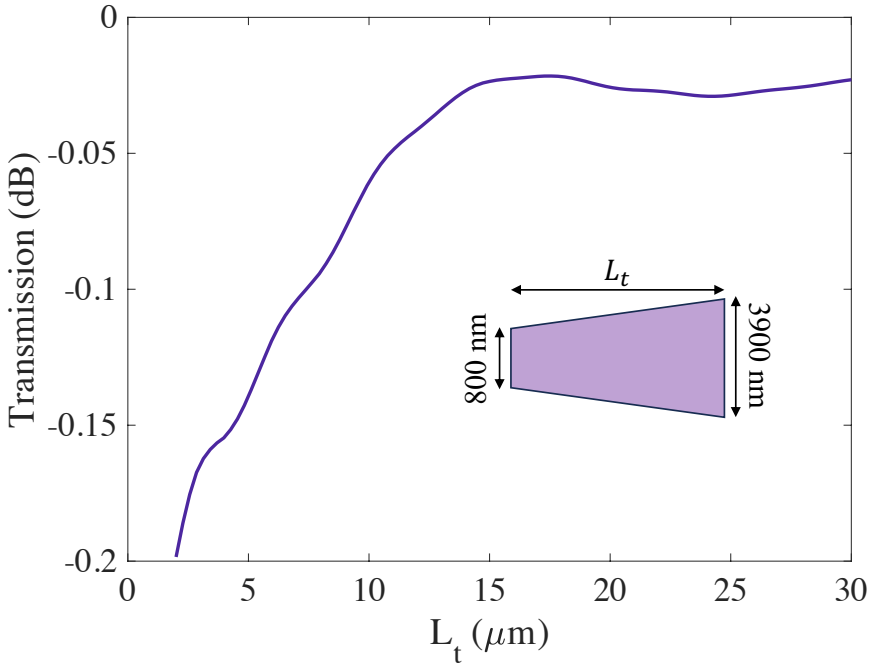


FIGURE 4.9 – Transmission of a linear taper versus its length

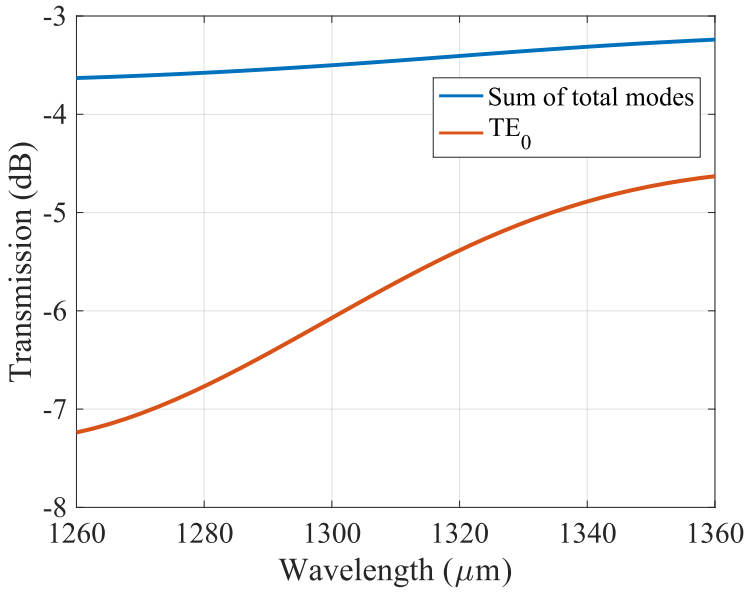


FIGURE 4.10 – Transmission of the fundamental mode and the total power in a PSO-optimized MMI with core width equal to 9μm. It could be observed that since the PSO-optimized geometry leads to the excitation of higher order modes, the transmission to the fundamental mode has significant amount of loss.

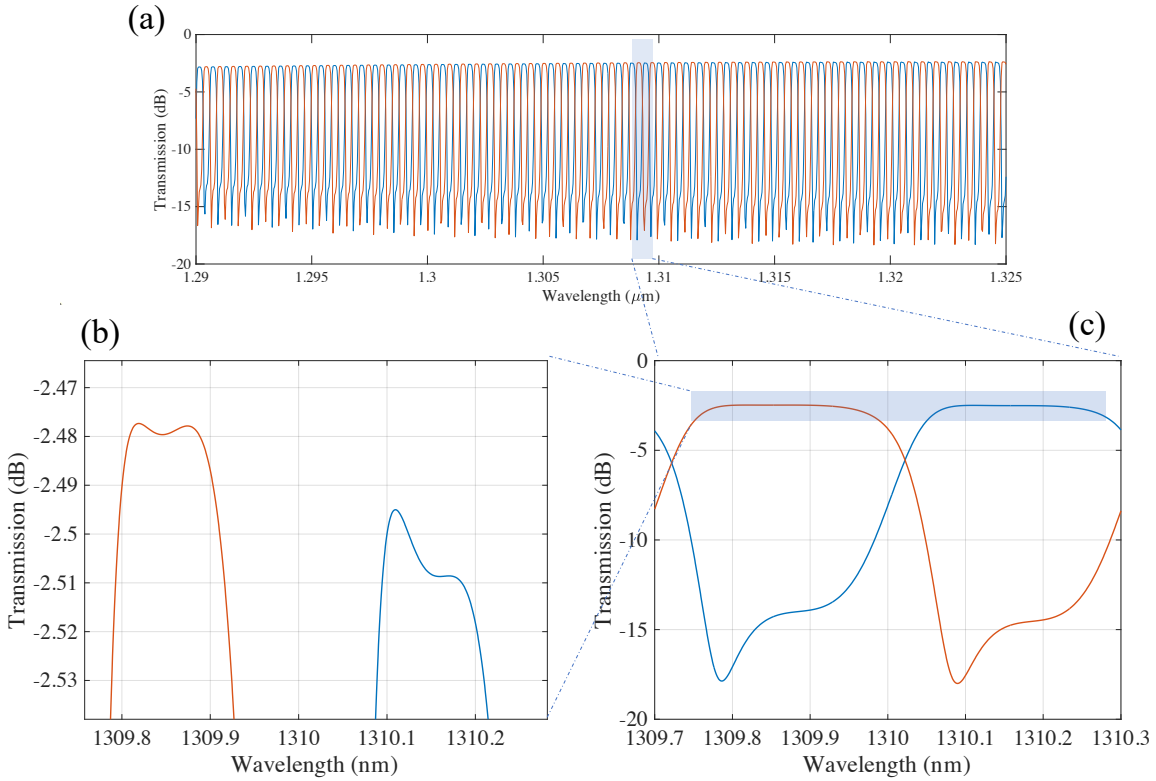


FIGURE 4.11 – INTERCONNECT simulation results of the fabrication tolerant interleaver.

The interleavers that were designed in the current and the previous chapters are submittd for fabrication in an electron-beam (e-beam) lithography foundary. In an e-beam processes, several factors contribute to fabrication variations [126; 127; 128; 129]. These variations can impact the final pattern or structure being created. The amount of electron beam exposure (dose) can vary across the substrate, leading to variations in the exposed patterns. Dose variations can be caused by factors such as uneven resist thickness or charging effects during the exposure. Charging of the substrate during e-beam exposure can result in variations. If the substrate or resist accumulates charge, it may deflect the electron beam, leading to non-uniform exposure and pattern distortions. Furthermore, imperfections in the electron optics of the lithography system can introduce variations in the beam shape and intensity [130].

Fabrication variations could be categorized in a hierarchical order [131]. These categories, from higher to lower hierarchies include : 1)Wafer-to-Wafer Variations, which refers to errors or variations that occur at the level of entire wafers. Example of these errors are differences in material properties, film thickness, or other parameters between different wafers in a batch. 2) Die-to-Die variations, which referes to errors or variations that occur between different dies on the same wafer. For example, Variations in dimensions, doping concentration, or other characteristics from one die to another on the same wafer. 3) Intra-Die variations, which are errors or variations that occur within a single die or chip. Local variations in feature

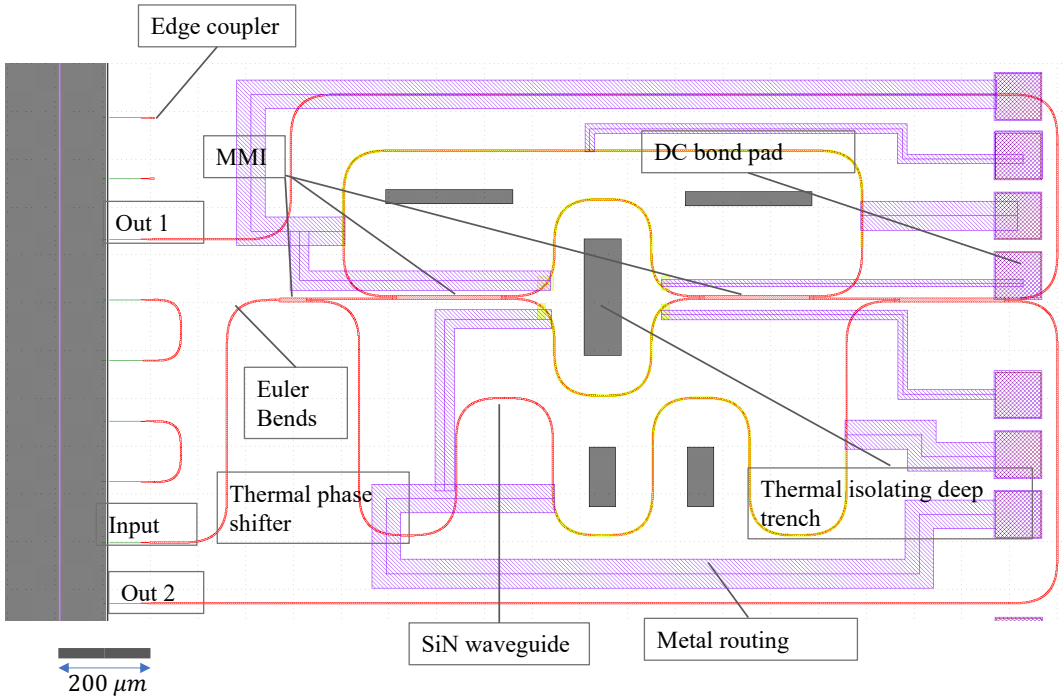


FIGURE 4.12 – Layout of the proposed fabrication tolerant design.

dimensions, material properties, or defects within the boundaries of a single die belong to this category of fabrication error. A through analysis of fabrication variation analysis is out of scope of the current text, however, our primary emphasis is directed towards investigation of device performance versus the variation in the width of the waveguide. In this regard, we choose ±30 nm variations in the width of the waveguide for both the single mode waveguide (width = 800 nm) used in Desing I (henceforth, the interleavers desinged in chapter 3 and 4 are referred to as Desing I and Design II, respectively) the waveguide used in Design II (width = 3000 nm). ±30 nm is consistent with the criteria set by the 6-σ standard, assuming σ (standard deviation of the width) is 5 nm [132; 133]

As could be observed from Fig. 4.2 and Fig. 4.3, ± 30 nm variation in the width of the waveguide leads to change in the effective index of the 3 μm and 0.8 μm waveguides by $\pm 2.28 \times 10^{-4}$ and $\pm 6.47 \times 10^{-3}$, respectively. A quick observation into these effective index variations reveals that the change in the effective index of the 3 μm waveguide is almost 28 times smaller than that of 0.8 μm waveguide. Fig. 4.13 shows the transmission response of the Desing I (Fig. 4.13(b)) and Design II. As could be observed from this figure, Design II has less been impacted by waveguide width variations.

Table 4.1 and 4.2 compare the interleaver FOMs as a function of waveguide width corner values for Design I and Design II, respectively. By comparing the values reported in these tables, it could be observed that the FOMs of Design II obtained using corner waveguide widths deviate relatively slightly from the FOMs obtained using the nominal waveguide width. This

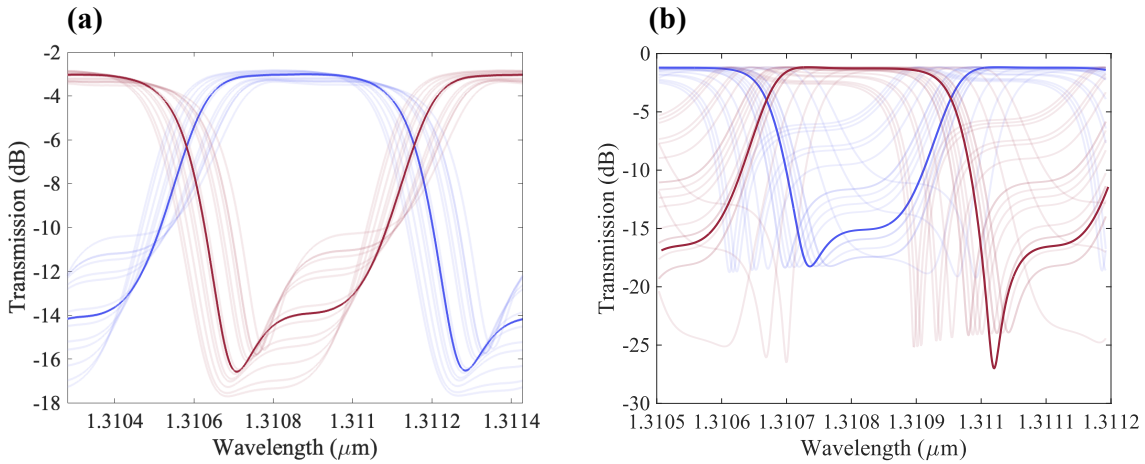


FIGURE 4.13 – Effect of variation in the width of the waveguide on the transmission of the (a) Design II and (b) Design I. Each transmission is associated to a waveguide width, which has been chosen randomly between -30 nm to +30 nm.

TABLE 4.1 – Corner analysis for Design I

Waveguide width	0.77 μm	0.8 μm	0.83 μm
3-dB bandwidth (GHz)	62	70	80
IL (dB)	1.2	1.27	1.2
ER (dB)	5.5	16	5
Passband ripples (dB)	1	0.03	1.1

TABLE 4.2 – Corner analysis for Design II

Waveguide width (μm)	2.97	3	3.03
3-dB bandwidth (GHz)	48	50	55
IL (dB)	2.9	2.5	2.9
ER (dB)	9	13	16
Passband ripples (dB)	0.112	0.012	0.1

is in contrast to the behaviour of the Design I obtained under corner waveguide width. One important point to notice is that, as shown in Table. 4.1, the value of the 3-dB bandwidth in the corners are increased with respect to the value of the standard waveguide width. This should not be interpreted as a improvement because while the bandwidth is increasing, at the same time other FOMs are significantly.

The following table summarizes the performance metrics of the interleavers in the literature.

Conclusion

In work, we have studied the principles of different building blocks required to implement a ring-assisted MZI interleaver. After that, we investigated the requirements to have a flat-top interleaver. Subsequently, the circuit level simulation of two proposed interleaver designs have been performed and the performance metrics of each design was reported. In the first design, we used PSO optimization algorithm to improve the performance of the MMIs used in the interleaver. Using the PSO optimization the MMI's final footprint is 26×3 μm^2 and the entire footprint of the interleaver was 600×800 μm^2. The ER, IL, passband bandwidth, ripple amplitude of this design is, respectively, 16 dB, 1.27 dB, 70 GHz, and 0.03 dB.

In the last chapter, we proposed a novel design for fabrication tolerant interleaver. In this design, wide waveguide widths have been exploited, which will result in reduced sensitivity to fabrication errors due to side wall roughness. In order to avoid the excitation of higher order modes in the wide waveguide, we used Euler bends. An Euler bend with effective radius of $102\mu m$ was used, which resulted in a small loss equal to 0.015 dB. A rectangular 50/50 splitting ratio MMI with core dimensions 9×220.74 μm^2 was considered in this design. After device level simulation and extracting the S-parameters using 3D-FDTD and exporting them to INTERCONNECT, the performance metrics of the device was extracted. The ER, IL, passband bandwidth, ripple amplitude of this design is, respectively, 13 dB, 2.5 dB, 50 GHz, and 0.012 dB.

The following table summarizes the key performance metrics of the interleavers designed in chapter 3 (Design I) and chapter 4 (Design II). From this table, it can be seen that Design II shows a promising performance in terms of passband ripples compared to other ones. In addition, between RAMZI structures, Design I and II show good performance in terms of IL.

The following items outlines several promising directions for future research in interleaver design. These proposed endeavors aim to not only address current limitations but also to push the boundaries.

1. **Thermal insensitive design**

   In light of the inherent phase sensitivity of the RAMZI interleavers, temperature fluctuations pose a significant challenge despite the low thermo-optical coefficient of SiN.

Ref	Material Platform	Frequency window (nm)	ER (dB)	IL (dB)	3-dB BW (GHz)	Ripples (dB)	Architecture	Year
[134]	SiN	1520-1620	22	8	100	-	RAMZI	2024
[135]	SOI	1530-1600	20	8	120	1.8	RAMZI	2010
[136]	SOI	1525-1650	15	1.5	150	0.5	RAMZI	2020
[137]	SOI	1549-1551	20	NR	31	1	Sagnac loop	2018
[138]	SOI	1570-1590	10	8	873	0.9	RAMZI	2008
[139]	SOI	1510-1580	10.5	10	1370	0.8	RAMZI	2008
[140]	SOI	1550-1590	20	8	255	1.1	8-shape CRR	2008
[141]	III-V/Si	1300-1311	10	2	69	0.75	RAMZI	2022
[142]	LN	1528-1605	10	17	10	3	MZI	2018
[143]	SOI	1510-1570	20	0.5	137	0.9	Sagnac loop	2017
[144]	SOI	1460-1620	20	0.3	3300	0.1	Interfering loop	2018
This work (I)	SiN	1260-1360	16	1.27	70	0.03	RAMZI	2023
This work (II)	SiN	1260-1360	13	2.5	50	0.012	RAMZI	2023

To mitigate the thermal sensitivity, strategies involving thermo-electric coolers and on-chip heaters are employed. These techniques aim to externally stabilize the device or locally compensate for temperature drift. However, such approaches introduce heightened complexity and elevate power consumption levels. The designs in this thesis remain susceptible to performance alterations induced by variations in temperature. Recognizing this, there emerges an imperative for the development of a thermally insensitive design. To the best of our knowledge, there is no published research on athermal interleaver. However, thermal insensitive MZI structures have been reported.

In this designs, the temperature dependence of the phase delay in the MZI arms is effectively neutralized. This results in a wavelength response that remains resilient to temperature variations, a balance achieved through the deliberate design of MZI arms featuring different widths [145; 146], different modes [147; 148; 149], or even different configurations [150; 151]. On the other hand, several groups reported athermal ring resonator [152; 153]. Using these primitive ideas, a potential structure for athermal interleaver could be developed.

2. **Higher order interleavers**

   In both interleaver that was designed in chapter 3 and 4, there was only one ring on the MZI. The number of rings in a RAMZI structure defines the order of the device. By increasing the order of the interleaver, i.e., exploiting more rings on each arm of the MZI, some of the FOMs of the devices improves, while insertion loss and footprint exacerbate. Up to our knowledge, there hasn't been reported any high order and high performance SiN-based interleaver. An example of high order silicon interleaver filter is reported in [154], however designing a similar device in SiN platform could be challenging in terms of footprint reduction.

3. **Improving the loss and footprint**

Even though SiN is inherently low in material loss, MMI couplers in the circuit can mainly contribute to the insertion loss. On the other hand small radius bends lead to small footprint of the device. However, by reducing the bend radii, loss increases. Hence, sophisticated bend structures and MMI couplers can potentially improve the loss of the interleaver. The loss issue becomes a bigger challenge when designing a higher order interleaver where there exist multiple number of MMIs and waveguide bends.

4. **Designing wide Euler bends to operate as an MMI**

   Even though higher order moded in an Euler bend have small amount of power, interference of those modes inside and Euler bend can effectively make a bent MMI. Optimizing this bent MMI structure can lead to even smaller bend radii. This fact could be observed in Fig. 4.6. In the left image of this figure, ther a tape is shown representing the loss for effective bend radii near 80 to 100 μm. A similar behaviour is observed in the results reported in [67]

Annexe A

Code listing

A.1 Listing 1

MATLAB code for generating coordinates of an Euler bend.

```
clear all
syms t



Lmax = 900;
Waveguide_Width = 3; #micrometers
h = Waveguide_Width;
L = linspace(0,Lmax,30000);
A = [33]; # Euler parameter
integ_res = 1000000; # accuracy of integration
for j=1:1
    j
x1(1)=0;y1(1)=0; # The orignin of the Euler bend is set to (0,0)

for ip=2:length(L)
        T = linspace(0, L(ip)/A(j), integ_res);
         DT = T(2) - T(1);
x1(ip) = A(j) * DT * sum(cos((T.^2)/2));
y1(ip) = A(j) * DT * sum(sin((T.^2)/2));
if (x1(ip)-x1(ip-1)) < (y1(ip)-y1(ip-1))
    break;
end
end


x2 = -y1 + x1(end)+y1(end);
```

```
y2 = -x1 + y1(end)+x1(end);

Reffective(j) = x1(end)+y1(end); # Effective radius of the Euler bend

X=[x1,flip(x2)];
Y=[y1,flip(y2)];

for i=2:length(X)
    dl(i-1)=sqrt((X(i)-X(i-1))^2 + (Y(i)-Y(i-1))^2);
end
lcurve(j) = sum(dl); # Length of the Euler bend
dX.(strcat('X_',num2str(j))) = X;
dY.(strcat('Y_',num2str(j))) = Y;

dataX=struct2cell(dX);
dataY=struct2cell(dY);

end

## Generation of a radial bend
th = linspace(3*pi/2,2*pi,length(X));
xunit = Reffective(1) * cos(th);
yunit = Reffective(1) * sin(th) +Reffective(1);

for i=2:length(xunit)
    dlunit(i-1)=sqrt((xunit(i)-xunit(i-1))^2 + (yunit(i)-yunit(i-1))^2);
end
##

## Plot and observe the Euler bend versus the radial bend
plot(X,Y,LineWidth=2)
hold on
plot(xunit,yunit,LineWidth=6)
legend("Euler bend","Radial bend")
set(gca,'fontsize', 20)
set(gca,'fontname','times')
xlabel('x (\mu{m})')
ylabel('y (\mu{m})')
axis equal
grid on


x1(end)-x1(end-1)
y1(end)-y1(end-1)
X1=dataX{1};Y1=dataY{1};
X1 = X1.';
Y1 = Y1.';
```

```
Xp = X1;
Yp = Y1;

alpha = zeros(1,length(X1)-1);

for i=2:length(Xp)
    alpha(i-1) = atan((Yp(i)-Yp(i-1))/(Xp(i)-Xp(i-1)));
end

for i=1:length(Xp)-1
    Xq(i) = Xp(i) + 0.5*h*sin(alpha(i));
end

for i=1:length(Xp)-1
    Yq(i) = Yp(i) - 0.5*h*cos(alpha(i));
end

for i=1:length(Xp)-1
    Xs(i) = Xp(i) - 0.5*h*sin(alpha(i));
end

for i=1:length(Xp)-1
    Ys(i) = Yp(i) + 0.5*h*cos(alpha(i));
end

Xq = Xq.';
Xs = Xs.';
Yq = Yq.';
Ys = Ys.';

Vsq(:,1) = cat(1,Xq,flip(Xs));
Vsq(:,2) = cat(1,Yq,flip(Ys));

pd = Vsq(:,2);
## This section elliminates the numerical artifacts
epsi = 0.2;

if ~mod(size(Vsq,1),2)
    for i=2:size(Vsq,1)/2
        if (pd(i)-pd(i-1))>epsi
           Ind1= i;
        else
        Ind1 = 0;

        end

    end
```

```
    for i=1+size(Vsq,1)/2:size(Vsq,1)
        if (pd(i)-pd(i-1))<-epsi
           Ind2= i;
        else
                    Ind2 = 0;

        end
    end

else
        for i=2:(size(Vsq,1)+1)/2
             if (pd(i)-pd(i-1))>epsi
           Ind1= i;
             else
                         Ind1 = 0;

             end
        end

    for i=(size(Vsq,1)+1)/2:size(Vsq,1)
        if (pd(i)-pd(i-1))<-epsi
           Ind2= i;
        else
                    Ind2 = 0;

        end
    end


end

if Ind1~=0 && Ind2~=0

Vsq(Ind1,:) = [];
Vsq(Ind2-1,:) = [];
end
##
Final coordinates for the Euler bend are saved in "Vsq"

Reff = Xp(end);
save("Vertices_R"+num2str(fix(Reff)+"_W"+num2str(fix(h))+"p"+num2str(10*(h-
    floor(h)))),"Vsq")
```

A.2 Listing 2

The following Lumerical script code calculates the coupling loss in an Euler bend. The code is written for Var-FDTD environment.

```
#This script calculates the amount of power coupled to the first 4 TE modes
#in an Euler bend. The X and Y arrays are the coordinates of an Euler bend.


switchtolayout;
clear;
deleteall;


NumberofEulerBends = 100;

# Number of frequency points
nf = 80;

# Waveguide width
W = 1e-6*[0.8];

# Preallocation of the arrays that our results are going to be saved in.
TE0TE0 = zeros(nf,100);
TE0TE1 = zeros(nf,100);
TE0TE2 = zeros(nf,100);
TE0TE3 = zeros(nf,100);

#for loop over Euler bend effective radia
for (i=NumberofEulerBends){
matlabload("dataX.mat");
X=dataX{i};
matlabload("dataY.mat");
Y=dataY{i};
V = 1e-6*transpose([X;Y]);


# The next three variables are used to define the positioning of the output
    monitors.
mon_x_span = 15e-6;
mon_y_span = 15e-6;
sosodis = 5e-6;

Reff = (X(end)+Y(end))/2;


# Oxide cladding
addrect;
set("name","SiO2");
set("x",0);
set("x span",800e-6);
```

```
set("y",0);
set("y span",800e-6);
set("z",0);
set("z span",10e-6);
set("material","SiO2 (Glass) - Palik");
set("render type","wireframe");

# The Euler bend waveguide
addwaveguide;
set("poles",V);
set("base width",W(1));
set("base height",0.4e-6);
set("Base Angle",83.5);
set("material","Si3N4 (Silicon Nitride) - Phillip");

wg=80e-6;
wgpoleI = [-wg,0;0,0];

# Input access waveguide
addwaveguide;
set("name","access_wg_I");
set("poles",wgpoleI);
set("base width",W(1));
set("base height",0.4e-6);
set("material","Si3N4 (Silicon Nitride) - Phillip");

wgpoleO = [Reff*1e-6,Reff*1e-6;Reff*1e-6,Reff*1e-6 + wg];
# Output access waveguide
addwaveguide;
set("name","access_wg_O");
set("poles",wgpoleO);
set("base width",W(1));
set("base height",0.4e-6);
set("material","Si3N4 (Silicon Nitride) - Phillip");


# Adding the solver and defining its position with respect to the Euler bend's
    effective radia.
xmin =-wg/10;
xmax =Reff*1e-6 + 2*mon_x_span;
ymin=-2*mon_y_span;
ymax=Reff*1e-6 + wg/10;
xcenter=(xmin+xmax)/2;
xp0 = -xcenter-wg/10+sosodis;
ycenter = (ymin+ymax)/2;
yp0 = -ycenter;
addvarfdtd;
set("simulation time",10000e-15);
```

```
set("x min",xmin);
set("x max",xmax);
set("y min",ymin);
set("y max", ymax);
set("z",0);
set("z span",4e-6);
set("x0",xp0);
set("y0",yp0);
set("mesh accuracy",4);


addpower;
set("name","Mon1");
set("monitor type", 6);
set("x",Reff*1e-6);
set("x span",2*mon_x_span);
set("y",Reff*1e-6 + wg/10 - sosodis);
set("z",0);
set("z span",4e-6);

addmodeexpansion;
set("name","expOutTE0");
set("monitor type", 1);
set("x",Reff*1e-6);
set("x span",2*mon_x_span);
set("y",Reff*1e-6 + wg/10 - sosodis);
set("z",0);
setexpansion("input", "Mon1");
setglobalmonitor("frequency points",nf);

addmodeexpansion;
set("name","expOutTE1");
set("monitor type", 1);
set("x",Reff*1e-6);
set("x span",2*mon_x_span);
set("y",Reff*1e-6 + wg/10 - sosodis);
set("z",0);
setexpansion("input", "Mon1");
setglobalmonitor("frequency points",nf);
set("mode selection","user select");
set("selected mode numbers",4);


addmodeexpansion;
set("name","expOutTE2");
set("monitor type", 1);
set("x",Reff*1e-6);
set("x span",2*mon_x_span);
```

```
set("y",Reff*1e-6 + wg/10 - sosodis);
set("z",0);
setexpansion("input", "Mon1");
setglobalmonitor("frequency points",nf);
set("mode selection","user select");
set("selected mode numbers",6);

addmodeexpansion;
set("name","expOutTE3");
set("monitor type", 1);
set("x",Reff*1e-6);
set("x span",2*mon_x_span);
set("y",Reff*1e-6 + wg/10 - sosodis);
set("z",0);
setexpansion("input", "Mon1");
setglobalmonitor("frequency points",nf);
set("mode selection","user select");
set("selected mode numbers",8);


addmodesource;
set("x",-((wg/10)-sosodis));
set("y min", -1*mon_y_span);
set("y max", 1*mon_y_span);
set("wavelength start",1260e-9);
set("wavelength stop",1360e-9);



run;


# Obtaining the net power from mode expansion monitor.
NN0= getresult("expOutTE0","expansion for input");
TE0TE0(:,i)=NN0.T_net;
NN1= getresult("expOutTE1","expansion for input");
TE0TE1(:,i)=NN1.T_net;
NN2= getresult("expOutTE2","expansion for input");
TE0TE2(:,i)=NN2.T_net;
NN3= getresult("expOutTE3","expansion for input");
TE0TE3(:,i)=NN3.T_net;

switchtolayout;
deleteall;
}
# Saving the data in a MATLAB file.
matlabsave("TransmissionTo_TE0",TE0TE0);
matlabsave("TransmissionTo_TE1",TE0TE1);
```

```
matlabsave("TransmissionTo_TE2",TE0TE2);
matlabsave("TransmissionTo_TE3",TE0TE3);
```

A.3 Listing 3

This script calculates the field profile across the core of an MMI. The script assumes a sinusoidal 2D mode inside the core and neglects the evanescent field in the cladding. As a first step of designing an MMI, it could be useful to estimate the length of the MMI for realizing a 3-dB coupler.

```
clear all
clc

Nmode = 40;  #number of available guiding modes in the core of MMI.
W = 9e-6;  # Width of the core of the MMI
x = linspace(0,W,500); # spatial grid over the core of the MMI



# effective index values for the first two modes of the core section
# corresponding propagation constants are calculated using the center
# wavelength of 1310 nm.

neff1 = 1.791184;
neff2 = 1.786943;
beta(1) = 2*pi*neff1/1310e-9;
beta(2) = 2*pi*neff2/1310e-9;
DelBeta = beta(1) - beta(2);


M =1;
N = 2;
Lc = pi/DelBeta;


LBL = 0e-6;
LBH = (3 * Lc * M/N);
SL = 1000;

#preallocation of the field
Eout_2D = zeros(length(x),SL);


Lsweep = linspace(LBL,LBH,SL);
```

```
for bbb=1:SL

App = 3.9e-6; # Access waveguide widths
off = 0e-6; # Offset of the access waveguides w.r.t to edge of the core







dx = x(2) - x(1);

L = Lsweep(bbb);

# As written in the following for-loop, the modes are approximated
# with sinusoidal functions
for i =1:Nmode
    E(:,i) = sin(pi*(i)*x/W);
end

f = sin(pi*(x-off)/App); # Input mode

for i=1:length(x)
if x(i)>App+off | x(i)<off
    f(i) =0;
end
end


# Calculation of the decomposition weithts (a is the same as $c_\nu$ in the
    text)
for i=1:Nmode
    a(i) = (2/W) * sum(dx * f.' .* E(:,i));#/sqrt(sum(dx * E(:,i) .* E(:,i)));
end


# Calculation of the propagation constants using self imaging theory
for i=2:Nmode
    beta(i) = beta(1) - (i-1)*(i+1)*(DelBeta/3);
end


# Propagation of each mode using the calculated propagation constant
for i=1:Nmode
    Epro(:,i) = E(:,i) * a(i) * exp(-1i*beta(i)*L);
end
```

```
# summation of the propagated fields at each longitudinal mesh
Eout = sum(Epro,2);
Eout_2D(:,bbb) = Eout;
end


for i=1:length(x)
if x(i)<(App/2)
    xpos = i;
end

end

imagesc(x*1e6,Lsweep*1e6,(abs(Eout_2D)))
set(gca,'fontsize',14)
xlabel("MMI length [um]")
ylabel("MMI core width [um]")
set(gca,'fontsize',20)
set(gca,'fontname','times')
colormap turbo
colorbar
```

A.4 Listing 4

```
# Transfer matrix model of ring-assisted MZI interleaver
# Waveguides are assumed to be lossless
# Effective and group index values are calculated for a 3 \mu{m} SiN
# waveguides
# Coupling from the upper arm of the MZI to the ring is assumed to be a
# point coupler

clear all
clc
close all
format long

c = physconst('lightspeed');
wavelength = linspace(1317e-9,1323e-9,1900);
FSR = 200e9;

neffw = 1.780950;  # Effective index of the waveguide calculated using
    Lumerical FDE
ngw = 2.047728;    # Group index of the waveguide calculated using Lumerical
    FDE
a = 1; # Lossless condition
lring =  2*c/ngw/FSR; # Ring's length
Coup_Co = 0.89; # MZI-ring point coupler coupling coefficient (power)
```

```
sigma = sqrt(1-Coup_Co);
DeL = lring*0.5; # Common arms of the MZI

l_com = 250e-6; # Length of each arm of the tunable coupler (l_1 and l_2 in
    the text)
r = sqrt(2)/2;  # r and t are the through and cross coefficient of a 3-dB
    couler.
t = sqrt(1-r^2);

# Lpi must be added to the ring's length
Lpi = 1310e-9/2/neffw; # waveguide length to produce \pi-phase shift


for i=1:length(wavelength)
    Phi_pi = exp(-1i*2*pi*neffw*1*Lpi./wavelength(i));
    Phi_comm = exp(-1i*2*pi*neffw*l_com/wavelength(i));
    Phi_rt = exp(-1i*2*pi*neffw*1*lring/wavelength(i)) * Phi_pi;

    Hring = 1 * (-a + sigma*Phi_rt)/(-a*sigma + Phi_rt);
    M1  = [t   exp(1i*pi/2)*r  ;  exp(1i*pi/2)*r   t];
    M2 = [Phi_comm , 0; 0 , Phi_comm];
    M3 = [ 1* Hring, 0; 0 , exp(1i*2*pi*neffw*1*DeL/wavelength(i))];
    M4 = [Phi_comm , 0; 0 , Phi_comm];
    M5 = M1;
    M = M1 * M2 * M3 * M4 * M5; # Total transfer matrix
    A = M * [1;0];
    X(i) = A(1);
    Y(i) = A(2);

end

f = c./wavelength;
omega = 2*pi*f;
T_pho = neffw*lring/(c);
normalized_Detun = omega * T_pho;

phaseImb=(angle(exp(1i*neffw*DeL*2*pi./wavelength)));
Phi_rt = exp(-1i*2*pi*neffw*1*lring./wavelength) * Phi_pi;

Hring = 1 * (-a + sigma*Phi_rt)./(-a*sigma + Phi_rt);
phaseRing = (angle(Hring));

figure ()
plot(1e-12*f,(phaseRing),LineWidth=2)
hold on
plot(1e-12*f,(phaseImb),LineWidth=2)
legend("Ring phase","Imbalance phase")
grid on
```

```
xlabel("Frequency (THz)")
ylabel("Phase (rad)")
set(gca,'fontsize',18)
set(gca,'fontname','Times')

figure()
plot(1e-12*f,((((abs(X)))).^2),LineWidth=2)
hold on
plot(1e-12*f,(((abs(Y))).^2),LineWidth=2)
grid on
xlabel("Frequency (THz)")
ylabel("Normalized transmission [#]")
set(gca,'fontsize',20)
legend("Bar","Cross")
set(gca,'fontname','Times')
```

A.5 Listing 5

```
    # The following scripts uses transfer matrix method for a balanced MZI
    tunable coupler

clear all
clc
close all

wavelength = linspace(1500e-9,1600e-9,1900);

neff = 1.76;
beta0 = neff*2*pi./wavelength;

# the following phase shift (phi0) is added to the upper arm of the MIZ
phi0 = 0.6484; # radian

l1 = 100e-6; # Length of the MZI arms.
l2 = l1 ;

a1 = 1; # Lossless waveguides a = exp(-\alpha * l) where alpha is assumed 0
a2 = 1; # Lossless waveguides a = exp(-\alpha * l) where alpha is assumed 0

t = sqrt(2)/2;
r = sqrt(1-t^2);

for i=1:length(wavelength)
phi1 = 2*pi*neff*l1/wavelength(i);
phi2 =phi0 + 2*pi*neff*l2./wavelength(i);

C1 = [t -1i*r;-1i*r t];
Len = [a1*exp(-1i*phi1),0; 0,a2*exp(-1i*phi2) ];
```

```
C2 = C1;
T = C2 * Len * C1 * [1;0];
Tbar(i)=T(1);
Tcross(i)=T(2);

end
plot(1e6*wavelength,(abs(Tbar).^2),1e6*wavelength,(abs(Tcross)).^2,LineWidth
    =2)
xlabel("Wavelength (\mu{m})")
ylabel("Transmission (#)")
set(gca,'fontname','times')
set(gca,'fontsize',20)
grid on
ylim([0 1]);
```

Bibliographie

[1] De Vries, A. (2023). The growing energy footprint of artificial intelligence. *Joule*, *7*(10), 2191–2194. https ://doi.org/10.1016/j.joule.2023.09.004

[2] Difrenza, R., Llinares, P., & Ghibaudo, G. (2003). The impact of short channel and quantum effects on the MOS transistor mismatch. *Solid-State Electronics*, *47*(7), 1161–1165. https ://doi.org/10.1016/s0038-1101(03)00033-9

[3] *Home - / TOP500*. (n.d.). https ://top500.org/

[4] Trueman, C. (2024, May 13). *Frontier remains world's most powerful supercomputer on TOP500 list; no longer only exascale machine*. All Content RSS. https ://www.datacenterdynamics.com/en/news/frontier-remains-number-one-on-fifth-consecutive-top500-list-no-longer-only-exascale-machine/

[5] Courtland, R. (2024b, February 13). *Gordon Moore : The Man Whose Name Means Progress*. IEEE Spectrum. https ://spectrum.ieee.org/gordon-moore-the-man-whose-name-means-progress

[6] Han, W., & Wang, Z. M. (2013). *Toward Quantum FinFET*. Springer Science & Business Media.

[7] Qu, J., Zhang, H., Xu, X., & Qin, S. (2011). Study of Drain Induced Barrier Lowering(DIBL) Effect for Strained Si nMOSFET. *Procedia Engineering*, *16*, 298–305. https ://doi.org/10.1016/j.proeng.2011.08.1087

[8] Minkenberg, C., Krishnaswamy, R., Zilkie, A., & Nelson, D. (2021). Co-packaged datacenter optics : Opportunities and challenges. *IET Optoelectronics*, *15*(2), 77–91. https ://doi.org/10.1049/ote2.12020

[9] Goodman, J., Leonberger, F., Kung, S., & Athale, R. (1984). Optical interconnections for VLSI systems. *Proceedings of the IEEE*, *72*(7), 850–866. https ://-doi.org/10.1109/proc.1984.12943

[10] Tsiokos, D., Kanellos, G. T. (2017). 2 - Optical interconnects : Fundamentals. In T. Tekin, R. Pitwon, A. Håkansson, N. Pleros (Eds.), Optical Interconnects for Data Centers (pp. 43–73). doi :10.1016/B978-0-08-100512-5.00002-4

[11] *WDM MSA : Defining laser standards for AI, HPC and high-density optics.* CW. (2023, March 5). https ://cw-wdm.org/

[12] Padmaraju, K., Zhu, X., Chen, L., Lipson, M., & Bergman, K. (2014). Intermodulation Crosstalk Characteristics of WDM Silicon Microring Modulators. *IEEE Photonics Technology Letters*, *26*(14), 1478–1481. https ://doi.org/10.1109/lpt.2014.2326621

[13] Rizzo, A. J., Dai, L. Y., Meng, X., London, Y., Patel, J., Glick, M., & Bergman, K. (2019). Ultra-low power consumption silicon photonic link design analysis in the AIM PDK. *SPIE*, *97*, 7. https ://doi.org/10.1117/12.2508514

[14] de Ridder, R. M., Roeloffzen, C. G. H. (2006). Interleavers. In H. Venghaus (Ed.), Wavelength Filters in Fibre Optics (pp. 381–432). doi :10.1007/3-540-31770-8-10

[15] Rizzo, A., Novick, A., Gopal, V., Kim, B. Y., Ji, X., Daudlin, S., Okawachi, Y., Cheng, Q., Lipson, M., Gaeta, A. L., & Bergman, K. (2023). Massively scalable Kerr comb-driven silicon photonic link. *Nature Photonics*, *17*(9), 781–790. https ://doi.org/10.1038/s41566-023-01244-7

[16] Horst, F., Green, W. M., Assefa, S., Shank, S. M., Vlasov, Y. A., & Offrein, B. J. (2013). Cascaded Mach-Zehnder wavelength filters in silicon photonics for low loss and flat pass-band WDM (de-)multiplexing. *Optics Express*, *21*(10), 11652. https ://-doi.org/10.1364/oe.21.011652

[17] Yi, Q., Zheng, S., Yan, Z., Cheng, G., Xu, F., Li, Q., & Shen, L. (2022). Silicon photonic flat-top WDM (de)multiplexer based on cascaded Mach-Zehnder interferometers for the 2 µm wavelength band. *Optics Express*, *30*(15), 28232. https ://-doi.org/10.1364/oe.467473

[18] Xing, Y., Dong, J., Khan, U., & Bogaerts, W. (2022b). Capturing the Effects of Spatial Process Variations in Silicon Photonic Circuits. *ACS Photonics*. https ://-doi.org/10.1021/acsphotonics.2c01194

[19] Lu, Z., Jhoja, J., Klein, J., Wang, X., Liu, A., Flueckiger, J., Pond, J., & Chrostowski, L. (2017). Performance prediction for silicon photonics integrated circuits with layout-dependent correlated manufacturing variability. *Optics Express*, *25*(9), 9712. https ://-doi.org/10.1364/oe.25.009712

[20] Chrostowski, L., Hochberg, M. (2015). Active Components. In Silicon Photonics Design : From Devices to Systems (pp. 215–216). chapter, Cambridge : Cambridge University Press.

[21] Wang, M., Chen, X., Khan, U., & Bogaerts, W. (2022). Programmable wavelength filter with double ring loaded MZI. *Scientific Reports*, *12*(1). https ://doi.org/10.1038/s41598-021-04598-6

[22] Jacques, M., Samani, A., El-Fiky, E., Patel, D., Xing, Z., & Plant, D. V. (2019). Optimization of thermo-optic phase-shifter design and mitigation of thermal crosstalk on the SOI platform. *Optics Express*, *27*(8), 10456. https ://doi.org/10.1364/oe.27.010456

[23] Watts, M. R., Sun, J., DeRose, C., Trotter, D. C., Young, R. W., & Nielson, G. N. (2013). Adiabatic thermo-optic Mach–Zehnder switch. *Optics Letters*, *38*(5), 733. https ://doi.org/10.1364/ol.38.000733

[24] Harris, N. C., Ma, Y., Mower, J., Baehr-Jones, T., Englund, D., Hochberg, M., & Galland, C. (2014). Efficient, compact and low loss thermo-optic phase shifter in silicon. *Optics Express*, *22*(9), 10487. https ://doi.org/10.1364/oe.22.010487

[25] Song, J., Fang, Q., Tao, S. H., Liow, T. Y., Yu, M. B., Lo, G. Q., & Kwong, D. L. (2008). Fast and low power Michelson interferometer thermo-optical switch on SOI. *Optics Express*, *16*(20), 15304. https ://doi.org/10.1364/oe.16.015304

[26] Van Campenhout, J., Green, W. M. J., Assefa, S., & Vlasov, Y. A. (2010). Integrated NiSi waveguide heaters for CMOS-compatible silicon thermo-optic devices. *Optics Letters*, *35*(7), 1013. https ://doi.org/10.1364/ol.35.001013

[27] Fang, Q., Song, J. F., Liow, T., Cai, H., Yu, M. B., Lo, G. Q., & Kwong, D. (2011). Ultralow Power Silicon Photonics Thermo-Optic Switch With Suspended Phase Arms. *IEEE Photonics Technology Letters*, *23*(8), 525–527. https ://doi.org/10.1109/lpt.2011.2114336

[28] Kudalippalliyalil, R., Murphy, T. E., & Grutter, K. E. (2020). Low-loss and ultra-broadband silicon nitride angled MMI polarization splitter/combiner. *Optics Express*, *28*(23), 34111. https ://doi.org/10.1364/oe.405188

[29] Sacher, W. D., Huang, Y., Lo, G., & Poon, J. K. S. (2015). Multilayer Silicon Nitride-on-Silicon Integrated Photonic Platforms and Devices. *Journal of Lightwave Technology*, *33*(4), 901–910. https ://doi.org/10.1109/jlt.2015.2392784

[30] Rahim, A., Ryckeboer, E., Subramanian, A. Z., Clemmen, S., Kuyken, B., Dhakal, A., Raza, A., Hermans, A., Muneeb, M., Dhoore, S., Li, Y., Dave, U., Bienstman, P., Thomas, N. L., Roelkens, G., Van Thourhout, D., Helin, P., Severi, S., Rottenberg, X., & Baets, R. (2016). Expanding the Silicon Photonics Portfolio With Silicon Nitride Photonic Integrated Circuits. *Journal of Lightwave Technology*, *35*(4), 639–649. https ://doi.org/10.1109/jlt.2016.2617624

[31] Zhang, Z., Yako, M., Ju, K., Kawai, N., Chaisakul, P., Tsuchizawa, T., Hikita, M., Yamada, K., Ishikawa, Y., & Wada, K. (2017). A new material platform of Si photonics for implementing architecture of dense wavelength division multiplexing on Si bulk wafer. *Science and Technology of Advanced Materials*, *18*(1), 283–293. https ://-doi.org/10.1080/14686996.2017.1301193

[32] Olivier, S., Malhouitre, S., Robin-Brosse, D., Garcia, S., Adelmini, L., Plantier, S., Hassan, K., Sciancalepore, C., Fowler, D., & Wilmart, Q. (2018). A hybrid SOI/SiN photonic platform for high-speed and temperature-insensitive CWDM optical transceivers. *Proceedings of the Silicon Photonics XIII*, *98911C*, 7. https ://doi.org/10.1117/12.2289633

[33] Boeuf, F., Baudot, C., Maitre, P., Vulliet, N., Planchot, J., Babaud, L., Carpentier, J., Messaoudene, S., Ben-Khedim, M., Orlandi, P., Durand, C., Fincato, A., & Cremer, S. (2018). Evaluation of Advanced Process Integration Options for next Generation of Si-Photonics platform. *Extended Abstracts of the 2018 International Conference on Solid State Devices and Materials*. https ://doi.org/10.7567/ssdm.2018.h-2-02

[34] Tsakyridis, A., Ruggeri, E., Kalfas, G., Oldenbeuving, R. M., Van Dijk, P. W. L., Roeloffzen, C. G. H., Leiba, Y., Miliou, A., Pleros, N., & Vagionas, C. (2021). Reconfigurable Fiber Wireless IFoF Fronthaul With 60 GHz Phased Array Antenna and Silicon Photonic ROADM for 5G mmWave C-RANs. *IEEE Journal on Selected Areas in Communications*, *39*(9), 2816–2826. https ://doi.org/10.1109/jsac.2021.3064649

[35] Baets, R., Subramanian, A. Z., Clemmen, S., Kuyken, B., Bienstman, P., Thomas, N. L., Roelkens, G., Van Thourhout, D., Helin, P., & Severi, S. (2016). Silicon Photonics : silicon nitride versus silicon-on-insulator. *Optical Fiber Communication Conference*. https ://doi.org/10.1364/ofc.2016.th3j.1

[36] Nejadriahi, H., Friedman, A., Sharma, R., Pappert, S., Fainman, Y., & Yu, P. (2020). Thermo-optic properties of silicon-rich silicon nitride for on-chip applications. *Optics Express*, *28*(17), 24951. https ://doi.org/10.1364/oe.396969

[37] Arbabi, A., & Goddard, L. L. (2013). Measurements of the refractive indices and thermo-optic coefficients of Si_3N_4 and SiO_x using microring resonances. *Optics Letters*, *38*(19), 3878. https ://doi.org/10.1364/ol.38.003878

[38] Komma, J., Schwarz, C., Hofmann, G., Heinert, D., & Nawrodt, R. (2012). Thermo-optic coefficient of silicon at 1550 nm and cryogenic temperatures. *Applied Physics Letters*, *101*(4), 041905. https ://doi.org/10.1063/1.4738989

[39] Wong, C. S., Tsang, H. K., Day, I. E., Harpin, A., Drake, J., & Asghari, M. (n.d.). Two-photon absorption and self-phase modulation in silicon waveguide at 1.5 m wavelength. *Technical Digest. CLEO/Pacific Rim 2001. 4th Pacific Rim Conference on Lasers and Electro-Optics (Cat. No.01TH8557)*, *1*. https ://doi.org/10.1109/cleopr.2001.967989

[40] Seiler, P. M., Georgieva, G., Winzer, G., Peczek, A., Voigt, K., Lischke, S., Fatemi, A., & Zimmermann, L. (2021). Toward coherent O-band data center interconnects. *Frontiers of Optoelectronics*, *14*(4), 414–425. https ://doi.org/10.1007/s12200-021-1242-0

[41] Wilmart, Q., Brision, S., Hartmann, J., Myko, A., Ribaud, K., Petit-Etienne, C., Youssef, L., Fowler, D., Charbonnier, B., Sciancalepore, C., Pargon, E., Bernabe, S., & Szelag, B. (2020). A Complete Si Photonics Platform Embedding Ultra-Low Loss Waveguides for O- and C-Band. *Journal of Lightwave Technology*, *39*(2), 532–538. https ://-doi.org/10.1109/jlt.2020.3030123

[42] Bucio, T. D., Lacava, C., Clementi, M., Faneca, J., Skandalos, I., Baldycheva, A., Galli, M., Debnath, K., Petropoulos, P., & Gardes, F. (2019). Silicon Nitride Photonics for the Near-Infrared. *IEEE Journal of Selected Topics in Quantum Electronics*, *26*(2), 1–13. https ://doi.org/10.1109/jstqe.2019.2934127

[43] Multi-Stage MA Architectures. (1999). In Optical Filter Design and Analysis (pp. 165–236). doi :10.1002/0471213756.ch4

[44] Bogaerts, W., De Heyn, P., Van Vaerenbergh, T., De Vos, K., Selvaraja, S. K., Claes, T., Dumon, P., Bienstman, P., Van Thourhout, D., & Baets, R. (2011). Silicon microring resonators. *Laser & Photonics Review*, *6*(1), 47–73. https ://-doi.org/10.1002/lpor.201100017

[45] Novarese, M., Garcia, S. R., Cucco, S., Adams, D., Bovington, J., & Gioannini, M. (2022). Study of nonlinear effects and self-heating in a silicon microring resonator including a Shockley-Read-Hall model for carrier recombination. *Optics Express*, *30*(9), 14341. https ://doi.org/10.1364/oe.446739

[46] Rizzo, A., Dave, U., Novick, A., Freitas, A., Roberts, S. P., James, A., Lipson, M., & Bergman, K. (2022). Fabrication-robust silicon photonic devices in standard sub-micron silicon-on-insulator processes. *Optics Letters*, *48*(2), 215. https ://-doi.org/10.1364/ol.476873

[47] Philipp, H. (1997). Silicon Nitride (Si3N4) (Noncrystalline). In *Elsevier eBooks* (pp. 771–774). https ://doi.org/10.1016/b978-012544415-6.50040-6

[48] Resende, J., Fuard, D., Cunff, D. L., Tortai, J., & Pelissier, B. (2020). Hybridization of ellipsometry and energy loss spectra from XPS for bandgap and optical constants determination in SiON thin films. *Materials Chemistry and Physics*, *259*, 124000. https ://-doi.org/10.1016/j.matchemphys.2020.124000

[49] Kim, S., Han, K., Wang, C., Jaramillo-Villegas, J. A., Xue, X., Bao, C., Xuan, Y., Leaird, D. E., Weiner, A. M., & Qi, M. (2017). Dispersion engineering and frequency comb

generation in thin silicon nitride concentric microresonators. *Nature Communications*, *8*(1). https ://doi.org/10.1038/s41467-017-00491-x

[50] Dyatlov, M., Dyatlov, M., Stoll, K., Delaye, P., Hu, J., Serna, S., Serna, S., Vivien, L., & Dubreuil, N. (2023, June 26). *Towards two-photon-absorption-free hybrid silicon nitride waveguides reaching silicon Kerr nonlinearity.* https ://opg.optica.org/abstract.cfm ?uri=cleo_europe-2023-cd_3_5

[51] Frey, B. J., Leviton, D. B., & Madison, T. J. (2006). Temperature-dependent refractive index of silicon and germanium. *Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE*. https ://doi.org/10.1117/12.672850

[52] Sze, S.M. and Ng, K.K. (2006). Physics and Properties of Semiconductors—A Review. In Physics of Semiconductor Devices (eds S.M. Sze and K.K. Ng). https ://-doi.org/10.1002/9780470068328.ch1

[53] Sun, T., & Xia, M. (2022). Low loss modified Bezier bend waveguide. *Optics Express*, *30*(7), 10293. https ://doi.org/10.1364/oe.452580

[54] Vogelbacher, F., Nevlacsil, S., Sagmeister, M., Kraft, J., Unterrainer, K., & Hainberger, R. (2019). Analysis of silicon nitride partial Euler waveguide bends. *Optics Express*, *27*(22), 31394. https ://doi.org/10.1364/oe.27.031394

[55] Song, J. H., Kongnyuy, T. D., Stassen, A., Mukund, V., & Rottenberg, X. (2016). Adiabatically Bent Waveguides on Silicon Nitride Photonics for Compact and Dense Footprints. *IEEE Photonics Technology Letters*, *28*(20), 2164–2167. https ://-doi.org/10.1109/lpt.2016.2585760

[56] Siew, S. Y., Li, B., Gao, F., Zheng, H. Y., Zhang, W., Guo, P., Xie, S. W., Song, A., Dong, B., Luo, L. W., Li, C., Luo, X., & Lo, G. (2021). Review of Silicon Photonics Technology and Platform Development. *Journal of Lightwave Technology*, *39*(13), 4374–4389. https ://doi.org/10.1109/jlt.2021.3066203

[57] Huang, Y., Song, J., Luo, X., Liow, T.-Y., & Lo, G.-Q. (2014). CMOS compatible monolithic multi-layer si_3n_4-on-soi platform for low-loss high performance silicon photonics dense integration. *Optics Express*, *22*(18), 21859. https ://doi.org/10.1364/oe.22.021859

[58] Darmawan, S., Lee, N. S., Lee, N. C., & Chin, N. M. (2005). A rigorous comparative analysis of directional couplers and multimode Interferometers based on ridge waveguides. *IEEE Journal of Selected Topics in Quantum Electronics*, *11*(2), 466–475. https ://-doi.org/10.1109/jstqe.2004.846521

[59] Chen, C. (2006). *Foundations for Guided-Wave Optics*. John Wiley & Sons.

[60] Haus, H. A. (1984). *Waves and Fields in Optoelectronics*. Prentice Hall.

[61] Bryngdahl, O. (1973). Image formation using self-imaging techniques. *Journal of the Optical Society of America, 63*(4), 416-419.

[62] Niemeier, T., & Ulrich, R. (1986). Quadrature outputs from fiber interferometer with 4 × 4 coupler. *Optics Letters*, *11*(10), 677. https ://doi.org/10.1364/ol.11.000677

[63] Bachmann, M., Besse, P. A., & Melchior, H. (1994). General self-imaging properties in N × N multimode interference couplers including phase relations. *Applied Optics*, *33*(18), 3905. https ://doi.org/10.1364/ao.33.003905

[64] Soldano, L., & Pennings, E. (1995). Optical multi-mode interference devices based on self-imaging : principles and applications. *Journal of Lightwave Technology*, *13*(4), 615–627. https ://doi.org/10.1109/50.372474

[65] Yao, C., Cheng, Y., Wu, J., Hong, X., Xu, K., Zhao, L., Li, W., Zuo, Y., Guo, H., Li, Y., Wang, W., & Lin, J. (2010). A rigorous comparative analysis of general and restricted interference couplers. *OECC 2010 Technical Digest*, 398–399. http ://yadda.icm.edu.pl/yadda/element/bwmeta1.element.ieee-000005588612

[66] Vogelbacher, F., Nevlacsil, S., Sagmeister, M., Kraft, J., Unterrainer, K., & Hainberger, R. (2019b). Analysis of silicon nitride partial Euler waveguide bends. *Optics Express*, *27*(22), 31394. https ://doi.org/10.1364/oe.27.031394

[67] Rizzo, A., Dave, U., Novick, A., Freitas, A., Roberts, S. P., James, A., Lipson, M., & Bergman, K. (2022b). Fabrication-robust silicon photonic devices in standard sub-micron silicon-on-insulator processes. *Optics Letters*, *48*(2), 215. https ://-doi.org/10.1364/ol.476873

[68] Xiao, Z., Luan, F., Liow, T., Zhang, J., & Shum, P. (2012). Design for broadband high-efficiency grating couplers. *Optics Letters*, *37*(4), 530. https ://-doi.org/10.1364/ol.37.000530

[69] Qin, N. K., Gao, N. D., Bao, N. C., Zhao, N. Z., Zhou, N. X., Lu, N. T., & Chen, N. L. (2012). High Efficiency and Broadband Two-Dimensional Blazed Grating Coupler With Fully Etched Triangular Holes. *Journal of Lightwave Technology*, *30*(14), 2363–2366. https ://doi.org/10.1109/jlt.2012.2198431

[70] Xu, D., Cheben, P., Dalacu, D., Delâge, A., Janz, S., Lamontagne, B., Picard, M., & Ye, W. N. (2004). Eliminating the birefringence in silicon-on-insulator ridge waveguides by use of cladding stress. *Optics Letters*, *29*(20), 2384. https ://-doi.org/10.1364/ol.29.002384

[71] Bozzola, A., Carroll, L., Gerace, D., Cristiani, I., & Andreani, L. C. (2015). Optimising apodized grating couplers in a pure SOI platform to 05 dB coupling efficiency. *Optics Express*, *23*(12), 16289. https ://doi.org/10.1364/oe.23.016289

[72] Zhang, J. Z. J., Yang, J. Y. J., Lu, H. L. H., Wu, W. W. W., Huang, J. H. J., & Chang, S. C. S. (2015). Polarization-independent grating coupler based on silicon-on-insulator. *Chinese Optics Letters*, *13*(9), 091301–091305. https ://doi.org/10.3788/col201513.091301

[73] Wang, Y., Shi, W., Wang, X., Lu, Z., Caverley, M., Bojko, R., Chrostowski, L., & Jaeger, N. a. F. (2015). Design of broadband subwavelength grating couplers with low back reflection. *Optics Letters*, *40*(20), 4647. https ://doi.org/10.1364/ol.40.004647

[74] Mu, X., Wu, S., Cheng, L., & Fu, H. Y. (2020). Edge couplers in silicon photonic integrated circuits : A review. *Applied Sciences*, *10*(4), 1538. https ://doi.org/10.3390/app10041538

[75] Papes, M., Cheben, P., Benedikovic, D., Schmid, J. H., Pond, J., Halir, R., Ortega-Moñux, A., Wangüemert-Pérez, G., Ye, W. N., Xu, D., Janz, S., Dado, M., & Vašinek, V. (2016). Fiber-chip edge coupler with large mode size for silicon photonic wire waveguides. *Optics Express*, *24*(5), 5026. https ://doi.org/10.1364/oe.24.005026

[76] Romero-Garcia, S., Marzban, B., Merget, F., Shen, B., & Witzens, J. (2014). Edge Couplers With Relaxed Alignment Tolerance for Pick-and-Place Hybrid Integration of III–V Lasers With SOI Waveguides. *IEEE Journal of Selected Topics in Quantum Electronics*, *20*(4), 369–379. https ://doi.org/10.1109/jstqe.2013.2292523

[77] Tao, H., Song, J., Fang, Q., Yu, M., Lo, G., & Kwong, D. (2008). Improving coupling efficiency of fiber-waveguide coupling with a double-tip coupler. *Optics Express*, *16*(25), 20803. https ://doi.org/10.1364/oe.16.020803

[78] Hatori, N., Shimizu, T., Okano, M., Ishizaka, M., Yamamoto, T., Urino, Y., Mori, M., Nakamura, T., & Arakawa, Y. (2014). A Hybrid Integrated Light Source on a Silicon Platform Using a Trident Spot-Size Converter. *Journal of Lightwave Technology*, *32*(7), 1329–1336. https ://doi.org/10.1109/jlt.2014.2304305

[79] Lai, Y., Yu, Y., Fu, S., Xu, J., Shum, P. P., & Zhang, X. (2017). Efficient spot size converter for higher-order mode fiber-chip coupling. *Optics Letters*, *42*(18), 3702. https ://doi.org/10.1364/ol.42.003702

[80] Yasuhara, K., Yu, F., & Ishigure, T. (2017). Polymer waveguide based spot-size converter for low-loss coupling between Si photonics chips and single-mode fibers. *Optical Fiber Communication Conference*. https ://doi.org/10.1364/ofc.2017.th2a.2

[81] Snyder, B., et al. (2018). Broadband, polarization-insensitive lensed edge couplers for silicon photonics. In *2018 IEEE 68th Electronic Components and Technology Conference (ECTC)* (pp. 841–847). San Diego, CA, USA. IEEE. doi : 10.1109/ECTC.2018.00130

[82] Jia, L., Li, C., Liow, T., & Lo, G. (2017). Efficient Suspended Coupler With Loss Less Than 1.4 dB Between Si-Photonic Waveguide and Cleaved Single Mode Fiber. *Journal of Lightwave Technology*, *36*(2), 239–244. https ://doi.org/10.1109/jlt.2017.2779863

[83] Larrea, R., Gutierrez, A. M., Griol, A., Brimont, A., & Sanchis, P. (2018). Fiber-to-Chip Spot-Size Converter for Coupling to Silicon Waveguides in the O-Band. *IEEE Photonics Technology Letters*, *31*(1), 31–34. https ://doi.org/10.1109/lpt.2018.2881334

[84] Madsen, C. K., & Zhao, J. H. (1999). *Optical Filter Design and Analysis*. https ://-doi.org/10.1002/0471213756

[85] Venghaus, H. (2006). Wavelength filters in fibre optics. In SPringer series in optical sciences. https ://doi.org/10.1007/3-540-31770-8

[86] Madsen, C. K. (2000). General IIR optical filter design for WDM applications using all-pass filters. *Journal of Lightwave Technology, 18*(6), 860–868. doi : 10.1109/50.848399

[87] Rizzo, A., Cheng, Q., Daudlin, S., & Bergman, K. (2020). Ultra-Broadband Interleaver for Extreme Wavelength Scaling in Silicon Photonic Links. *IEEE Photonics Technology Letters*, *33*(1), 55–58. https ://doi.org/10.1109/lpt.2020.3044262

[88] Luo, L., Ibrahim, S., Nitkowski, A., Ding, Z., Poitras, C. B., Yoo, S. J. B., & Lipson, M. (2010). High bandwidth on-chip silicon photonic interleaver. *Optics Express*, *18*(22), 23079. https ://doi.org/10.1364/oe.18.023079

[89] Mathews, I., Abdullaev, A., Lei, S., Enright, R., Wallace, M. J., & Donegan, J. F. (2015). Reducing thermal crosstalk in ten-channel tunable slotted-laser arrays. *Optics Express*, *23*(18), 23380. https ://doi.org/10.1364/oe.23.023380

[90] Moscoso-Mártir, A., Tabatabaei-Mashayekh, A., Müller, J., Nojić, J., Setter, R., Nielsen, M., Sandomirsky, A., Rockman, S., Mentovich, E., Merget, F., Garreau, A., Lelarge, F., & Witzens, J. (2018). 8-channel WDM silicon photonics transceiver with SOA and semiconductor mode-locked laser. *Optics Express*, *26*(19), 25446. https ://-doi.org/10.1364/oe.26.025446

[91] Moscoso-Mártir, A., Müller, J., Hauck, J., Chimot, N., Setter, R., Badihi, A., Rasmussen, D. E., Garreau, A., Nielsen, M., Islamova, E., Romero-García, S., Shen, B., Sandomirsky, A., Rockman, S., Li, C., Azadeh, S. S., Lo, G., Mentovich, E., Merget, F., . . . Witzens, J. (2017). Silicon Photonics Transmitter with SOA and Semiconductor Mode-Locked Laser. *Scientific Reports*, *7*(1). https ://doi.org/10.1038/s41598-017-14347-3

[92] Hu, H., & Oxenløwe, L. K. (2021). Chip-based optical frequency combs for high-capacity optical communications. *Nanophotonics*, *10*(5), 1367–1385. https ://doi.org/10.1515/nanoph-2020-0561

[93] Gaeta, A. L., Lipson, M., & Kippenberg, T. J. (2019). Photonic-chip-based frequency combs. *Nature Photonics*, *13*(3), 158–169. https ://doi.org/10.1038/s41566-019-0358-x

[94] Marin-Palomo, P., Kemal, J. N., Karpov, M., Kordts, A., Pfeifle, J., Pfeiffer, M. H. P., Trocha, P., Wolf, S., Brasch, V., Anderson, M. H., Rosenberger, R., Vijayan, K., Freude, W., Kippenberg, T. J., & Koos, C. (2017). Microresonator-based solitons for massively parallel coherent optical communications. *Nature*, *546*(7657), 274–279. https ://-doi.org/10.1038/nature22387

[95] Levy, J. S., Gondarenko, A., Foster, M. A., Turner-Foster, A. C., Gaeta, A. L., & Lipson, M. (2009). CMOS-compatible multiple-wavelength oscillator for on-chip optical interconnects. *Nature Photonics*, *4*(1), 37–40. https ://doi.org/10.1038/nphoton.2009.259

[96] Kippenberg, T. J., Gaeta, A. L., Lipson, M., & Gorodetsky, M. L. (2018). Dissipative Kerr solitons in optical microresonators. *Science*, *361*(6402). https ://-doi.org/10.1126/science.aan8083

[97] Lundberg, L., Karlsson, M., Lorences-Riesgo, A., Mazur, M., Torres-Company, V., Schröder, J., & Andrekson, P. A. (2018). Frequency Comb-Based WDM Transmission Systems Enabling Joint Signal Processing. *Applied Sciences*, *8*(5), 718. https ://-doi.org/10.3390/app8050718

[98] Hermans, A., Van Gasse, K., & Kuyken, B. (2022). On-chip optical comb sources. *APL Photonics*, *7*(10). https ://doi.org/10.1063/5.0105164

[99] Suh, M., Yang, Q., Yang, K. Y., Yi, X., & Vahala, K. J. (2016). Microresonator soliton dual-comb spectroscopy. *Science*, *354*(6312), 600–603. https ://-doi.org/10.1126/science.aah6516

[100] Newman, Z. L., Maurice, V., Drake, T., Stone, J. R., Briles, T. C., Spencer, D. T., Fredrick, C., Li, Q., Westly, D., Ilic, B. R., Shen, B., Suh, M., Yang, K. Y., Johnson, C., Johnson, D. M. S., Hollberg, L., Vahala, K. J., Srinivasan, K., Diddams, S. A., . . . Hummon, M. T. (2019). Architecture for the photonic integration of an optical atomic clock. *Optica*, *6*(5), 680. https ://doi.org/10.1364/optica.6.000680

[101] Liang, W., Eliyahu, D., Ilchenko, V. S., Savchenkov, A. A., Matsko, A. B., Seidel, D., & Maleki, L. (2015). High spectral purity Kerr frequency comb radio frequency photonic oscillator. *Nature Communications*, *6*(1). https ://doi.org/10.1038/ncomms8957

[102] Xu, X., Tan, M., Corcoran, B., Wu, J., Nguyen, T. G., Boes, A., Chu, S. T., Little, B. E., Morandotti, R., Mitchell, A., Hicks, D. G., & Moss, D. J. (2020). Photonic Perceptron Based on a Kerr Microcomb for High-Speed, Scalable, Optical Neural Networks. *Laser & Photonics Review*, *14*(10). https ://doi.org/10.1002/lpor.202000070

[103] Kim, B. Y., Okawachi, Y., Jang, J. K., Yu, M., Ji, X., Zhao, Y., Joshi, C., Lipson, M., & Gaeta, A. L. (2019). Turn-key, high-efficiency Kerr comb source. *Optics Letters, 44*(18), 4475–4478.

[104] Reed, G. T., Mashanovich, G., Gardes, F. Y., & Thomson, D. J. (2010). Silicon optical modulators. *Nature Photonics*, *4*(8), 518–526. https ://-doi.org/10.1038/nphoton.2010.179

[105] Li, G., Krishnamoorthy, A. V., Shubin, I., Yao, J., Luo, Y., Thacker, H., Zheng, X., Raj, K., & Cunningham, J. E. (2013). Ring Resonator Modulators in Silicon for Interchip Photonic Links. *IEEE Journal of Selected Topics in Quantum Electronics*, *19*(6), 95–113. https ://doi.org/10.1109/jstqe.2013.2278885

[106] Xu, Q., Schmidt, B., Shakya, J., & Lipson, M. (2006). Cascaded silicon micro-ring modulators for WDM optical interconnection. *Optics Express*, *14*(20), 9431. https ://-doi.org/10.1364/oe.14.009431

[107] Guha, B., Preston, K., & Lipson, M. (2012). Athermal silicon microring electro-optic modulator. *Optics Letters*, *37*(12), 2253. https ://doi.org/10.1364/ol.37.002253

[108] Müller, J., Merget, F., Azadeh, S. S., Hauck, J., García, S. R., Shen, B., & Witzens, J. (2014). Optical Peaking Enhancement in High-Speed Ring Modulators. *Scientific Reports*, *4*(1). https ://doi.org/10.1038/srep06310

[109] Padmaraju, K., Zhu, X., Chen, L., Lipson, M., & Bergman, K. (2014b). Intermodulation Crosstalk Characteristics of WDM Silicon Microring Modulators. *IEEE Photonics Technology Letters*, *26*(14), 1478–1481. https ://doi.org/10.1109/lpt.2014.2326621

[110] Kim, Y., Jin, T., & Bae, Y. (2021). A comparative simulation study on lateral and L-shaped PN junction phase shifters for single-drive 50 Gbps lumped Mach–Zehnder modulators. *Japanese Journal of Applied Physics*, *60*(5), 052002. https ://doi.org/10.35848/1347-4065/abeedd

[111] Bahadori, M., Bergman, K., Nikdast, M., Rumley, S., Dai, L. Y., Janosik, N., Van Vaerenbergh, T., Gazman, A., Cheng, Q., & Polster, R. (2018). Design Space Exploration of Microring Resonators in Silicon Photonic Interconnects : Impact of the Ring Curvature. *Journal of Lightwave Technology*, *36*(13), 2767–2782. https ://-doi.org/10.1109/jlt.2018.2821359

[112] Rizzo, A., Daudlin, S., Novick, A., James, A., Gopal, V., Murthy, V., Cheng, Q., Kim, B. Y., Ji, X., Okawachi, Y., Van Niekerk, M., Deenadayalan, V., Leake, G., Fanto, M., Preble, S., Lipson, M., Gaeta, A., & Bergman, K. (2022). Petabit-Scale Silicon Photonic Interconnects With Integrated Kerr Frequency Combs. *IEEE Journal of Selected Topics in Quantum Electronics*, *29*(1), 1–20. https ://doi.org/10.1109/jstqe.2022.3197375

[113] London, Y., Van Vaerenbergh, T., Rizzo, A. J., Sun, P., Hulme, J., Kurczveil, G., Seyedi, A., Wang, B., Zeng, X., Huang, Z., Rhim, J., Fiorentino, M., & Bergman, K. (2019). Energy Efficiency Analysis of Comb Source Carrier-Injection Ring-Based Silicon Photonic Link. *IEEE Journal of Selected Topics in Quantum Electronics*, *26*(2), 1–13. https ://doi.org/10.1109/jstqe.2019.2934121

[114] Rizzo, A. (2022). *Highly parallel silicon photonic links with integrated Kerr frequency combs* (Doctoral dissertation, Columbia University). ProQuest Dissertations and Theses. https ://doi.org/10.5555/AAI29215973

[115] Kim, J., Kim, J., Yoon, J., Yoon, H., Park, H., & Kurt, H. (2022). Experimental demonstration of inverse-designed silicon integrated photonic power splitters. *Nanophotonics*, *11*(20), 4581–4590. https ://doi.org/10.1515/nanoph-2022-0443

[116] B, M., Saleem, H. I., & Sadasivan, V. (2022). Inverse Design of Silicon Photonic Modulators. *Journal of Lightwave Technology*, *40*(20), 6939–6945. https ://-doi.org/10.1109/jlt.2022.3189098

[117] Zhang, E., Zhang, S., Yang, T., Zhu, X., Chen, L., Zhang, Y., Yang, X., & Zhang, L. (2023). Improved Particle Swarm Optimization With Less Manual Intervention for Photonic Inverse Design. *IEEE Photonics Technology Letters*, *35*(24), 1355–1358. https ://-doi.org/10.1109/lpt.2023.3325635

[118] Skarda, J., Su, L., Yang, K. Y., Vercruysse, D., Sapra, N. V., & Vučković, J. (2019). From inverse design to implementation of practical photonics. In *Proceedings of the 2019 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)* (pp. 1-4). IEEE. https ://doi.org/10.1109/ICCAD45719.2019.8942167

[119] Pond, J., & Kawano, M. (2010). Virtual prototyping and optimization of novel solar cell designs. *Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE*. https ://doi.org/10.1117/12.873114

[120] Mitsunori, K., & James, P. (2010). *Design optimization of photonic crystal organic solar cells using the FDTD method in combination with particle swarm optimization*. Article Information, J-GLOBAL. https ://jglobal.jst.go.jp/en/detail ?JGLOBAL_ID=201002288639785900

[121] Mutitu, J. G., Shi, S., Chen, C., Creazzo, T., Barnett, A., Honsberg, C., & Prather, D. W. (2008). Thin film silicon solar cell design based on photonic crystal and diffractive grating structures. *Optics Express, 16*(20), 15238-15248. https ://doi.org/10.1364/OE.16.015238

[122] Shokooh-Saremi, M., & Magnusson, R. (2010). Leaky-mode resonant reflectors with extreme bandwidths. *Optics Letters*, *35*(8), 1121. https ://doi.org/10.1364/ol.35.001121

[123] Magnusson, R., Shokooh-Saremi, M., & Johnson, E. G. (2010). Guided-mode resonant wave plates. *Optics Letters*, *35*(14), 2472. https ://doi.org/10.1364/ol.35.002472

[124] Gad, A. G. (2022). Particle swarm optimization algorithm and its applications : A systematic review. *Archives of Computational Methods in Engineering, 29*, 2531–2561. https ://doi.org/10.1007/s11831-021-09694-4

[125] Chrostowski L, Hochberg M. Modulators. In : *Silicon Photonics Design : From Devices to Systems*. Cambridge : Cambridge University Press ; 2015 :217-258. doi :10.1017/CBO9781316084168.007

[126] Zhao, X., Lee, S., Choi, J., Lee, S., Shin, I., Jeon, C., Kim, B., & Cho, H. (2014). Dependency analysis of line edge roughness in electron-beam lithography. *Microelectronic Engineering*, *133*, 78–87. https ://doi.org/10.1016/j.mee.2014.11.017

[127] Pala, N., Karabiyik, M. (2012). Electron Beam Lithography (EBL). In : Bhushan, B. (eds) Encyclopedia of Nanotechnology. Springer, Dordrecht. https ://doi-org.acces.bibl.ulaval.ca/10.1007/978-90-481-9751-4-344

[128] Ren, L., & Chen, B. (2004). Proximity effect in electron beam lithography. In *Proceedings of the 7th International Conference on Solid-State and Integrated Circuits Technology* (Vol. 1, pp. 579-582). IEEE. https ://doi.org/10.1109/ICSICT.2004.1435073

[129] Kruit, P., & Steenbrink, S. W. H. K. (2006). Shot noise in electron-beam lithography and line-width measurements. *Scanning*, *28*(1), 20–26. https ://-doi.org/10.1002/sca.4950280104

[130] Darcie, A. (2022). *Diagnosing electron beam lithography quality using optical ring resonators* (T). University of British Columbia. Retrieved from https ://open.library.ubc.ca/collections/ubctheses/24/items/1.0417524

[131] Xing, Y., Dong, J., Khan, U., & Bogaerts, W. (2022c). Capturing the Effects of Spatial Process Variations in Silicon Photonic Circuits. *ACS Photonics*. https ://-doi.org/10.1021/acsphotonics.2c01194

[132] Han, Q., Robin, D., Gervais, A., Ménard, M., & Shi, W. (2022). Phase errors and statistical analysis of silicon-nitride arrayed waveguide gratings. *Optics Express*, *30*(24), 42784. https ://doi.org/10.1364/oe.467841

[133] Han, Q., Robin, D., Gervais, A., Menard, M., & Shi, W. (2022). Statistical analysis of silicon-nitride arrayed waveguide gratings. *2022 IEEE Photonics Conference (IPC)*, 1–2. https ://doi.org/10.1109/ipc53466.2022.9975494

[134] Sabri, L., Ahdab, R. E., Nabki, F., & Ménard, M. (2024). Broadband SiN Interleaver with a Ring Assisted MZI using a Tapered MMI Coupler. *IEEE Photonics Journal*, *16*(5), 1–5. https ://doi.org/10.1109/jphot.2024.3444825

[135] Luo, L., Ibrahim, S., Nitkowski, A., Ding, Z., Poitras, C. B., Yoo, S. J. B., & Lipson, M. (2010). High bandwidth on-chip silicon photonic interleaver. *Optics Express*, *18*(22), 23079. https ://doi.org/10.1364/oe.18.023079

[136] Rizzo, A., Cheng, Q., Daudlin, S., & Bergman, K. (2020). Ultra-Broadband Interleaver for Extreme Wavelength Scaling in Silicon Photonic Links. *IEEE Photonics Technology Letters*, *33*(1), 55–58. https ://doi.org/10.1109/lpt.2020.3044262

[137] Zhou, N., Zheng, S., Long, Y., Ruan, Z., Shen, L., & Wang, J. (2018). Reconfigurable and tunable compact comb filter and (de)interleaver on silicon platform. *Optics Express*, *26*(4), 4358. https ://doi.org/10.1364/oe.26.004358

[138] Song, J., Fang, Q., Tao, S. H., Yu, M. B., Lo, G. Q., & Kwong, D. L. (2008). Passive ring-assisted Mach-Zehnder interleaver on silicon-on-insulator. *Optics Express*, *16*(12), 8359. https ://doi.org/10.1364/oe.16.008359

[139] J. Song, Q. Fang, S. H. Tao, M. B. Yu, G. Q. Lo, and D. L. Kwong, "Proposed silicon wire interleaver structure," Opt. Express, vol. 16, no. 11, p. 7849, May 2008, doi : 10.1364/OE.16.007849.

[140] J. Song et al., "Effective thermo-optical enhanced cross-ring resonator MZI interleavers on SOI," Opt. Express, vol. 16, no. 26, p. 21476, Dec. 2008, doi : 10.1364/OE.16.021476.

[141] Cheung, S., Yuan, Y., Peng, Y., Kurczveil, G., Srinivasan, S., Hu, Y., Descos, A., Liang, D., & Beausoleil, R. G. (2022). Demonstration of a 17 × 25 Gb/s Heterogeneous III-V/Si DWDM Transmitter based on (De-) Interleaved Quantum Dot Optical Frequency Combs. *Journal of Lightwave Technology*, 1–10. https ://-doi.org/10.1109/jlt.2022.3196914

[142] Li, X. P., Chen, K. X., & Wang, L. F. (2018). Compact and electro-optic tunable interleaver in lithium niobate thin film. *Optics Letters*, *43*(15), 3610. https ://-doi.org/10.1364/ol.43.003610

[143] Jiang, X., Zhang, H., Zhang, Y., Qiu, C., & Su, Y. (2018). Compact CWDM interleaver based on an interfering loop containing a one-dimensional Fabry–Perot cavity. *Optics Letters*, *43*(5), 1071. https ://doi.org/10.1364/ol.43.001071

[144] Jiang, X., Yang, Y., Zhang, H., Peng, J., Zhang, Y., Qiu, C., & Su, Y. (2017). Design and Experimental Demonstration of a Compact Silicon Photonic Interleaver Based on an Interfering Loop With Wide Spectral Range. *Journal of Lightwave Technology*, *35*(17), 3765–3771. https ://doi.org/10.1109/jlt.2017.2720188

[145] Uenuma, M., & Motooka, T. (2009). Temperature-independent silicon waveguide optical filter. *Optics Letters*, *34*(5), 599. https ://doi.org/10.1364/ol.34.000599

[146] Guha, B., Gondarenko, A., & Lipson, M. (2010). Minimizing temperature sensitivity of silicon Mach-Zehnder interferometers. *Optics Express*, *18*(3), 1879. https ://doi.org/10.1364/oe.18.001879

[147] Dwivedi, S., D'heer, H., & Bogaerts, W. (2013). A Compact All-Silicon Temperature Insensitive Filter for WDM and Bio-Sensing Applications. *IEEE Photonics Technology Letters*, *25*(22), 2167–2170. https ://doi.org/10.1109/lpt.2013.2282715

[148] Guan, X., & Frandsen, L. H. (2019). All-silicon interferometer with multimode waveguides for temperature-insensitive filters and compact biosensors. *Optics Express*, *27*(2), 753. https ://doi.org/10.1364/oe.27.000753

[149] Xing, P., & Viegas, J. (2015). Broadband CMOS-compatible SOI temperature insensitive Mach-Zehnder interferometer. *Optics Express*, *23*(19), 24098. https ://doi.org/10.1364/oe.23.024098

[150] Xing, P., & Viegas, J. (2016). Subwavelength grating waveguide-integrated athermal Mach-Zehnder interferometer with enhanced fabrication error tolerance and wide stable spectral range. *Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE*, *9752*, 97520U. https ://doi.org/10.1117/12.2214101

[151] Hiraki, T., Fukuda, H., Yamada, K., & Yamamoto, T. (2015). Small Sensitivity to Temperature Variations of Si-Photonic Mach–Zehnder Interferometer Using Si and SiN Waveguides. *Frontiers in Materials*, *2*. https ://doi.org/10.3389/fmats.2015.00026

[152] Guha, B., Kyotoku, B. B. C., & Lipson, M. (2010). CMOS-compatible athermal silicon microring resonators. *Optics Express*, *18*(4), 3487. https ://doi.org/10.1364/oe.18.003487

[153] Teng, J., Dumon, P., Bogaerts, W., Zhang, H., Jian, X., Han, X., Zhao, M., Morthier, G., & Baets, R. (2009). Athermal Silicon-on-insulator ring resonators by overlaying a polymer cladding on narrowed waveguides. *Optics Express*, *17*(17), 14627. https ://doi.org/10.1364/oe.17.014627

[154] Wang, M., Chen, X., Khan, U., & Bogaerts, W. (2022b). Programmable wavelength filter with double ring loaded MZI. *Scientific Reports*, *12*(1). https ://doi.org/10.1038/s41598-021-04598-6